



Service

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Interaction Center

The interaction center provides multichannel collaboration and communication tools to enable efficient and consistent customer service.

Interaction center (IC) agents can handle inbound and outbound service transactions using the phone, e-mail, fax, chat, and letter communication channels. They can process business transactions, such as service orders and enhance their productivity by using alerts and a knowledge search. All relevant account information is available to them in the IC, such as account data, service order status, and product-related information.

The IC provides agents with the tools they need for the processes that comprise their daily work. Generally, the IC supports the following processes:

- Communication process

This business process involves the communication between an IC agent and a business partner (customer, employee, and so on) in real time, for example using the telephony channel. Since IC agents can also work asynchronously, processing e-mails, service requests, or workflow items from the agent inbox, the communication process is not necessarily performed at all times.

- Interaction process

The interaction process is the central IC process, which often encompasses the communication process and the business transaction process, and is performed whenever an agent uses the IC to communicate or interact – either directly with a business partner or asynchronously by working on a business transaction belonging to a business partner.

The interaction process starts in the interaction mode with either the identification and confirmation of the business partner or the selection of a business transaction, e-mail, workflow, or other item belonging to a business partner. The interaction process includes all steps that the agent takes to work on a customer request. The interaction process can optionally result in the creation of one or more business transactions.

- Business transaction process

Some simple customer or employee requests can certainly be resolved by providing a short answer or other information, not making use of a business transaction. However, frequently an interaction will result in one or more follow-on business transactions, such as a service request. In such cases, each business transaction is linked to the interaction record, providing a connection between the business process and the interaction process.

To start the interaction center, you use the WUI transaction and select a corresponding business role, such as **Service IC Agent**.

Related Information

[Interaction Center WebClient](#)

[E-Mail Response Management System](#)

[Integrated Communication Interface](#)

Interaction Center WebClient

Use

The interaction center (IC) WebClient is a thin-client, highly optimized desktop application for interaction center agents. It provides a comprehensive set of interaction center (IC) functions and supports various processes required to answer customer queries and

work in corresponding business transactions.

Features

- [Account Identification](#)

A customer search function that agents use to identify customers in the system. You can set up this function to identify customers even before agents accept contacts, based on automatic number identification (ANI) or the address of an incoming e-mail.

- [Account Overview](#)

An overview that agents use to get complete information for an entire account, including individual account details, notes, roles, planned activities, and business partner relationships.

- [Interaction Record](#)

A customer-focused overview of the current interaction and customer history.

- Interaction History

A comprehensive interaction history that provides one view of a customer. Search and overview of business transactions for a customer. Agents can use various search criteria to find the business transactions they are looking for. Agents can record data both during and immediately after an interaction.

- Knowledge Article Search

Interaction center agents can search for knowledge articles, which can include documents, links to Web sites, and screenshots, to answer customer queries.

- [Agent Inbox](#)

A central worklist that allows agents to process incoming and outgoing mails, planned activities, and other work items.

- [E-Mail](#)

An interface that agents use for processing incoming and outgoing e-mails, as well as completing unfinished e-mails.

- [Chat](#)

An interface that agents use for processing incoming chat sessions.

- [Scratch Pad](#)

An electronic notepad that agents use to record miscellaneous information during an interaction. Notes from the scratch pad can be imported into fields in the IC WebClient.

- [Alerts](#)

Functions that combine system resources intelligently to guide agents during customer interactions.

- Queue Status

A feature that agents use to quickly display the channels and queues they are assigned to, based upon configuration in the communication management software.

Specifics for Business Transactions in the Interaction Center

Use

This topic summarizes the specifics that apply to the processing of business transactions in the interaction center (IC).

Features

- The information from a confirmed account is automatically copied to any newly created business transaction.
- The business transactions that you use in the IC have a tiled layout. Alternatively, you can configure an overview layout for these business transactions in Customizing for the UI framework.

Settings

- Business transaction profile in IC

You can make the following settings in Customizing for [Service](#) under [Interaction Center WebClient](#) [Business Transaction](#) [Define Business Transaction Profiles](#):

- Define which transaction types are available to the agent to create business transactions, and which transaction type is to be the default
- Define whether dialog boxes are allowed and displayed automatically (automatic display is only supported by the interaction record)
- Define the transaction type for transaction data and the additional business transactions separately

- Automatic routing

Set up automatic routing. For information about the necessary settings in the rule modeler, see [Interaction Center WebClient](#) [Business Transaction](#) [Define Automatic Routing](#).

Sales Transactions

Get an overview on how to create and change sales orders and sales quotations, display item proposals, as well as display and print sales transactions.

To use the sales transactions, you use the WUI transaction and select a corresponding role, such as [Sales Professional](#).

[Entering Sales Documents on the WebClient UI](#)

[Item Proposals in Sales Transactions](#)

[Searching for, Displaying and Printing Sales Transactions on the WebClient UI](#)

Entering Sales Documents on the WebClient UI

Use

You use this function to create and change sales orders and sales quotations.

You carry out your presales activities such as Opportunities and Activities on the WebClient UI, and then, for example, process a rush order without having to exit.

This function is of particular use to sales employees who work mainly with the WebClient UI and therefore do not require the full range of functions offered by Sales Quotation and Order Management. For example, sales assistants can use this function to

Prerequisites

You can directly create and change a sales document on the WebClient UI once you have carried out the following tasks:

- You have assigned the relevant sales documents to a profile in Customizing under [▮▮Service ▸ Transactions ▸ Settings for Sales Transactions on WebClient UI ▸ Define Profiles for Sales Transactions ▮](#).
- You have defined copy control of transaction types in Customizing for [Service](#) under [▮▮Transactions ▸ Settings for Sales Transactions on WebClient UI ▸ Copy Control for Opportunities and Sales Transactions ▮](#).
- If you want to use payment cards in sales orders, you have made settings in Customizing under [▮▮Sales and Distribution ▸ Billing ▸ Payment Cards ▮](#).

Features

When you maintain a sales document using the WebClient UI, the system enters data directly in SAP. This means that you can decide yourself which UI you use to process the sales order.

Creating Sales Quotations and Sales Orders as Follow-Up Transactions from an Opportunity

The sales representative or sales assistant has determined a potential sales opportunity, and has therefore created an opportunity. If the customer then requests a quotation, the sales employee can immediately create a sales quotation. To do so, the sales employee opens the opportunity that he or she created earlier, and creates the sales quotation as a follow-up transaction from the opportunity.

When you do so, the following data is copied to the sales quotation:

- Business partners
- Sales organization
- Distribution channel
- Division
- Sales office
- Sales group
- Product configuration
- Items
- Product
- Quantity
- Notes
- Unit

You can only display the following fields:

- Product
- Product ID

- Transfer
- Item number

You can navigate to the sales document using the transaction history in the follow-up activity.

Offering Alternative Items in Sales Quotations

In addition to the item requested by the customer, the sales employee can also offer an alternative item in the sales quotation. In order to offer the alternative item, when the sales employee copies the standard item and the alternative item from the opportunity to the sales quotation, the sales employee must make sure that the alternative item comes directly after (by item number) the standard item in the list of items.

Creating a Follow-Up Activity from a Sales Document

You can create a follow-up activity from a sales document. You can then navigate to the sales document using the transaction history of the follow-up activity. The system shows the follow-up activity in the following:

- Transaction history of sales orders and sales quotations
- Document flow shown in SAP GUI (transactions VA02, VA22, and VA42)

Correcting Errors

The system handles errors according to [Sales and Distribution](#) rules. This means that if errors occur, you cannot save the sales order. You can only continue processing the sales order after the errors have been corrected.

i Note

The assignment of digital payment cards is currently not supported on the WebClient UI.

Activities

Certain restrictions apply to this function. Therefore, it should only be used by sales employees who work with sales documents occasionally.

Example

You create a sales quotation as a follow-up transaction for an opportunity. The system copies data directly from the opportunity to the sales quotation.

Item Proposals in Sales Transactions

Use

- Item proposals are used as input help in sales orders and quotations.
- Item proposals are available on the WebClient UI for Service, including the Interaction Center (IC).

Prerequisites

You have set up the item proposals in sales transactions in Customizing by choosing **►Service ► Transactions ► Settings for Sales Transactions on WebClient UI ► Product Proposals ►**. Under the node **Past Orders**, you make the settings for item proposals from previous orders.

Features

Item proposals are displayed directly in the **Items** assignment block.

You can automatically display item proposals, or manually trigger the display. To manually trigger the display of proposals, set the indicator **Past Orders on Request** in the Customizing of product proposals, in the activity **Assign Method Schema to Transaction Type**.

Item proposals appear at the end of the item list as suggested items, that is without item numbers. You can transfer the suggested items as final order items, by entering an order amount. Only then can the system carry out pricing and an availability check for the items.

- **Item Proposals from Past Orders**

The system proposes items from the previous sales transactions for this customer, that is products and order quantities.

- You can define the transaction types and the number of transactions in a certain time frame that the system draws from to determine proposed items, for example the last 10 sales orders from the past three months.
- You can define how the system calculates the proposed quantities: from the last order, or as an average quantity from the previously defined number of orders.
- You can select the required transactions using a popup, or the system can immediately (after you have entered the customer data) display all products from the past transactions.

Searching for, Displaying and Printing Sales Transactions on the WebClient UI

Searching for and Displaying Sales Transactions on the WebClient UI

If you want to provide your customer with information on the current status of a sales order that has already been created, you can use the **Search** function to locate transactions quickly, and forward the necessary information to the customer immediately. In addition, this function allows you to enter changes requested by the customer quickly and easily in the sales transaction.

Displaying sales transactions related to the relevant account

You can display sales transactions, that is, sales quotations and sales orders related to the relevant account on the **Account** overview page, under the corresponding assignment blocks. This allows you to gain an overview of all existing sales transactions for a specific customer, for example, in preparation for a customer visit.

Sales representatives can also create new sales transactions, edit or delete existing sales transactions from these assignment blocks. When you create a new sales transaction from here, the **Sold-to Party** field is automatically filled with the account from the **Account** overview page. Once you save and return to the **Account** overview page, the newly created sales transaction appears in the relevant assignment block.

Data retrieval parameters are delivered for each of these assignment blocks, which enable you to filter which data you want to display on the corresponding assignment blocks. For example, you may only want to display sales orders with posting dates for last month.

The following data retrieval parameters are available for the corresponding assignment blocks:

- Quotations: [Posting Date](#), [Transaction Type](#), [Valid From/To](#)
- Sales Orders: [Posting Date](#), [Transaction Type](#)

For more information on data retrieval parameters, see [Configurable Data Retrieval](#).

Printing Sales Transactions

You can print sales transactions, that is, quotations and sales orders from the WebClient UI. This allows sales representatives to preview, save or print sales transactions in order to check them, share them with their team, or file them to their documents. It is not intended for sales representatives to send transactions directly to their customers; this will continue to be done from the standard [Sales and Distribution](#) UI.

Once you have saved the sales transaction on the WebClient UI, you can use the [Preview Output](#) pushbutton for a print preview with a PDF viewer. You can also save locally or print out the preview using a PDF viewer. To print sales transactions, you can choose the output type set up in SAP [Sales and Distribution](#), including PDF-based print forms, SAPscript and Smart Forms.

For more information, see [PDF-Based Print Forms](#)

Specifics for Interaction Center: Communication Management Software

Use

Communication management software (CMS) refers to software products that manage communication channels and handle individual contacts with customers through different channels, such as phone, e-mail, and chat. The Integrated Communication Interface (ICI) allows the CMS to communicate with the Interaction Center WebClient.

For more information about ICI, see [Integrated Communication Interface](#).

For information about the features that are available in the telephony channel, see [Communication Management Software](#).

For information about the features that are available in the chat and e-mail channels, see [Chat](#) and [E-Mail](#) in the Interaction Center WebClient documentation.

Prerequisites

- You have made the settings in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Define Communication Management Software Profiles](#) .
- Optionally, you have made the settings for checking the continued communication between the communication session, the CMS, and the Web browser in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Define Heartbeat Profile](#) .

In this Customizing activity, you can make settings for the following checking methods:

- Heartbeat without watchdog
- Heartbeat and watchdog
- CMS session availability

Business Process Push

Use

You can use this function to push business processes to external routers or communication management software (CMS) for distribution to interaction center (IC) agents based on their real-time availability. For example, the following business processes can be pushed:

- Business transactions, such as service orders
- Faxes and letters received through SAPconnect or the ArchiveLink scenario

The ID of the business process and a parameter that specifies the type of business process are transferred to the external router or CMS.

In addition, for each business transaction, you can define a set of additional parameters to represent, for example, status and priority. In order for the business process to be pushed and prioritized correctly in the CMS, you define rule policies to set the values for the parameters before these values are transferred to the CMS.

Prerequisites

- You have made the settings for business process push in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Business Process Push](#) > [Define Settings for Business Process Push](#).
- You have assigned parameters to those agent inbox item types that you want to be pushed to external routers or CMS in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Business Process Push](#) > [Assign Parameters for Business Process Push](#).
- You have defined rule policies to set the values for the parameters. For more information, see [Rule Policies](#).
- You have made the settings for pushing faxes in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Agent Inbox](#) > [Settings for Asynchronous Inbound Processing](#) > [Define Receiving E-Mail/Fax Settings](#).
- You have made the settings for pushing letters in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Business Process Push](#) > [Define Letter Settings](#).

Features

Business transactions, such as service orders, are automatically transferred to an external router or CMS if the following conditions are met:

- The business transaction is saved and has ended, or is saved and escalated.
- The values of the parameters that have been defined in Customizing for the business transaction correspond to the values of the parameters that are set in the rule policies.

For example, the business transaction must have status [In Process](#) and priority [High](#) to be pushed to an external router or CMS.

The business transaction is locked and the current employee responsible for the business transaction is removed. No changes to the business transaction and its status are possible until the transaction has been pushed to an IC agent.

When an IC agent is alerted that an inbound business transaction is routed, the business transaction ID and type are shown on the screen. The IC agent can either accept or reject the business transaction. The IC agent who accepts it is assigned as an employee responsible and can process that business transaction and change its status.

- Rule policies

You define rule policies to set the values for the additional parameters for business transactions and cases before these values are transferred to the CMS. You define rule policies in the [Order Routing](#) and the [Intent Driven Interaction \(IC WebClient\)](#) contexts using attributes, conditions, and actions.

❖ Example

You can create the following rule policy in the [Order Routing](#) context:

Conditions:

If Priority Is Not Equal To Very High

Actions:

- Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA
- Send business process push parameter

You can create the following rule policy in the [Intent Driven Interaction \(IC WebClient\)](#) context:

Conditions:

If Current Event Equals Order Was Saved

Actions:

- Set business process push parameter (Item Type = SALES ORDER; Parameter Name = PRI
- Set business process push parameter (Item Type = SALES ORDER; Parameter Name = STA

Conditions:

If Current Event Equals Interaction Ended

Actions:

Send business process push parameter

- Pushing of faxes and letters received through SAPconnect or the ArchiveLink scenario

If the Customizing settings for pushing faxes and letters have been made, faxes and letters received through SAPconnect or the ArchiveLink scenario are pushed to IC agents based on their real-time availability. The number at which the fax arrives is the indicator for pushing the fax.

An IC agent is alerted that an inbound fax or letter is pushed. The IC agent can then either accept or reject it. Once the IC agent accepts the fax or letter, the [Identify Account](#) page and the [Fax Preview](#) or [Letter Preview](#) is displayed.

ERMS E-Mail Push

Use

You can use this function to push e-mails to the communication management software (CMS) for distribution to an interaction center (IC) agent.

When e-mail is routed to an agent, the [Accept](#) and [Reject](#) context area buttons allow the agent to decide whether or not to accept the inbound message. If the agent accepts the e-mail, the system searches for the business partner based on the e-mail address. The agent can then use standard e-mail functions in the e-mail editor view.

Prerequisites

You have specified **Inbox** as the e-mail provider for the e-mail profile in Customizing for **Service** under **IC WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.

Activities

- Create rules in the **ERMS** context to push e-mail to the CMS for distribution

❖ Example

You can create the following rule policy in the **ERMS** context:

Condition:

If E-Mail Sender Equals jdoe@customer.com

Action:

Push E-Mail to IC WebClient Agent

- Configure a CMS system ID that points to a CMS system that supports action routing

Create an entry for the CMS system ID **ERMS_ACTION**. You do this in the **SAP Easy Access** menu under **Service** > **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Communication Management Software** > **Interface Settings** > **Maintain Communication Management Software Connections**.

More Information

[E-Mail Response Management System](#)

Queue Status

Use

The queue status, located in the bottom right hand corner of the interaction center, allows you to quickly display the channels and queues you are assigned to.

Features

The queue status:

- Displays the current date and time (default display) or queue.

If queue information is available, then an arrow is shown after the date. You can click on the arrow to show which queue the current incoming contact was sent to.

Clicking on the arrow again returns the current date and time.

- Displays a tool tip with a quick view of multichannel information.

The queue status tooltip displays the channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and all queues to which you are currently assigned.

The queue status tool tip is only available if the connection to the relevant system is currently available.

- Displays the [Agent Dashboard](#).

If you click on the queue status, the agent dashboard is displayed.

Agent Dashboard

Use

You can use the agent dashboard to view information relevant to your current interaction center (IC) session. It is useful when you need more information about what is displayed in the [queue status](#). You can access the agent dashboard by clicking the queue status at the bottom right corner of the browser window.

Features

The agent dashboard displays the following information:

- Your user ID and business role
- Channels and queues

You see all channels you are working in, for example, e-mail, chat, and telephony, specific phone numbers and e-mail addresses from which you can receive inquiries, and the queues to which you are assigned.

This information is only available if:

- There is a connection to a communication management software system, and
- The communication management software passes queue and channel information to the interaction center.
- System information

System information is used primarily by system administrators for troubleshooting. It includes:

- Information on the domain in which the servers are operating (for example, wdf.sap.corp)
- Name and description of the communication management software being used

Specifics for Interaction Center: Toolbar

Use

The toolbar provides an interface between your communication management software (CMS) and the Interaction Center (IC) WebClient. The toolbar contains pushbuttons and icons for communication functions that apply to the various communication channels.

You can modify the content of the toolbar to better meet your business requirements. For example, you can specify that different sets of pushbuttons are available in the toolbar depending on the connection of an agent to the CMS and the status of the agent's interaction with a contact.

Integration

The following processes are examples of how you can use the pushbuttons in the Interaction Center toolbar:

- Communication process

For a phone call, you start the process by choosing the pushbutton **Dial** or **Accept**, and you end the process by choosing the pushbutton **Hang Up**.

- Interaction process

Includes account maintenance and the interaction record. Additionally, business processes such as sales and service can also be handled in the interaction center. You start the process by confirming the account and you end the process by choosing the pushbutton **End**.

- Communication process and interaction process

The two processes can take place simultaneously and do not necessarily start and end in the same order, for example in the following cases:

- Multiple phone calls

You can dial and end multiple phone calls during the same interaction. This occurs, for example, when a phone line is disconnected during a customer interaction and needs to be dialed again.

- Multiple interactions

Multiple interactions can happen during a single phone call. If the agent receives and accepts the next call whilst wrapping up the previous interaction, they can end the first interaction by choosing the **End** pushbutton despite already being in the next active phone call.

Prerequisites

- Your system administrator has set up the CMS and ensured that the user names there match those in the IC. For more information, see [Specifics for Interaction Center: Communication Management Software](#) and the documentation for the CMS you are using.
- You have made the settings for the toolbar in Customizing for **Service** under **Interaction Center WebClient** **Basic Functions** **Communication Channels** **Define Toolbar Profiles**.

Features

You use the toolbar to work in different channels. The system determines the pushbuttons that appear in the toolbar, depending on the following:

- The communication channel in which you are currently working
- Customizing settings

Example

Pushbuttons in the IC toolbar

Channel	Name	Definition
Universal (available for all channels)	Accept	Accepts the contact. In telephony, Accept is similar to picking up the receiver.
	Reject	Sends the contact back to its original queue.

Channel	Name	Definition
	Transfer or Blind Transfer (in the telephony toolbar)	In telephony, puts caller on hold while you transfer the call to another agent's extension without first consulting with that agent.
	End	Ends the interaction (indicating that the agent has finished wrap up activities), saves all objects created or changed during the interaction, and clears customer information from the screen and from the business data context.
	Hang Up	Similar to hanging up the receiver. Ends a phone call that was started with Dial or Accept .
	Clear Interaction	Discards the interaction record and resets the account identification. i Note Discarding other transactions such as sales or service transactions is not possible in this way. Depending on the process status, this may be done by choosing Cancel if it is part of the transaction toolbar. If you want to clear the entire context of the session including the business data context, you have to choose the Clear Interaction pushbutton followed by the End pushbutton.
	Reset CTI	Resets the context area of the browser session to synchronize it with the communication session state. The different states can for instance be caused by network issues.
	Dial Pad	Displays a dial pad that looks like the numbers on a telephone, which you can use to make a call.
Telephony	Hold	Puts the active call on hold.
	Toggle	Switches between a call on hold and an active call, automatically placing the active call on hold and vice versa.
	Retrieve	Takes a call off hold and makes it active once again.
	Consult	Puts customer call on hold while you call another agent. After you are finished consulting with the other agent, the agent call is dropped and the customer call is active once again.

Channel	Name	Definition
	Warm Transfer	Puts caller on hold while you consult with another agent before transferring the call to the agent's extension.
	Conference	Puts customer on hold while you call another agent. Then you activate both calls at the same time so that all three parties can participate in a discussion.
	Wrap Up	Indicates to the system that you need time to wrap up the call. Wrap up time ends when you choose the End pushbutton.
E-Mail	Pushbuttons specific to the e-mail function, such as Reply , New , or Forward , appear on the e-mail screen dynamically, as and when they are required.	Represent standard e-mail functions.
Chat	Leave	Finishes the chat.

Work Modes

Name	Use
Ready	When you choose Ready , the CMS system sends incoming contacts to you.
Not Ready	When you choose Not Ready , the CMS system does not send incoming contacts to you.

Scratch Pad

A temporary workspace in the interaction center that you use as an electronic notepad to store miscellaneous information during an interaction.

Customers usually provide a lot of information during the first few minutes of conversation, before you have a chance to access the relevant transaction. Most agents have separate notepads to take notes and then enter the data into the system later.

The scratch pad replaces the need to write manual notes. It allows you to electronically store information so it can be copied into specific functions that require this data.

The scratch pad allows you to focus on recording and listening to what the customer is relaying. Customers do not have to hold their conversation until you follow through the sequence of the interaction, gather the preliminary information, such as contact information, and then navigate to the appropriate area in the IC WebClient to record the information. This also saves customers from having to provide information more than once.

The scratch pad is located in the top left corner of the IC, under the title bar. It is available throughout the interaction, and can be copied into different IC functions, such as the interaction record or knowledge search by choosing **Import Scratch Pad**. All of the data on the scratch pad is imported, but you can still edit, add, or delete it, if necessary.

Data remains on the scratch pad for the entire interaction until you either select **End** from the toolbar, or select **Clear**. Closing the scratch pad does not delete the contents.

Alerts and Messages

Use

These functions display texts on your page to alert you about items or situations that require your immediate attention. There are three types:

- Alerts

Alerts are predefined by system administrators or Interaction Center (IC) managers to be triggered by certain events in the system.

- Messages

There are three types of messages: informational, warning, and error.

Features

- Placement on page

Alerts appear in the alerts area and system messages appear in the system messages area.

- Alerts

- If more than two alerts appear, an arrow appears, which allows you to page through alerts in pairs.
- All alerts disappear when you choose **End**.

Example

Alerts

Marcus is a customer service representative at an interaction center. He accepts a call from Ms. Anna Glenn and confirms her identity in the system. The following alert appears:

Service order 00225

Marcus asks Ms. Glenn if she is calling regarding this order. She says that she would like to know the status of the service order. Marcus simply chooses the link to the service order and can inform Ms. Glenn of its current status.

Creating Alerts

Use

You can create alerts in that use system information to provide agents with information they may need during customer interactions.

i Note

No preconfigured alerts are delivered. Interaction center managers define alerts in the **Alert Editor** and assign them through rule policies. Alerts are created independent of any rule policy context.

Alerts can only be triggered in the **Intent-Driven Interaction** context. This context allows you to automate specific interaction center functions, for example, triggering and terminating alerts, via business rules. Interaction center alerts are not relevant for other rule policy contexts.

Prerequisites

- You have completed the Customizing for **Service** under ► **Interaction Center WebClient** ► **Additional Functions** ► **Intent-Driven Interactions** ► **Service Manager** ► **Configure Alerts** ►.
- You have completed the Customizing for **Service** under ► **Interaction Center WebClient** ► **Additional Functions** ► **Intent-Driven Interaction** ► **Define Intent-Driven Interaction Profiles** ►.
- If you want to use user-triggered alerts, you have selected this alert type in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Communication Processing Profiles** ►.

Features

The following types of alerts are available in the IC:

- Real-time alerts

With these alerts, the system continually polls the server to check for alerts.

- User-triggered alerts

With these alerts, the system waits until the user triggers an event to perform a server round-trip to check for alerts.

This type of alert allows you to show alerts without using server polling and can be used with asynchronous processing.

When creating alerts, you can select a theme that affects the appearance of alerts by changing, for example, the font style, font color, and the alert icon. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Themes must be defined in Customizing before they are available for selection when creating an alert. You can define themes in Customizing for **Service** under ► **Interaction Center WebClient** ► **Additional Functions** ► **Define Themes for Alerts** ►.

Broadcast Messaging

Use

Interaction Center (IC) supervisors can send messages to IC agents in real time using broadcast messaging. They can also create and maintain distribution lists for broadcast messaging and display sent messages. IC agents can receive messages, but cannot send messages.

IC supervisors can access broadcast messaging under ► **Managing Operations** ► **Broadcast Messaging** ►.

Prerequisites

You have made the settings for broadcast messaging in Customizing for **Service** under ► **Interaction Center WebClient** ► **Additional Functions** ► **Define Broadcast Messaging Profiles** ►.

Features

- Distribution lists

To create or maintain distribution lists, click [Maintain Distribution Lists](#). Broadcast messaging profiles contain an organizational ID. When you create distribution lists, the names that appear in the input help for [Add User](#) come from the organizational unit entered in the selected broadcast messaging profile. You can also copy an existing distribution list, delete a user from a distribution list, or delete a distribution list. The default distribution list is the one that appears first alphabetically.

To share distribution lists, the owner of the list can flag other users within the list as administrators. As an administrator, you can add and remove users from the list, use the list to send broadcast messages, and you will be able to view any broadcast messages that were sent to this list in the [Sent Log](#).

- Themes

You can define themes for broadcast messaging in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Additional Functions](#) > [Define Themes for Broadcast Messages](#) .

Once a theme has been defined in Customizing, it is available for selection when creating a broadcast message. The theme affects the appearance of broadcast messages by changing, for example, the font style and color. You can also preview the theme. Once you select a theme, an example text is displayed with the theme applied.

Interaction Record

Use

The interaction record is a screen in the interaction center (IC) that contains data for the current interaction with an identified business partner. IC agents can log current interactions in this screen, and postprocess them if necessary.

Interaction with a customer always includes an interaction record, which can then be linked to another business transaction, such as a service process. The interaction record can, however, also be saved without creating another business transaction.

i Note

From a technical point of view, the interaction record is created as a business transaction.

Once an agent has identified an account, the system navigates to the interaction record. In the interaction record, the agent logs the interaction, records central data from the call, and postprocesses the interaction when the call is finished.

To increase system performance at high volume call centers, you can use the lean interaction record. The lean interaction record is a modified version of the full interaction record.

Prerequisites

- You have defined a date profile and assigned this to your transaction type to use interaction time recording in Customizing for [Service](#) under [Basic Functions](#) > [Date Management](#) and [Transactions](#) > [Basic Settings](#) > [Define Transaction Types](#) .
- You have activated the rescheduling view and optionally enabled automatic text entry for call rescheduling in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Business Transaction](#) > [Define Business Transaction Profiles](#) .
- **Using Activity Reasons:**
 - You have made the settings for activity reasons in Customizing for [Service](#) under [Transactions](#) > [Settings for Activities](#) > [Define Activity Reasons](#) .

i Note

The interaction reason corresponds to the activity reason.

- o You have assigned the **Reason for Activity** subject profile in your transaction type to the **Business activity** transaction type in Customizing at the header level.

You can make settings for transaction types in Customizing for **Service** under **Transactions > Basic Settings > Define Transaction Types**.

- **Using Multilevel Categorization:**

You have made the following settings in Customizing for **Service**:

- o Activate categorization for the interaction record under **Interaction Center WebClient > Business Transaction > Define Business Transaction Profiles**.
- o Make the necessary settings under **Transactions > Basic Settings > Define Transaction Types**.
- o Define the subject profile for the category modeler under **Basic Functions > Catalogs, Codes and Profiles**.
- o Make the necessary categorization settings for the interaction record under **Interaction Center WebClient > Business Transaction > Define Categorization Profiles**.

See [Multi-Level Categorization](#) for more information.

- If you want to use the lean interaction record instead of the full interaction record, you have assigned a business transaction profile that contains transaction type 0100 to the business role. You do this in Customizing for **Service** under **UI Framework > Business Roles > Define Business Role**.

Activity Clipboard

Use

Interaction Center (IC) agents can use this function to view a list of the business objects involved in a current or past interaction. It is located in the [Interaction Record](#) or the navigation area.

i Note

The activity clipboard in the interaction record visualizes past and current interactions.

The activity clipboard in the navigation area visualizes current interactions.

The objects that were created or changed during those interactions are listed including navigation links.

Integration

This function is integrated with:

- Interaction Record
- Interaction History
- Business data context

In the Interaction Center (IC), the activity clipboard is one view of the business data context.

Prerequisites

Depending on where you want the activity clipboard to appear, you have to make different settings in Customizing:

Activity Clipboard in the Interaction Record

- You have made the necessary settings in Customizing to define the display of the activity clipboard so that the business partner appears in the activity clipboard list. Your system administrator can either use the default activity clipboard profile, or define which objects are relevant for linking and should appear in the activity clipboard. You do this in Customizing for [Service](#), by choosing ► [Interaction Center WebClient](#) ► [Basic Functions](#) ► [Define Activity Clipboard Profiles](#) ►.
- Both the programming of the business object itself and your Customizing settings must be in place for a business object to appear in the activity clipboard.

Activity Clipboard in the Navigation Area

You have made additional settings in Customizing for [Service](#):

- Define a direct link group of the type EE under ► [UI Framework](#) ► [Technical Role Definition](#) ► [Define Navigation Bar Profile](#) ► [Define Direct Link Groups](#) ►.
- Assign the direct link group to a navigation bar profile under ► [UI Framework](#) ► [Technical Role Definition](#) ► [Define Navigation Bar Profile](#) ► [Define Navigation Bar Profiles](#) ► [Assign Direct Link Groups To Nav. Bar Profile](#) ►.
- Assign the navigation bar profile to your business role profile under ► [UI Framework](#) ► [Business Roles](#) ► [Define Business Role](#) ►.

Features

If a business object is accessed repeatedly, it only appears once on the activity clipboard, in the order in which it was first accessed.

- By default the data is displayed in three columns, however, in Customizing you can choose the number of columns to be displayed with the maximum being five columns. The first column contains the name of the object or activity. Your system administrator can define what information appears in the other columns in Customizing for the activity mentioned above. By default, the [Object ID](#) and the [Process Type](#) are displayed.
- You can use the links to navigate to objects that appear in the activity clipboard, such as leads or sales tickets.

Account Identification/Account Fact Sheet

Use

IC agents can use this part of the interaction center (IC) to identify accounts in different ways, such as by name, company, or product, and to obtain an overview of an account and contact. If an account does not yet exist in the system, or if it must be updated, IC agents can also create and change accounts from this screen. In addition, they can quick create an account with just the basic data.

Prerequisites

- You have set up one of the following account identification profiles:
 - The standard account identification profile in Customizing for [Service](#) under ► [Interaction Center WebClient](#) ► [Master Data](#) ► [Define Account Identification Profiles](#) ►.

- The profile that enables confirmation of multiple business partners at once in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Master Data](#) > [Define Account Identification Profiles for Multiple Business Partners](#).
- In your account identification profile, you have made settings for the following:
 - Contact attached data (if you want to use this feature)
 - Configurable context area (if you want to use this feature)
 - Embedded object components
 - Search scenario
 - Search results filtered for account roles and contact person roles
 - Default contact type for contact searches using search fields for contact communication details ([Telephone](#), [E-Mail](#), [Fax](#))
 - International address versions (if you want to use this feature)
 - Employee search (if you want to use this feature)

- If you are using contact attached data (CAD), you have mapped the contact attached data from the communication management software to account identification using an XML file.

You have created a function profile in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Define Profiles for Contact Attached Data](#).

Additionally, if you want to use CAD attributes when creating alerts or rules, you have enabled the relevant CAD attributes for fact gathering. For more information, see Customizing for [Service](#) under [Interaction Center WebClient](#) > [Basic Functions](#) > [Communication Channels](#) > [Enable CAD Attributes for Fact Gathering](#).

- If you are using international address versions, you must have activated them. You do this in Customizing for [Service](#) under [SAP NetWeaver](#) > [Application Server](#) > [Basis Services](#) > [Address Management](#) > [International Settings](#) > [Activate International Address Versions](#).

Mapping from logon language to address version is available in the default implementation of Business Add-In ADDR_LANGU_TO_VERSIONMAP_LANGUAGE_TO_VERSION, or an implementation exists that maps the logon language to that version.

- If you are using employee search, you have defined employees as business partners.
- If you are using the standard account identification profile, you have defined any object components that you want to be displayed in the upper right area of the screen in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Customer-Specific System Modifications](#) > [Define Object Components](#).

If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you define in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Master Data](#) > [Define Account Identification Profiles for Multiple Business Partners](#).

- You have defined an interaction history profile in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Master Data](#) > [Define Interaction History Profiles](#).
- If you are using the briefcase, you have made the necessary setting in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Master Data](#) > [Define Briefcase](#).
- If you are using the enhanced search, you have made the necessary setting in Customizing for [Service](#) under [Master Data](#) > [Business Partner](#) > [Search](#) > [Activate Wildcard Search for Accounts and Contacts](#).
- If you want to ensure that all rules and events that are triggered when the [Business Partner Confirmed](#) (BPConfirmed) event is raised are also triggered when a call is transferred, you have added the [Business Partner Confirmed for Call](#)

Transfer (BPConfirmed_Trans) event to the relevant rule policy in the rule modeler. Once you have done so, an account that is confirmed by an agent is confirmed automatically if the call is transferred to a second agent, and the system does not create a new interaction record when the call is transferred. For more information about creating rule policies, see [Rule Modeler](#).

Features

- Automatic number identification (ANI)

If your telephony switch system has ANI, the system can identify accounts even before IC agents accept contacts.

- Business context data (BCD)

When IC agents confirm an account or product, the system displays all relevant information for the account or product in the activity clipboard. This information is also recorded in the BDC so that it can be reused by other functions.

For more information, see [Activity Clipboard](#).

- Contact Attached Data

When IC agents confirm an account, the system can incorporate contact attached data from the communication management software (CMS). This information is displayed on the contact attached data screen.

You can also enable contact attached data attributes for fact gathering so that these attributes can be used when creating alerts or rules. Using these attributes, you can, for example, retrieve a service request number from the CAD attributes of an incoming call and set up a rule that displays an alert providing this information to the IC agent once the account has been confirmed. The IC agent can then click the service request number in the alert to navigate directly to the service order. When using CAD attributes for this purpose, you must ensure that the data that is passed has the same CAD group name and CAD attribute name that have been defined in Customizing.

- Configurable Context Area

The context area provides background information for the current interaction. If you would like to display different account information in the context area, ask your system administrator to configure the context area accordingly.

- Embedded Object Components

If you are using the standard account identification profile, you can replace the upper right area of the **Account Identification** screen with your own views that enable you to search for business partner-related business objects. You can define object components so that the agent can search for:

- Objects based on the confirmed business partner
- Business partners based on the confirmed objects

If you are using the account identification profile for multiple business partners, you can define object components within the search tabs you have defined.

- International Address Versions

If international address versions are activated in your system, you can use the default character set or the international character set to search for business partners (provided a character set is defined for your logon language).

The system automatically searches in both the default character set and the international character set. For example, in Japan, agents can search for the caller's name using the furigana version of the name. As a result, all names with the same pronunciation are displayed, irrespective of the character set.

International address versions are available for all countries for which you have maintained an international address version.

- Search Scenario

You can customize search fields and search views that appear as tabs on the **Account Identification** screen. For each search, you can define the title of the tab and the type of search (for example, business partner or object component).

You can also define the sequence of relationship types that appear in the search criteria.

- Filter search results for account roles and contact person roles

You can set the system to filter search results for account searches by account roles and by contact person roles.

- Define default contact type for searches using the contact communication data search criteria

For contact searches using the fields for contact communication details (**Telephone, E-Mail, Fax**), you can set a default contact type in Customizing that determines whether the search criteria are used to search for a contact of the type account, contact person, or both. IC agents can change this setting in the WebClient UI.

- Use contact name in account search

When searching for accounts in the IC, entering data in both the contact name fields and the account fields will generate a combined search result and provide more precise results.

i Note

You cannot search for accounts using the account search fields in combination with the search fields for contact communication details (**Telephone, E-Mail, Fax**). The contact communication details take priority over search criteria entered in the other search fields. If one of the search fields for contact communication details is filled, the system performs a contact search based on the communication details only.

- Employee Search

You can configure your interaction center to service internal employees. IC agents can use the employee search to look up employee details such as cost center, personnel area, employee group, and employee subgroup. They can also look up telephone numbers and e-mail addresses, or, when they receive a telephone call or e-mail, the system can identify the employee by ANI or from the incoming e-mail, and display the employee's details.

- Multiple Business Partners

You can configure your interaction center so that IC agents can confirm more than one business partner at once. You can also define a list of partner functions that can be used to confirm business partners and determine the sequence in which partner functions are displayed. You can then map the partner functions of confirmed business partner to the existing partner functions in business transactions and the interaction record.

- Account Fact Sheet

The account fact sheet gives an overview of the confirmed account. You can personalize views of the fact sheet. For more information, see [Fact Sheet](#).

- Briefcase

You can set up a briefcase that contains a collection of briefing cards that give you an overview of relevant information, such as addresses and contacts. Each briefing card can be assigned to more than one briefcase.

- Account Quick Create

You can quick create an account from every business object such as an opportunity or an activity. When you save the new account, the data you have entered in the fields for the business object is automatically filled in for the new account.

- Enhanced Search

You can search for an account and contact with the following search criteria:

- Region by Country

The search criterion refines your search for accounts by filtering the correct region of the respective country.

- E-Mail, Telephone, and Website

You can enter an asterisk (*) at the beginning or the end of a search string to perform a wildcard search. For e-mail, you can also put the asterisk within a search string.

Chat

Use

Interaction center (IC) agents can use the chat function to process incoming chat messages. The chat function in the IC includes standard chat functions, such as joining chat sessions, sending messages, and transferring chat sessions. Agents can transfer chat sessions to other queues or agents. Agents can also see additional details such as chat subject, location where the customer initiated the chat, and shopping cart items. Further features are described below.

Integration

Chat is integrated with the following IC functions:

- Toolbars

Agents use the channel-specific toolbar that is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- Interactive scripting

Agents can use scripts to guide them through the chat session. The script is available directly next to the chat text area, so agents can easily and quickly access the information without navigating to a different screen.

- Knowledge search

Agents can use the knowledge search to find solutions to problems or other issues reported by customers.

- Telephony

Agents can remain in the chat channel and make a short phone call by using **Dial** in the toolbar.

- Interaction record

Depending on Customizing settings, the chat transcript can be saved and later accessed in the activity clipboard of the interaction record.

Prerequisites

- The Integrated Communication Interface (ICI) is configured (see [Integrated Communication Interface](#)).
- You have defined the chat profiles in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Chat Profiles** ►.
- You have assigned chat profiles to agents through business roles.
- If you want to use standard responses, you have defined the standard response group in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication Channels** ► **Define Standard Response Groups** ►.
- If you want to add additional knowledge bases for use with chat, you have made the settings for knowledge base implementation in Customizing for **Service** under ► **Interaction Center WebClient** ► **Basic Functions** ► **Communication**

- If you want a list of available queues when transferring chat sessions, [Communication Management Software](#) settings are defined.

Features

- Account identification

The customer search is automatically started when an incoming chat session is routed to an available agent. The customer's e-mail address or other business partner details are used to determine if customer data exists. If the account is confirmed, then chat transcripts can be saved to the confirmed business partner's interaction record. For more information, see [Account Identification/Account Fact Sheet](#).

- Standard responses

IC agents can insert standard responses into the chat text. They can choose from a predefined list of commonly used responses, or can use the standard response screen to search and preview other responses.

- Knowledge base integration

You can define which knowledge bases IC agents can access while using the chat function. You can, for example, integrate knowledge bases such as interactive scripts, knowledge articles, mail forms (standard responses), solution database (problems and solutions), or chat transcripts.

- Multiple chat sessions

IC agents can process multiple chat requests simultaneously to maximize efficiency.

i Note

For more information about settings for multiple chat, contact your communication management software (CMS) vendor.

When you use multiple chat, several parallel HTTP sessions are created on the server in Service. This affects the sizing on the server side.

If you are using multiple chat, the client switch scenario is not supported.

- Web shop integration

IC also supports chat in Live Web Collaboration in Web Channel. You can integrate IC with your Web shop and with Internet Customer Self-Service (ICSS) so that customer requests for assistance via chat are forwarded directly to the interaction center.

E-Mail

Use

You can use this function to process incoming and outgoing e-mails. This function is integrated with the following functions in the interaction center (IC):

- Toolbar

IC agents use the channel-specific toolbar, which is available for telephony, e-mail, and chat. The toolbar options change dynamically depending upon what channel is selected.

- Account identification

Upon receipt of an e-mail, the customer search is automatically initiated based upon the **From** e-mail address. For more information, see [Account Identification/Account Fact Sheet](#).

- Information search

Results from the knowledge search can be added to e-mails as attachments or as text in the e-mail message. For more information, see [Knowledge Articles](#).

- Agent inbox

IC agents can access e-mail through their inbox and process them. For more information, see [Agent Inbox](#).

- Interaction record

IC agents have access to previously sent or saved e-mails in the confirmed business partner's interaction record. For more information, see [Interaction Record](#).

- Telephony

IC agents can make a phone call by using the **Dial** function in the application toolbar while remaining in the e-mail channel.

Prerequisites

- You have made settings to display fields (for example, date and from fields) and editor buttons in Customizing for **Service** under **UI Framework > Technical Role Definition > Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.
- You have made settings for e-mail in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels**. For example, you have made the following settings:
 - You have defined toolbar properties for e-mail in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Toolbar Profiles**.
 - You have defined standard response groups in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Standard Response Groups**.
 - You have defined outgoing e-mail address groups in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define Outgoing E-Mail Address Groups**.
 - You have defined e-mail profiles in Customizing for **Service** under **Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles**.

Features

- Standard responses

IC agents can reuse use predefined text. For more information, see [Standard Responses](#).

- Multiple incoming e-mail addresses

Interaction centers can use more than one incoming e-mail address. Each address is assigned to a group of IC agents. IC agents can be assigned to more than one address.

- Multiple outgoing e-mail addresses

Interaction centers can also use more than one outgoing e-mail address. IC agents are assigned to one default outgoing e-mail address based upon their e-mail profile as defined in Customizing.

- Unfinished e-mails

IC agents can process unfinished e-mails if they need to deal with another transaction before completing an e-mail response.

Smart Responses

Use

Smart responses are [standard responses](#) with the ability to include additional dynamic information not limited to simple business partner attributes or system information.

Text is populated with business values and inserted into the e-mail content at runtime. This reduces the amount of time agents have to spend searching for and compiling information in a response. Agents are unaware if the response they choose is a standard response or a smart response.

❖ Example

A customer sends an e-mail to the interaction center asking what his account balance is. After the agent confirms the business partner and initiates a reply to the e-mail, the agent selects a standard response. Variables such as account balance and account number are populated with the business partner's values in the response.

i Note

Interaction center agents cannot insert smart responses manually into an e-mail using the **Standard Response** dropdown.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function in the role of an interaction center manager under ► **Knowledge Management** ► **Mail Forms** ►.
- You have defined [Multilevel Categorization](#). (Not relevant for Auto Acknowledgements)
- You have defined the [rule policy](#) with the **Send Auto Acknowledgement**, **Auto Respond**, and/or **Auto Prepare** actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.
- You have defined the data source classes in Customizing for **Service** by choosing ► **Interaction Center WebClient** ► **Customer-Specific System Modifications** ► **Smart Responses** ► **Define Data Source Classes** ►.

Features

Like standard responses, you can insert placeholders in the mail form that can be filled in by data related to the e-mail or to the business partner of the e-mail. You can define how the values for business and transactional data are retrieved and placed in mail forms. You can use data from either the fact base attributes or from the business data context (BDC).

Smart responses can be sent automatically to customers with or without agent intervention based upon multilevel categorization. They can be inserted into the e-mail text or attached to the e-mail. For more information, see the actions and parameters table in [ERMS Rule Policies](#) for the **Auto Respond** and **Auto Prepare** actions. They also can be used in auto acknowledgements and response proposals.

Standard Responses

Use

IC agents can use standard responses to help automate e-mail processing. Standard responses use predefined text configured from mail forms that are available to IC agents in the e-mail editor. Standard responses provide consistent handling and messaging to customers regardless of which IC agent processes the e-mail.

On the e-mail editor screen, two assignment blocks offer standard responses.

1. The **Standard Responses** assignment block provides a list of standard responses from four different sources:
 - Standard response groups as defined in the **Define Standard Response Groups** Customizing
 - Standard responses are available in the e-mail editor when you select a relevant category
 - Standard responses from a manual search done by the IC agent
 - Standard responses that the IC agent has saved in their favorites.
2. The **Standard Response Tree** assignment block displays standard responses based on the assigned multilevel categorization schema. Standard responses are also available in chat text. For more information about e-mail and chat, see [E-Mail](#) and [Chat](#). Multilevel categorization and Customizing settings help determine which standard responses are available.

Prerequisites

- You have defined standard response groups in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Standard Response Groups**.
- You have assigned the standard response groups to e-mail profiles in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define E-Mail Profiles**.
- If you want to use standard responses for chat, you must assign the standard response groups to chat profiles in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Chat Profiles**.
- If you want to determine the default language for your standard response texts, you must make settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **BAdIs for Communication Channels** > **BAdI: Language Determination for Standard Response Texts**.
- If you want to filter the standard responses to limit their use across channels, you must make settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **BAdIs for Communication Channels** > **BAdI: Filter for Standard Response Texts**.
- If you want to use the **Standard Response Tree** assignment block, you must have your e-mail editor profile to the EMAIL_PROFILE parameter for the e-mail application area. You can choose any categorization schema without the restriction of using an email profile when you use the CRM_IC_STD_RESP_SCHEMA business add-in (BAdI) definition, within enhancement spot CRM_IC_STD_RESPONSE. These standard responses will be in addition to those defined in the **Define Standard Response Group** Customizing.

Features

IC agents can do the following:

- Perform a search and preview standard response texts before choosing a standard response
- Choose the language of standard responses after searches
- Sort and filter standard responses

- Manage favorite standard responses
- View the source of standard response proposals (for example, from favorites or searches)

You can use variables in the mail form to insert placeholders for business partners. When IC agents choose the standard response or insert the standard response into an e-mail, the system fills these placeholders with values. For example, you can populate the customer's name in the salutation.

More Information

- [Auto Suggest](#)
- [Smart Responses](#)
- [Actions and Parameters](#)
- [ERMS Rule Policies](#)

Fax and Letter

Use

Interaction center (IC) agents can create, edit, and receive fax messages and letters in the interaction center. IC agents can also use an internal HTML editor to process faxes and letters.

Prerequisites

- You have made settings to display fields (for example, subject, date, and from fields) and editor buttons in Customizing for **Service** under **UI Framework** > **Technical Role Definition** > **Define Parameters**. If you make fields visible, IC agents can personalize by choosing to show or hide those fields. If you set no parameters, the system shows all fields.
- If you would like the **To** field of fax messages to be filled automatically, you have maintained the business partner (BP) data with a fax number that includes the country code.
- If you want to use business process push (BPP) with fax and letter, you have to specify the communication management software (CMS) connection by executing the program on the **SAP Easy Access** screen, under **Interaction Center** > **Interaction Center WebClient** > **Administration** > **Communication Management Software** > **Interface Settings** > **Maintain Communication Management Software Connections**. For more information about BPP, see [Business Process Push](#).

Caution

You must enter the following values:

- Under **Comm. Mgmt Software System ID** enter **FAX_LETTER**.
- Under **Connection** enter the RFC connection to your CMS system.

Features

Agents can search for incoming fax messages and letters in the agent inbox and display them. They can also view old fax messages or letters for a confirmed account in the interaction record.

i Note

This is custom documentation. For more information, please visit [SAP Help Portal](#).

Once a fax has been sent or a letter has been finished, it can no longer be edited.

Using the HTML editor, IC agents can:

- Create, send, and print fax messages and letters
- Display information for the current interaction, such as:
 - Reason
 - Priority
 - Status
 - Name
- Add and display attachments
- Change the default header data for fax messages and letters

Activities

Creating Letters

Before IC agents can process letters in the interaction center, you must upload them to the interaction center by doing the following:

1. Scan the letters.
2. Save them to your hard drive.
3. Go to transaction `oawd`.
4. Expand the node and double-click the highlighted line that appears.
5. Select **Drag & Drop**.
6. In the dialog box that appears, drag and drop the relevant files from your local drive.

Agent Inbox

Use

You can use the agent inbox (inbox) to call up e-mails, business transactions, and various other business objects for display or processing. The inbox is a central worklist that the entire team can use to work on incoming objects.

Prerequisites

- You have made the basic settings required to use the inbox. For more information, see [Basic Settings for the Agent Inbox](#).
- If you want to display an alert showing the number of incomplete e-mails linked to a particular account, you have created a rule policy and a rule to set up the alert. The rule policy must include the **Business Partner Confirmed** (BPConfirmed) event and the rule must include the **Show Number of Incomplete E-Mails** (AH_NO_CUST_MAIL) action.

For more information about creating rule policies, see [Rule Modeler](#) and [ERMS Rule Policies](#).

Features

Business Roles

You can use the inbox in business roles with the following profile types:

- [IC WebClient Business Role](#)
- [CRM WebClient Business Role](#)

i Note

The following inbox functions are not available for business roles with the profile type [CRM WebClient Business Role](#):

- Interact
- Link
- Next item
- Displaying e-mails, fax messages, and letters

Inbox Items

You can retrieve, access, and process the following items in the inbox:

- Business transactions in Service

You can call up business transactions, such as service orders.

- Workflow work items

You can process SAP workflow work items, including those that you have customized for use in the inbox. These work items can originate from the current system (SAP S/4HANA or SAP S/4HANA Cloud Private Edition) or from another SAP system.

- E-mails, fax messages, and letters

- Outbound correspondence

The inbox supports outbound e-mails, fax messages, and letters created in the interaction center (IC) and linked to an interaction record. The inbox also supports outbound e-mails created by the E-Mail Response Management System (ERMS).

- Inbound e-mails, fax messages, and letters

E-mails and fax messages that were received via SAPconnect appear as workflow work items. Inbound letters are also available in the inbox as work items.

A workflow forwards inbound e-mails to the appropriate group of agents.

- Multiple incoming and outgoing e-mail addresses

Agents can be assigned to various incoming addresses. All items sent to the agent's various addresses appear in the same inbox. Agents can also be assigned to more than one outgoing e-mail address. A default address is proposed, or the agent can select a different address.

- ERMS routing

Depending on the rule policies you define, you can use a general task for an ERMS decision to allow e-mail routing to all agents or you can specify individual agents or agent groups to allow e-mail routing only to those agents.

- o Number of incomplete e-mails linked to confirmed account

You can set up the inbox so that once an account is confirmed, an alert is displayed that shows the number of incomplete e-mails linked to the confirmed account. IC agents can click the alert to navigate directly to the inbox, where only the incomplete e-mails from the confirmed contact person of the account are displayed. If no contact person has been confirmed, only e-mails from the e-mail address linked with the account are listed in the inbox.

Inbox Search

For information about the inbox search features, see [Inbox Search](#).

Inbox Result List

For information about the inbox result list features, see [Inbox Result List](#).

Basic Settings for the Agent Inbox

Use

System administrators use this process to set up the agent inbox in the Interaction Center (IC) WebClient.

Prerequisites

- The SAPconnect interface is configured.

For more information about SAPconnect, search for “SAPconnect (BC-SRV-COM)” on the SAP Help Portal at <http://help.sap.com>.

- E-mail addresses are created.
- If you want to edit scanned letters in your agent inbox, you have defined an ArchiveLink scenario.

For more information, see the following Customizing activities under **SAP NetWeaver > Application Server > Basis Services > ArchiveLink**:

- o **Basic Customizing > Edit Document Classes** and **Edit Document Types**
- o **Customizing Incoming Documents > Workflow Scenarios > Assign Document Type to a Workflow**

Process

Step	Description
Create business partner master data for each employee, with an employee role.	Maintain all inbound e-mail addresses that the agent should be responsible for. Assign a user ID.
Maintain settings for recipient distribution	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Maintain Recipient Distribution .

Step	Description
Maintain automatic workflow Customizing	In Customizing for Service, choose Basic Functions > SAP Business Workflow > Maintain Standard Settings for SAP Business Workflow. For more information, see SAP Business Workflow in the Interaction Center.
Activate event linkage for workflow template WS 14000164 and assign agents for e-mail handling	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Assign Agents for Fax and Letter Processing. For more information, see Assigning Agents for E-Mail Handling.
Forward to employees/groups	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Forwarding > Define Recipient Profile for Forwarding. Use this transaction to define the content of the dropdown box Forward To in the result list of the agent inbox.
Define quick searches	In the SAP Easy Access menu, choose Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches. Use this transaction to define the content of the dropdown box Quick Search in the agent inbox search.
Maintain Customizing activities	In Customizing for Service, choose Interaction Center WebClient > Agent Inbox. For more information, review the Implementation Guide (IMG) documentation for each activity.

More Information

[Agent Inbox in the Interaction Center WebClient](#)

Assigning Agents for Fax and Letter Processing

Use

System administrators use this procedure to make the following settings for the agent inbox (inbox) in the Interaction Center (IC) WebClient:

- Activate event linking.

Workflow can define how incoming fax messages and letters are routed and processed. Activating event linking activates the workflow you defined and allows it to be used for the agent inbox. You only have to activate event linking once.

- Assign agents for fax and letter processing.

i Note

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see [ERMS Rule Policies](#).

Procedure

Activating Event Linking

1. In the **SAP Easy Access** menu, choose ► **Service** ► **Interaction Center** ► **Interaction Center WebClient** ► **Administration** ► **Agent Inbox** ► **Assign Agents for Fax and Letter Processing** ►.
2. In the **Event Linkage** column, choose **Activate event linking**.
3. Under **IC WebClient: Inbound Process AUI** expand the workflow template WS 14000164 (**IC: Mail Inbox Handling**).
4. Select the icon in the **Activate/deactivate** column.

Once selected, the icon should display as **Activated**.

Result

Workflow is activated for the agent inbox.

Define Rules for Fax and Letter Processing

Use

System administrators follow this procedure to define routing rules for incoming fax messages and letters in the agent inbox (inbox) in the Interaction Center (IC) WebClient.

The IC WebClient inbox uses workflow to route incoming fax messages and letters to agents.

Additional routing rules use business partner attributes to route e-mails to agents who specialize in particular areas (see [Additional Routing Rules for Fax and Letter Processing](#)).

i Note

The incoming e-mails are routed using E-Mail Response Management System (ERMS). For more information, see [ERMS Rule Policies](#).

Process

As a system administrator, you define routing rules in the following steps:

1. Create rules.
2. Assign rules to agents or grouping of agents.

Creating Rules

1. In the **SAP Easy Access** menu, choose ► **Service** ► **Interaction Center** ► **Interaction Center WebClient** ► **Administration** ► **Agent Inbox** ► **Define Rules for Fax and Letter Processing** ►.
2. Choose **Create responsibility**.

3. Enter the following:

- Object abbreviation
- Description
- Start date
- End date

4. Choose **Continue**.

5. Enter a priority.

6. Enter values for each business partner field that you want to use to further define routing.

To enter more than one range for the same field, select the field and choose **New row**.

Enter * (asterisk) for the fields that you do not want to define.

7. Save your data.

Assigning Rules to Agents

1. On the **Responsibilities for Rule** screen, select the rule that you created.
2. Choose **Insert agent assignment**.
3. Select an object type.
4. Search for and select one or more objects of the type that you specified in the previous step.
5. Choose **Create**.

If you want to delete an assignment, select a rule on the **Responsibilities for Rule** screen and choose **Delete agent assignment**.

Example

You want to use the business partner language Spanish as an additional routing field. You make the following settings:

Name	From	To
Address	*	<no entry>
Country	*	<no entry>
Language	ES	<no entry>
Name	*	<no entry>
Region	*	<no entry>
Postal code	*	<no entry>

Additional E-Mail Routing Rules

Use

The Interaction Center (IC) WebClient agent inbox uses workflow to route incoming mails to agents. In the standard workflow, after a customer sends an incoming mail to the interaction center, the system retrieves the address to which the customer sent the mail. It determines which agents are assigned for processing incoming mails for that address. This determination is based on the e-mail addresses entered in the employee's business partner master data.

In addition, you can use certain business partner data to define which agents or groups of agents should process the mails (for example, to route e-mails to agents who specialize in particular areas). The routing rules can be assigned to object types such as users, positions, and organizational units.

Prerequisites

Routing rules for e-mails have been defined and assigned to agents or groups or agents (see [Defining Routing Rules for E-Mails](#)).

Features

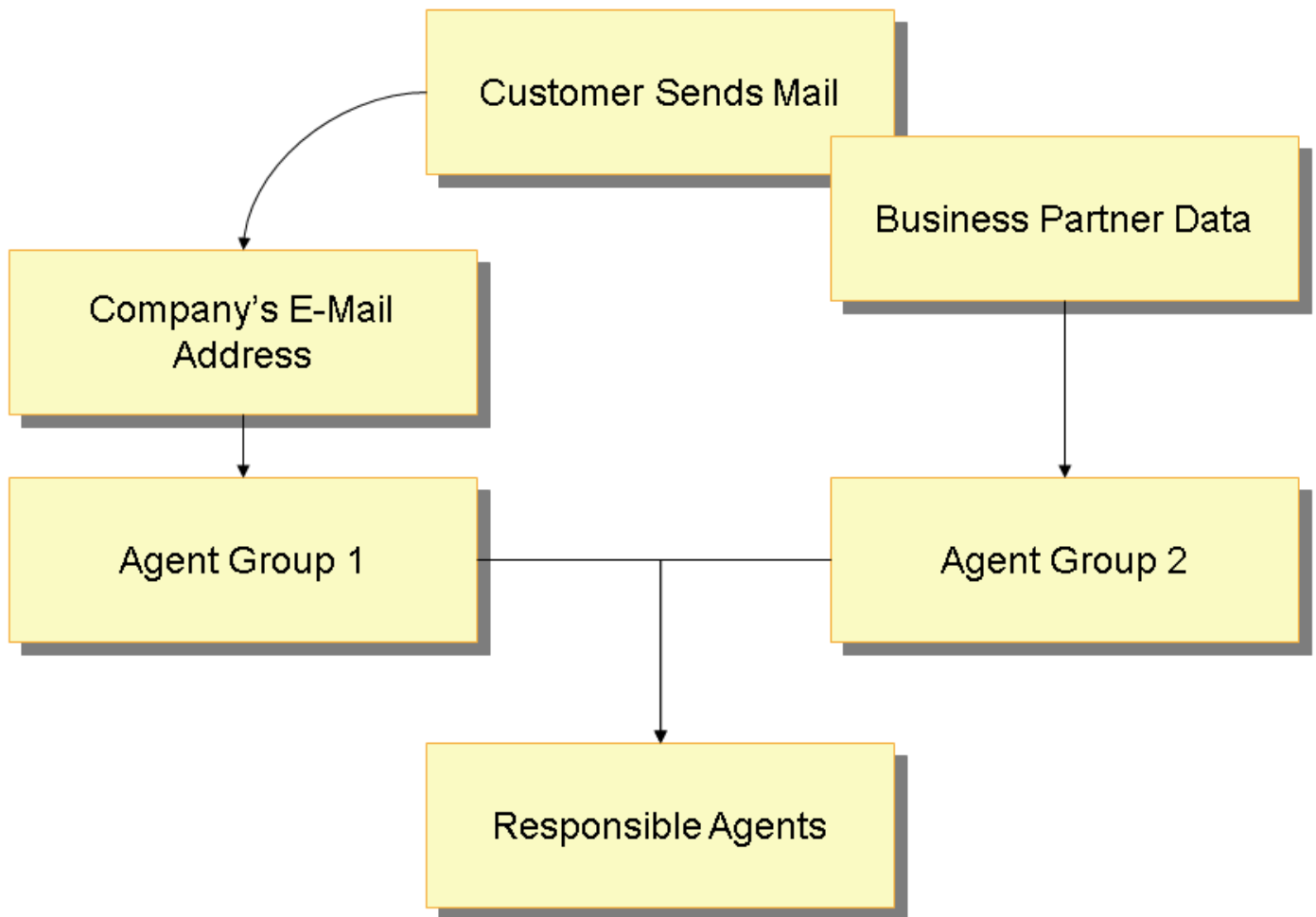
The following business partner data can be used to define how mails are routed:

- Address
- Country
- Language
- Name
- Region
- Postal code

To use other fields, you need to create a new rule. From the **SAP Easy Access** menu, choose ► **Architecture and Technology** ► **ABAP Workbench** ► **Development** ► **SAP Business Workflow** ► **Definition Tools** ► **Rules for Agent Assignment**. ► **Create** ►. The **Responsibilities** tab allows the creation of rule definitions and assignments to agents.

Activities

The figure below shows how the system determines which agents process incoming mails and is followed by an explanation:



1. With additional routing rules, the system tries to determine the business partner by the customer's e-mail address. If the business partner is determined, attributes from the business partner master data can be used to further define which agents process the mail.

i Note

If the business partner cannot be determined, the additional routing rules are not used. In this case, agents who are assigned to process mails for that incoming address or number receive the mail (see [Defining Routing Rules for E-Mails](#)).

2. The system checks which agents are assigned to the incoming mail address (**Agent Group 1**), as specified in the inbound distribution.
3. The system checks which agents are assigned to process the specific business partner attributes (**Agent Group 2**) if the business partner address or number is found. This depends upon the defined rules and which agents are assigned to them.
4. The routing depends on how many agent groups exist:
 - o Two agents groups

The mail is routed only to agents who are in both of the groups (**Agent Group 1** and **Agent Group 2**). If no agents are found in both groups (agents in **Agent Group 1** and **Agent Group 2**, but none belonging to both groups) the mail is routed to all agents in each group.
 - o One agent group

If no agents are in **Agent Group 1**, the mails are routed to agents in **Agent Group 2**. Alternatively, if there are no agents in **Agent Group 2**, the mails are routed to agents in **Agent Group 1**.

Example

Routing works as follows when two agent groups exist:

1. Mr. Smith sends an e-mail to service@company.com.

Julie and David (**Agent Group 1**) are assigned to process mails sent to this e-mail address.

2. The system identifies Mr. Smith's business partner data based upon his outgoing e-mail address.

3. The system determines that an additional rule can be used based upon Mr. Smith's country (**US**).

Michael and Julie (**Agent Group 2**) are assigned to an e-mail rule for US customers.

4. The system determines that only Julie should receive the incoming mail because she is the only agent in both of the agent groups.

If Julie was not in both of the groups, then all of the agents, David, Michael, and Julie, would receive the incoming mail.

Define Recipient Profile for Forwarding

Use

Use this procedure to define to which person(s) you may forward items for further processing from the agent inbox in the Interaction Center (IC) WebClient. Inbox items may be forwarded to an individual employee or a group.

Prerequisites

Business partners specified as recipients must be defined.

Procedure

In the **SAP Easy Access** menu, choose **Service > Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Forwarding > Define Recipient Profile for Forwarding**.

Quick Searches

Use

You can define searches that can be used as options for finding business objects in the agent inbox (inbox).

Prerequisites

Features

You can define quick searches in the **SAP Easy Access** menu, under **Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches**.

SAP delivers one Quick Search (0001) that is defined to select **My Open Items** sorted first by priority, then by main category. You can define additional quick searches based on your own selection criteria.

You can use an asterisk (*) as a wildcard to specify the title of a business object that has been derived from the business transaction.

You can also define a default quick search for each business role in Customizing for **Service** under **Interaction Center WebClient** **Agent Inbox** **Define Inbox Profiles**.

Note that the inbox supports the following time periods for quick searches: creation period, change period, and due period.

Example

You can define quick searches to retrieve item groups such as:

- All overdue items
- All items for which you are responsible
- All high-priority items

Inbox Search

Use

The agent inbox (inbox) search provides a broad range of search attributes and search features that you can use to tailor the inbox search according to your business requirements.

Prerequisites

You have set up and configured the inbox. For more information, see [Agent Inbox in the Interaction Center WebClient](#).

Features

Search Attributes

For information about the available search attributes, see [Search Attributes](#).

Search View: Standard and Advanced

The standard search provides a fixed arrangement of search fields that the user cannot change.

The advanced search provides selectable search attributes, the multiple use of search attributes, and the use of saved searches.

You can specify whether you want to use the standard search or the advanced search. You do this in Customizing for **Service** under **Interaction Center WebClient** **Agent Inbox** **Define Inbox Profiles**.

Lean Search

During a lean search, the system determines only the required data and does not store any data. It only determines the object attributes that are relevant for the search result list. This enables a faster search and faster processing times when working with the result list. The remaining attributes are determined when the object is used, for example, when the object is opened in the display mode or in the edit mode.

Predefined Searches

Quick Searches

You can create predefined search queries that appear as selection criteria in the **Quick Search** field. For example, you can define a search that agents can use to retrieve all high-priority items. For more information, see [Quick Searches](#).

Quick searches are assigned to business roles through the inbox profile.

Saved Searches

The saved searches function is only available in the advanced search. Saved searches are user-defined.

Search Attribute Validity Check

The following attribute validity check prevents unclear search results:

If a main category cannot be combined with a selected search attribute, the main category is excluded from the search. For example, a search involving the main category **E-Mail** in combination with the **Checklist ID** search attribute is not valid, and the search is not performed.

This system behavior also applies if only one main category has been selected or if no main category has been selected. In the latter situation, the validity check is performed for all main categories that are defined for the corresponding inbox profile in Customizing, and the search is performed for all main categories that have passed the validity check.

Search Traceability

A search processing log consolidates various detailed search messages, such as which search attribute is not supported by which main category, in a structured way. The processing log can be accessed using a corresponding button in the result toolbar.

In addition to the entries in the processing log, the following general messages are displayed on the UI, as appropriate:

- Search has been only partially performed
- Search could not be performed

This enables the user to understand the search behavior and to tailor future searches accordingly, or to contact the system administrator, as required.

Predefined Inbox Search Tables

You can use predefined inbox search tables that contain the inbox search attributes and inbox result attributes that are relevant to specific inbox item types. Predefined inbox search tables are available for the following inbox item types:

- **Inbox Item: OneOrder**
- **Inbox Item: Workitem - Inbound Email, Fax and Letter**
- **Inbox Item: Outbound Correspondence**

For information about the prerequisites and features, see [Predefined Inbox Search Tables](#).

Search Attributes

Use

You can use various search attributes and search attribute combinations to locate the required agent inbox (inbox) items.

Prerequisites

For information about the necessary settings for the inbox search, see [Inbox Search](#).

Features

Available Search Attributes

Area	Search Attribute	Function
Predefined searches	Quick Search	Predefined search criteria (for example, My Open Items) You can configure these searches on the SAP Easy Access screen under Interaction Center > Interaction Center WebClient > Administration > Agent Inbox > Define Quick Searches. You must assign the relevant quick searches to the corresponding inbox profile, and you can define a default quick search. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.
	Saved Searches	The saved searches function is available in the advanced inbox search.
Inbox item	Main Category	Inbox item type as defined in the following Customizing activities in Customizing for Service under Interaction Center WebClient > Agent Inbox: Interaction Search Definitions > Define Item Types for Searches Define Inbox Profiles
	Object ID	ID of the inbox item
	Description	Description of the inbox item
Partner information	Created By	User who created the inbox item
	Account	Account for which the inbox item has been created
	Assigned To	Assigned agent or agent group To use the Assigned To search attribute to search for business transactions and cases, the relevant user or group must be maintained as a business partner in the system. The business transaction and case search is based on the business partner in combination with the inbox partner mapping (inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). The search for work items (inbound e-mails, fax messages, and letters), outbound correspondence (outbound e-mails, fax messages, and letters), social media posts, and workflow work items is based on the assigned users. Assigned To: Me The search returns the inbox items that are assigned to the current user. The search returns the following results: - Business transactions and cases to which the current user is assigned as the employee responsible - Work items for which the current user is the employee responsible (assigned agent)

Area	Search Attribute	Function
		• Work items that are available for processing by the user and have not yet been picked by another user • Outbound correspondence that was last changed by the user • Social media posts for which the current user is the employee responsible (assigned agent) Assigned To: My Groups The search returns the inbox items that are assigned to the selected groups. The search returns the following results: • Business transactions and cases to which the selected groups are assigned as the groups responsible • Work items to which the users of the selected groups are assigned as the processor • Work items that are available for processing by the users of the selected groups • Outbound correspondence that was last changed by users of the selected groups • Social media posts to which the users of the selected groups are assigned as the processor Assigned To: <no entry> For work items, the Assigned To: My Groups search is performed. Note that the search for workflow work items always returns the workflow work items assigned to the current user. Therefore, the Assigned To search attribute is not relevant for workflow work items.
	Employee Responsible	Assigned employee responsible The specifics for the Assigned To: Me search attribute also apply to the Employee Responsible search attribute. The Employee Responsible search attribute cannot be combined with the search attribute Assigned To.
Time	Creation Period Change Period Due Period	Predefined periods of time for searching inbox items (for example, Last 3 days) In the standard search, you can use the respective From/To fields to enter a user-defined date or date range. You can define the time periods in Customizing for Service under Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Time Periods for Searches. You can define which time periods are available for the respective search attributes in the inbox search. You do this in Customizing for Service under Interaction Center WebClient > Agent Inbox > Define Inbox Profiles.
	Created On Changed On Due On	Date or date range These search attributes are only available in the advanced search.
Importance	Status	Inbox status, such as Open, In Process, Completed

Area	Search Attribute	Function
		You can define the inbox statuses in Customizing for Service under Interaction Center WebClient Agent Inbox Inbox Search Definitions Define Inbox Status ✎ You must map the system statuses and user statuses of the inbox item types to the inbox statuses. You do this in Customizing for Service under Interaction Center WebClient Agent Inbox Map Item Attributes to Inbox Attributes Map Item Status to Inbox Status ✎.
	Priority	Inbox priority, such as High, Medium, Low You must map the inbox item type priorities to the inbox priorities. You do this in Customizing for Service under Interaction Center WebClient Agent Inbox Map Item Attributes to Inbox Attributes Map Item Priorities to Inbox Priorities ✎.
Multilevel categorization	Category ID	ID of category used in multilevel categorization For more information, see Multilevel Categorization.
Checklist	Checklist ID	Not applicable
E-mail addresses	Sender Address Recipient Address	E-mail addresses that can be used to search for the following inbox items: - Work items (inbound e-mails, fax messages, and letters) If you are not using the predefined inbox search table for work items, entries in the Sender Address field are restricted to 20 characters (use an asterisk in case of longer e-mail addresses). - Outbound correspondence (Outbound e-mails, fax messages, and letters)
Sorting	Sort By	Sort attributes by which the search result is sorted A maximum of two sort attributes can be used for each search.
Generic functions	Clear	Clears the search attribute values
	Reset	Sets the inbox search criteria and the inbox result list back to the initial state: - The Quick Search field is cleared. - The search criteria is set back to the predefined combination of search fields (relevant to the advanced search only), and the field values are cleared. - The Maximum Number of Results field is set back to the default. - The result list is cleared, and the sorting and the filtering are set back to the initial state. Note that personalization settings for the result list and the selected display type (table or tree) are not set back.
	Maximum Number of Results	Maximum number of search results You can define the maximum number of search results to improve the search performance, for example. You can define the default values and range of user entries for the Maximum Number of Results field. You do this in Customizing for Service under Interaction Center WebClient Agent Inbox Define Inbox Profiles ✎.

Search Attributes not Enabled for Multiple Use:

- [Category ID](#)
- [Assigned To: Me](#) and [Assigned To: My Groups](#) cannot be combined with other [Assigned To](#) search criteria. You cannot for example perform a search using [Assigned To: Me](#) in combination with [Assigned To: My Groups](#).
- [Sort By](#)

Predefined Inbox Search Tables

Use

The standard system contains predefined inbox search tables, which contain the inbox search attributes and inbox result attributes relevant for a specific inbox item type. You can improve the search performance by using predefined inbox search tables. For each of the following inbox item types, a separate table is available:

- [Inbox Item: Workitem - Inbound Email, Fax and Letter](#)

Inbox search table for work items (CRMD_AUI_WI_IDX)

- [Inbox Item: Outbound Correspondence](#)

Inbox search table for outbound correspondence (CRMD_AUI_OUTCORR)

i Note

For outbound correspondence, the predefined inbox search table is mandatory for both of the abovementioned inbox item types.

- [Inbox Item: Inbox One Order item](#)

For this inbox items type a predefined search table is not required because the search is optimized already. However, in case of Customizing changes, this inbox item type must be updated. For more information, see the *Features* section.

Prerequisites

General

- You have defined the business transactions and work items that you want to use in the inbox in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Agent Inbox](#) > [Inbox Search Definitions](#) > [Define Item Types for Searches](#) .
- You have assigned partner functions for use in the inbox in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Agent Inbox](#) > [Map Item Attributes to Inbox Attributes](#) > [Map Business Transactions to Responsible Employees and Groups](#) .
- You have defined the inbox status mapping in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Agent Inbox](#) > [Map Item Attributes to Inbox Attributes](#) > [Map Item Status to Inbox Status](#) .

Inbox Search Table for Work Items

- You have made the settings in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Agent Inbox](#) > [Settings for Asynchronous Inbound Processing](#) > [Assign Standard Tasks to Communication Methods](#) .
- You have made the settings in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Agent Inbox](#) > [Define Inbox Profiles](#) > [Work Item Table](#) .

Inbox Search Table for Outbound Correspondence

- If you want to adjust the data record of an outbound correspondence item before it is stored in the inbox search table for outbound correspondence, you make the settings in Customizing for [Service](#) under [Interaction Center WebClient](#) [Agent Inbox](#) [Business Add-Ins](#) [BAdI: Outbound Correspondence Database Update](#).

Features

Data Load

To use the predefined inbox search tables, you first need to upload the data for the relevant main categories. You do this in the SAP GUI, on the [SAP Easy Access](#) screen under [Interaction Center](#) [Interaction Center WebClient](#) [Administration](#) [Agent Inbox](#) [Process Data for Inbox-Specific Searches](#) (transaction CRM_IC_INDEX_LOAD).

Only the main categories that are mapped to the corresponding inbox item types in Customizing activity [Define Item Types for Searches](#) are available in the program [Process Data for Inbox-Specific Searches](#).

i Note

If you change any of the Customizing settings detailed in the [Prerequisites](#) section, you must update the corresponding inbox search table accordingly.

In this case, you must also update the entries for the inbox item type [Inbox One Order Item](#) to fill various header fields in the corresponding business transaction header table, for example CRMS4D_SERV_H for service orders.

Process Data for Inbox-Specific Searches: "Inbox Item Type"

- Choose the inbox item type ONEORDER [Inbox One Order Item](#) to update the following header fields in the corresponding business transaction header table, for example CRMS4D_SERV_H (only in the case of Customizing changes):
 - INBOX_STATUS
 - INBOX_ASSIGNED_TO
 - INBOX_GROUP_ASSIGNED
 - INBOX_ACCOUNT
 - INBOX_CONTACT_PERSON
- Choose the inbox item type WORKITEM [Inbox Workitem](#) to process the data for the inbox search table for work items. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.
- Choose the inbox item type OUTCORRESP [Outbound Correspondence Item](#) to process the data for the inbox search table for outbound correspondence. The upload is required initially and if the Customizing settings under *Prerequisites* are changed.

Process Data for Inbox-Specific Searches: Displayed Figures

- [Source Entries](#) (Not available for outbound correspondence)

The number of entries in the tables, which the system uses to update the corresponding predefined inbox search table

- [Target Entries](#)

The number of entries in the predefined inbox search table that is being updated

Process Data for Inbox-Specific Searches: Processing Options

- **Update**

Loads the data for the selected main categories from the source tables to the corresponding inbox search table

Note that for the selected main categories, all previous entries in the corresponding inbox search table are removed and new entries are provided.

- **Remove**

Removes the selected main categories from the corresponding inbox search table

- **Calculate**

Updates the displayed figures

- **Reset Status**

Resets the status to **Not Started**

You can use the **Reset Status** function if you have canceled a job and the status of the corresponding main category is still **In Process**.

The functions above are also available if multiple main categories are selected, unless one of the main categories cannot be processed due to the status.

Process Data for Inbox-Specific Searches: Availability of the “Update” and “Remove” Functions

The following table provides an overview of the availability of the **Update** and **Remove** functions and the corresponding status:

Context	Status of Corresponding Main Category	Update	Remove
One of the selected main categories is no longer supported according to the settings in Customizing for Service under Interaction Center WebClient Agent Inbox Inbox Search Definitions Define Item Types for Searches	Not Supported	Not enabled	Enabled
One of the selected main categories is already included in another data load.	In Process	Not enabled	Not enabled
You have canceled a job and reset the status to Not Started using the Reset Status function.	Not Started	Enabled	Enabled
The initial data load for one of the selected main categories has not yet been performed.	Not Started	Enabled	Enabled
One of the selected main categories has already been removed from the inbox search table.	Not Started	Enabled	Enabled

Context	Status of Corresponding Main Category	Update	Remove
The data for one of the selected main categories has been uploaded to the inbox search table.	Completed	Enabled	Enabled

Delta Update

The automatic delta update is limited to the main categories that you have uploaded, or are in the process of uploading, to the relevant search tables:

- Inbox search table for work items

When a work item is saved

- Outbound correspondence

- Outbound e-mails

When an e-mail is saved, sent, or deleted

- Outbound faxes and outbound letters

When a fax or letter is saved or deleted and the session is ended

- For outbound correspondence, you are not required to upload the data for the relevant main categories to enable the delta update.
- The system status is used to calculate whether a business transaction is overdue.

Specifics for the Inbox Search Table for Outbound Correspondence

The system fills the following fields from the interaction record using your settings in Customizing for [Service](#) under [Interaction Center WebClient](#) > [Business Transaction](#) > [Assign Partner Functions to Business Transactions](#) :

- ACCOUNT
- CONTACT

If more than one interaction record is assigned to an outbound correspondence item, the interaction record that the fields are supplied from (during data load) is undefined.

During data load, the system loads the outbound correspondence items linked to an interaction record during an interaction center session or during ERMS processing. For more information, see [Interaction Record](#).

Caution

If you delete outbound correspondence items from the inbox search table, you can use the [Process Data for Inbox-Specific Searches](#) transaction to restore items that are linked to an interaction record. However, you cannot use this transaction to restore items that are not linked to an interaction record. For example, you cannot restore outbound e-mails sent by ERMS when they were not created with an interaction record.

Inbox Result List

Use

The agent inbox (inbox) result list displays the inbox search results in table or tree format and provides the functions required to process the inbox items.

Prerequisites

- You have set up and configured the inbox. For more information, see [Agent Inbox in the Interaction Center WebClient](#).
- If applicable, you have configured the result toolbar in Customizing for **Service** under [Interaction Center WebClient > Agent Inbox > Define Inbox Toolbar Profiles](#).

Features

Result List in Tree Format or Table Format

You can display the result list in tree format or table format:

- The tree format provides context information, such as related business transactions.
- The table format provides better performance, sorting and filtering functions, and the use of table charts.

You can define whether the result list is displayed in tree format or in table format as a default. You do this in Customizing for **Service** under [Interaction Center WebClient > Agent Inbox > Define Inbox Profiles](#).

The user can switch between the tree format and table format in the inbox, as required.

Maximum Number of Hits

You can define the maximum number of search results.

For more information, see Customizing for **Service** under [Interaction Center WebClient > Agent Inbox > Define Inbox Profiles](#).

Advanced Warning for Due Date

You can define an advanced warning for each inbox item type, with reference to the due date.

For more information, see Customizing for **Service** under [Interaction Center WebClient > Agent Inbox > Inbox Search Definitions > Define Item Types for Searches](#).

Configurable Result Toolbar

You can configure the result toolbar according to your business needs for interaction center (IC) business roles (business role profile type **IC WebClient Business Role**) and for other core business roles in Service:

- Result toolbar buttons

You can define the following for each button:

- Description
- Label
- Tooltip
- Icon

- Availability of button for use in IC business roles, in other business roles for core or both (providing the button function is available in the corresponding business role)

You can also add user-defined buttons.

- Result toolbar profiles

You can define toolbar profiles for IC business roles and for other core business roles. You can define the following for each profile:

- Which buttons are available
- Button order
- Button grouping

You can group buttons using separators.

- Whether the label, icon, or both are displayed for the button

If you do not define toolbar profiles for your business roles, the system uses default profiles.

Available Toolbar Buttons and Corresponding Functions:

Button Description	Button Function
Complete	The statuses of the selected inbox items are set to Completed. The Complete button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters) • Workflow work items Whether a workflow work item can be completed depends on the corresponding workflow task. • Outbound text messages
Delete	The selected inbox items are deleted. The Delete button is only supported for the following inbox items: • Work items (inbound e-mails, fax messages, and letters)
Display	The system navigates to the details page for the inbox item in the display mode: • No interaction record is created. This also applies if the user switches from the display to the edit mode from the object. (For workflow work items and work items (e-mails, fax messages, letters), switching from the display mode to the edit mode is not supported.) • The account is not confirmed and not published in the context area. When switching to the edit mode from the object, the system defines the user who is currently logged on as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups) and confirms the business partner. The Display button is not supported for work items (inbound e-mails, fax messages, and letters) for business roles with the profile type WebClient Business Role.
Edit	The following steps are performed:

Button Description	Button Function
	• The system navigates to the details page for the selected inbox item in the edit mode. • The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). • No interaction record is created. • The business partner is confirmed and published in the context area. If other business transactions are opened during the editing process, always the business partner of the business transaction that was most recently opened is confirmed and published in the context area. • The business transaction is linked to the activity clipboard. • All subsequently opened business transactions are also linked to the activity clipboard, unless the user chooses the End button. Choosing the End button clears the activity clipboard. • If the user chooses Interact, all business transactions that were linked to the activity clipboard during the editing process are linked to the interaction record. • Editing during an interaction If the user edits other business transactions during an interaction, the business partners for the other business transactions are not confirmed or published in the context area. The Edit button is only supported for business transactions.
Forward	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the Forward button, a dropdown menu is displayed that contains the recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile. If the recipient profile contains more than 10 entries, the 10th item of the button dropdown menu is a More entry that launches an input help. The input help contains all remaining recipients that you have defined in the recipient profile for forwarding. The forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Forward (Extended)	Defines the forwarding target (employee or group) as the employee responsible or group responsible, respectively When the user chooses the Forward button, a dropdown menu is displayed that contains the following entries (maximum of 10 entries): • The recipients that you have defined in the recipient profile for forwarding that is assigned to the inbox profile • The More entry that launches an input help The input help contains the remaining recipients that you have defined in the recipient profile for forwarding. • The Other Employees entry that launches the employee search • The Other Service Teams entry that launches the business partner search The extended forwarding function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Interact	The following steps are performed by the system:

Button Description	Button Function
	• The system navigates to the details page of the inbox item. • The account and contact are confirmed and published in the context area. • An interaction record is created. • The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). • In the case of work items (e-mails, fax messages, letters) and workflow work items, the system navigates to the account identification where a preview of the work item or workflow work item is displayed. The Interact button is not supported for the following inbox items: • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) • Workflow work items The Interact button is not supported for business roles with the profile type WebClient Business Role.
Link	If the user selects a second inbox item during an interaction and chooses the Link button, the second inbox item is linked to the current interaction record. The Link button is not supported for the following inbox items: • Outbound correspondence (outbound e-mails, fax messages, letters, and text messages) • Workflow work items The Link button is not supported for business roles with the profile type WebClient Business Role.
Next Item	The system navigates to the next inbox item that is available for processing and locks the item for other users. Once the user has chosen the Next Item button, the button is deactivated until the user chooses the End button. If the Next Item button is used for outbound correspondence (outbound e-mails, fax messages, and letters), the system navigates to the next outbound correspondence item that is in process and to which the current user is assigned as the employee responsible. In this case, the outbound correspondence item is not locked for other users. The Next Item button is not supported for business roles with the profile type WebClient Business Role.
Preview (Toggle)	Displays a preview of the currently selected e-mail, fax message, and letter The preview is displayed below the table chart area. The preview does not provide follow-up buttons, such as Reply and Forward.
Reference	Sets a reference to the currently selected inbox item If the user chooses the Reference button, a dropdown menu is displayed that contains the relevant entries from the activity clipboard. If the user chooses an object from the dropdown menu, the currently selected inbox item is referenced from the object (the inbox item is added to the object's business context). Note that the Reference button only works if the BUSINESS_CONTEXT function profile is assigned to the business role. In Customizing for the business context, the following is defined: • The inbox item types to which a reference can be added Only these inbox item types are available in the dropdown menu of the Reference button.

Button Description	Button Function
	Which inbox item types can be added as a reference For more information, see Customizing for Service under Transactions > Settings for Service Requests > Define Settings for Business Context > The Reference function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type WebClient Business Role.
Reference (Extended)	Unlike the dropdown menu for the Reference function, the dropdown menu for the Reference (Extended) function contains entries from the Recent Items. The last entry in the dropdown menu is the Other Transactions entry that launches a transaction search. Apart from this, the Reference and Reference (Extended) functions are identical. The Reference (Extended) function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages) and for business roles with the profile type IC WebClient Business Role.
Refresh	The system performs the most recent search again, based on the selected search criteria or based on a quick search. The Refresh button enables the user to conduct the most recent search again without expanding the Search Criteria view and choosing the Search button, or selecting the quick search again.
Reserve	The user that is currently logged on is defined as the employee responsible (according to the partner mapping in the inbox Customizing activity Map Business Transactions to Responsible Employees and Groups). The Reserve button is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation	Resets the reservation of the inbox item and removes the employee responsible Only the employee responsible assigned to the inbox item can reset the reservation of the inbox item. The Reset Reservation function is not supported for the following inbox items: - SAP ERP business transactions - Outbound correspondence (outbound e-mails, fax messages, letters, and text messages)
Reset Reservation (Extended)	Resets the reservation of inbox items on behalf of other users The Reset Reservation (Extended) function is not supported for outbound correspondence (outbound e-mails, fax messages, letters, and text messages).
Separator	The separator is used to group toolbar buttons in the result toolbar profile.
Search Processing Log	Provides access to the search processing log. For more information, see Inbox Search (Search Traceability section).
Show Table View (Toggle)	Displays the result list in table format
Show Tree View (Toggle)	Displays the result list in tree format

Multiple Selection of Inbox Items

The following functions are available when multiple inbox items are selected (unless the function is generally not available for one of the selected inbox items):

- [Delete](#)
- [Reserve](#)
- [Reset Reservation](#)
- [Complete](#)
- [Link](#)
- [Forward](#)
- [Reference](#)

The following functions are not available when multiple inbox items are selected:

- [Edit](#)
- [Display](#)
- [Interact](#)
- [Preview On](#)

Business Transaction Routing

Use

You can use this function to forward business transactions and cases in the interaction center (IC) manually or automatically.

Prerequisites

You have made the necessary settings for automatic routing in the BSP application rule editor.

For more information, see Customizing for [Service](#) under [Interaction Center WebClient](#) > [Business Transaction](#) > [Define Automatic Routing](#).

Features

You can forward business transactions and cases to agent groups as well as to individual agents. Both manual and automatic routing are available:

- Manual routing

You can manually enter a responsible employee or a responsible group as the business partner.

The transaction or case can be selected in the agent inbox of the assigned responsible employee as an open item for processing.

- Assigning a responsible employee

You can either enter an employee directly or using a search.

If you start a search and a responsible group is assigned, the system searches for agents in the responsible group. If there is no responsible group assigned, the system takes all agents into account during the search.

- Assignment of a responsible group

You can enter the responsible group directly or using a search. If you start a search and a responsible employee is assigned, the system searches for all organizational units to which this employee is assigned in the organizational model. If no responsible employee is assigned, the system searches for all organizational units that are assigned to the business role. The organizational units are displayed in the search results according to the structural authorization of the agent.

- Automatic routing

You trigger automatic routing by choosing [Escalate](#).

You can include up to five of the following attributes in the routing:

- Priority
- Status
- Confirmed product
- Confirmed customer
- Categorization
- Transaction type
- Business role

You can also push business transactions and cases to external routers or communication management software (CMS) for distribution to IC agents based on their real-time availability. For more information, see [Business Process Push](#).

E-Mail Response Management System

Use

E-Mail Response Management System (ERMS) is a tool for managing large amounts of incoming e-mail. Instead of routing all incoming e-mails into one queue, ERMS provides services for automatically processing and organizing incoming e-mail. There are several automated activities that reduce the need for manual intervention by interaction center agents, thus substantially increasing efficiency and processing accuracy, by helping agents to process e-mail in less time and allowing them to provide the same quality response regardless of agent skill level. ERMS provides tools for agents to efficiently and consistently respond to messages, and for managers to administer, monitor, and report on the whole e-mail process.

SAP delivers preconfigured ERMS settings and functions that you can extend using custom coding.

Integration

- Organizational Management

Defines how e-mails are routed and to what groups or organizations and how they are going to handle and process e-mails. You can route e-mails to an organizational unit or responsible group, a position, or even an individual user. The e-mails are routed to the agents' inboxes where they are processed.

- SAP Business Workflow

Enables automated processing and routing of e-mails. For more information, see [Workflow for ERMS](#).

- SAPconnect

Receives inbound and sends outbound e-mails. Once e-mails are received, SAPconnect creates SAP office documents, which are then processed by SAP Business Workflow.

- Multilevel Categorization
- Machine Learning

You can connect ERMS with [SAP Service Ticket Intelligence](#) to automatically categorize incoming e-mails.

- E-Mail Response Management System Analytics

Allows you to generate reports and analyze your ERMS processing on historical data. You can analyze historical ERMS e-mail volume, handling time, response time and service level.

Features

Manager Features and Tools for Productivity

The WebClient UI is available for modeling rules and categories, and for monitoring and viewing analytics.

- [Rule Policies](#)
- Category Modeler
- [ERMS Simulator](#)
- [E-Mail Workbench](#)

Agent Features and Tools for Productivity

- [Agent Inbox in the Interaction Center WebClient](#)

Central location in which agents receive and respond to e-mails. E-mails are integrated with other communication channels.

- [E-Mail Threading](#)
- Automatic Business Object Creation

Interaction records, along with appropriate interaction reasons, and service tickets can be created and associated to incoming e-mails automatically. For more information, see the [Create Interaction Record](#) and [Create Service Ticket](#) actions detailed in the actions and parameters table in [ERMS Rule Policies](#).

Automated Features

- [ERMS Responses](#)
- E-Mail Filtering

Certain incoming e-mails that are not relevant for agent processing, such as out-of-office replies, spam, or bounced e-mails, can be filtered or deleted based upon the e-mail subject or content.

- [Rules](#)

These identify and determine the automated processing of e-mails.

- E-Mail Routing

Incoming e-mails can be routed to particular queues or agent groups that are most capable of responding to the issue in the e-mail. For example, you can route e-mails that contain foreign languages to queues in which agents are fluent in that language.

- Content Analysis and Fact Gathering

The system analyzes attributes and content regarding each incoming e-mail and stores it in a fact base, enabling the use of the data for making decisions in respect to categorizing, routing, and automatic responses.

- [E-Mail Escalation](#)

E-mails can be escalated if they are not processed within a specific period of time. For more information, see [Defining E-Mail Escalation](#).

- Web Forms

In addition to accepting incoming e-mails, it is also possible to accept information created from a web form on a website.

- Support of Multiple Incoming and Outgoing E-Mail Addresses

You can define multiple incoming e-mail addresses. You can also define multiple outgoing addresses that agents can change when they are responding to e-mails.

More Information

[Setting Up ERMS](#)

ERMS Rule Policies

Use

You can create rule policies in the e-mail response management system (ERMS) context to handle and evaluate incoming e-mails.

Rules are evaluated every time an incoming e-mail is received if the policy in which they belong is invoked during rule processing.

To create ERMS rules and policies, you use objects configured in the ERMS repository.

Prerequisites

- You have defined attributes, actions, and action parameters in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.
- You have created authorization groups in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.
- You have assigned the authorizations groups to authorization object CRM_ERMS_P in **Role Maintenance** (transaction PFCG).
- You have assigned the authorization groups to policies in the WebClient UI in Service. You can see (and in some cases select) authorization groups in the WebClient UI under **Rule Policy Details**, after you select a policy (the top node of the policy structure). The possible values are the entries you maintained in the Customizing activity above.

Activities

- Select attributes

Most attributes are self explanatory. Only those needing further explanations are listed below.

Attribute	Use
ERMS: Content Analysis Accuracy	A value between 0 and 1 describing the accuracy of the content analysis result. Accuracy = 1 indicates the result is highly focused. All matching categories form a single path in the category hierarchy. Accuracy = 0 indicates the result is contradictory. At least two matching categories point to different paths in the category hierarchy.
ERMS: Best Minus Next Best Score	Distance in score between the category with highest score and the category with second highest score. Each matching category has a score which is a value between 0 and 1 describing how good the content-queries of the category match the given textual content.
ERMS: Number of Top Categories	Number of matching categories that all have the highest score.
ERMS: Best Category Score	Highest score among all matching categories.
ERMS: Top Scoring Category	Matching category with the highest score.
Escalation Date	Date when the escalation of an e-mail occurs. The date is calculated based upon either rule evaluation, Customizing, or service level agreements (SLA) at the time the e-mail arrives.
Escalation Time	Time when the escalation of an e-mail occurs. The time is calculated based upon either rule evaluation, Customizing, or SLA at the time the e-mail arrives.
Escalation Type	Depends on the property METHOD of the service UT_ESCALDETERM. If the METHOD is set to 1, the Escalation Type is SLA. If the METHOD is set to 2, the Escalation Type is determined as the shortest time out of one of the following: rule evaluation, Customizing, or SLA.
Response Time Duration (Hours)	Used for escalations. Represents the initial response time in hours.

- Set actions and parameters

This covers the standard delivered actions and their corresponding parameters.

Action	Use	Parameters
Add Attribute Value	Adds a value to a multi-value attribute	Attribute The attribute that a value is added to. Value The value that is added.

Action	Use	Parameters
Auto Prepare	Same as Auto Respond, except the response is not sent automatically. The agent can review the content first, make any necessary changes, and then send the response. This action is triggered when an agent selects Reply to an incoming e-mail.	Mail Form Name of the mail form that determines the content of the outgoing e-mail. Outgoing E-Mail Address Address used for the outgoing e-mail's sender address. If you do not use this parameter, the incoming e-mail address that the e-mail was originally sent to, will be used. Category Schema Schema to determine which Top Scoring Category is used. Inline Determines whether standard responses and solutions tied to the category are inserted directly into the e-mail content instead of as an attachment. If both are inserted into the e-mail content, then the standard responses are shown first in the order they are attached to the category. The solutions are then added, also in the order they are attached to the category.
Auto Respond	Sends an automated response with proposed answers and/or solutions, to an incoming e-mail, without agent intervention. The solutions are inserted into the e-mail as attachments and are determined from the Top Scoring Category of the incoming e-mail. Not to be confused with the Send Auto Acknowledgement action.	Category Schema Mail Form Outgoing E-Mail Address See descriptions above from Auto Prepare. Create Interaction Record Creates an interaction record, after the response is sent, which includes both the incoming and the outgoing e-mails. Route To (On Exception) Organizational unit that the incoming e-mail is sent to if no solutions are attached to the Top Scoring Category. Inline See description above from Auto Prepare.
Create Interaction Record	Creates an interaction record in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to interaction record, so the original e-mail can be viewed at a later time, if necessary.	Description Description saved in the interaction record.

Action	Use	Parameters
Create Service Ticket	Creates a service ticket in the background, only if there is a business partner associated to the incoming e-mail address. The incoming e-mail is linked to service ticket, so the original e-mail can be viewed at a later time, if necessary.	Description Description saved in the service ticket. You can configure the transaction type (for example, TSRV or TSVO) for this service ticket. You do this in Indirectly Called Services and Properties in Customizing for Service by choosing E-Mail Response Management System Service Manager Define Service Manager Profiles.
Delete E-Mail	Deletes an incoming e-mail. Can be combined with the Create Interaction Record action, so the interaction record shows an e-mail was received and was also deleted.	(No parameters)
Forward E-Mail		Forward To Forward From
Invoke Policy	Temporarily suspends the processing of the current policy and continues with the processing of the specified policy. After the specified policy is finished processing, the action stops processing the original policy, unless the processing was stopped previously by the Stop Further Rule Processing action. This action can be nested, that is a policy that is invoked may invoke other policies. If a policy has multiple variants, then the active variant is invoked.	Policy Name of the rule policy that is invoked
Push E-Mail	Allows e-mail routing through the communication management software instead of directly going to an inbox. E-mails that are pushed to agents using this action are also routed to the Agent Inbox by the ERMS default routing, unless another routing destination is specifically maintained. Therefore, we recommend directly specifying a Route E-Mail action after the Push E-Mail action, allowing all pushed e-mails to be monitored by a supervisor or special organizational unit, and avoiding incorrect routing of push e-mails to the default routing that are already completed.	Organizational Object
Route E-Mail	Directs the incoming e-mail to an agent or group of agents. The e-mail is then accessible from the agent inbox.	Organizational Object You can enter an agent's system user name or an organizational unit. If you enter an organizational

Action	Use	Parameters
		unit, then the e-mail can be processed by any agent who belongs to that organizational unit.
Route to Case Processor	Directs the incoming e-mail to the agent who is assigned as the case Processor in the Case Overview , only if the incoming e-mail is related to an existing case and the case number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no case associated to the incoming e-mail or if there is no processor assigned to the case.
Route to Service Ticket Responsible	Directs the incoming e-mail to the agent who is assigned as the Employee Responsible in the Service Header , only if the incoming e-mail is related to an service ticket and the service ticket number is still in the e-mail text.	Route To (On Exception) Organizational unit that the e-mail is routed to if there is no service ticket number associated to the incoming e-mail or if there is no employee responsible assigned to the service ticket.
Send Auto Acknowledgement	Sends an automatic notification that the e-mail was received.	Mail Form Outgoing E-Mail Address See descriptions from Auto Prepare . Create Interaction Record Creates an interaction record to record that an acknowledgement was sent after receiving the e-mail. (The outgoing e-mail acknowledgement text is not tied to the interaction record.)
Set Attribute Value	Allows you to set an attribute value in the fact base if certain conditions are met. This attribute can then be used in rule evaluation or passed on to actions.	Attribute Value
Show Number of Incomplete E-Mails	Searches for incomplete e-mails linked to an identified account.	(No parameters)
Stop Further Rule Processing		(No parameters)

- Set rule policy authorizations

You can define your own groupings for authorization, for example, by department or by organization.

Authorization Field	Description	Values
ACTVT	Activity	The following values affect authorizations for policies: 01: Create or Generate 02: Change 03: Display 06: Delete 43: Release (For releasing draft rules)

Authorization Field	Description	Values
ERMS_AUGR	Authorization Group	The values you defined for the authorization groups in Customizing activity Define Repository .
ERMS_CTXT	Context	The context in the WebClient UI in Service as defined in Customizing activity Define Repository .

More Information

[Rule Modeler](#)

[Rule Policies](#)

[Rules](#)

[Conditions](#)

[Actions and Parameters](#)

[Conflict Resolution](#)

[Rule Policy Authorizations](#)

ERMS Simulator

Use

The **ERMS Simulator** allows an interaction center (IC) manager to simulate the results of new or changed rules contained in rule policies that are used for e-mail handling. IC managers also see a detailed view of the resulting actions and categorizations for those e-mails. An IC manager can run the simulation in the background and have the result sent to an e-mail address.

The functions in the **ERMS Simulator** are available in the WebClient UI.

Prerequisites

- You have created [rule policies](#).
- You have set up ERMS and already processed incoming e-mails, using ERMS.

More Information

[Setting Up ERMS](#)

E-Mail Workbench

Use

The **E-Mail Workbench** is an overview of e-mail processing in an interaction center, which you can also use as a means to mass process e-mails.

You can access the **E-Mail Workbench** in the WebClient UI by choosing **IC Manager > E-Mail Workbench**.

You can perform manual operations for e-mails, such as searching, routing, forwarding, re-categorizing, and deleting e-mails. You can also view the specific rules that were invoked for a particular e-mail and process [escalated e-mails](#).

You can view and in some cases edit the following elements of an e-mail:

- E-mail text
- Attachments
- [Rules](#) for processing the e-mail
- Associated categories, that you can add or remove. Automated processing can assign categories to e-mails as determined from Multilevel Categorization. You can change these categories if the assignment is not accurate or complete.
- The Fact Base
- Related e-mails

Prerequisites

- You have [set up](#) and actively used ERMS to process incoming e-mails.
- You have set **UT_ERMS_REPLICAT** as your ERMS service manager profile for **Directly Called Services**. You do this in Customizing for **Service** by choosing **E-Mail Response Management System > Service Manager > Define Service Manager Profiles**.

ERMS Responses

Use

There are several types of E-Mail Response Management System (ERMS) responses you can use to automate e-mail processing. This automated processing saves agents time in compiling and searching for information and ensures consistent handling and messaging to customers. You can use mail forms to configure the e-mail content for the responses.

Prerequisites

- You have defined mail forms in the WebClient user interface. You can access this function under **Knowledge Management > Mail Forms**.
- You have defined multilevel categorization. (Not relevant for Auto Acknowledgements)
- You have defined a rule policy with the **Send Auto Acknowledgement**, **Auto Respond**, and/or **Auto Prepare** actions.
- You have assigned mail forms to actions in the WebClient UI or to categories through multilevel categorization.

Features

The following is a list of responses you can determine for ERMS:

- Auto Acknowledgement

When an e-mail is received, the system can send an automatic acknowledgment informing the sender that the e-mail was received. If the business partner is identified, you can use business partner attributes, such as first and last name, to personalize the response.

- Auto Response

The system can send a solution or a response automatically, without intervention from agents, based upon the e-mail content.

- Auto Prepare

Response proposals are similar to auto responses. However, instead of automatically sending the response, the response is reviewed by agents for accuracy. Agents can send, modify, or delete the proposal.

More Information

[ERMS Rule Policies](#)

E-Mail Threading

Use

You can use e-mail threading for service requests and incidents to link e-mails to the associated business transaction, keeping all communication and related e-mails together for easy access to the information, regardless who handles the business transaction. E-mail threading also enables the e-mails to be routed to the person responsible for that business transaction if the customer sends the e-mail back to the interaction center.

Mail forms are used to create a tracking text.

Prerequisites

- You have created tracking text in WebClient UI under **Knowledge Management > Mail Forms**. You can add the **Email Response Management** context attribute to the **Text Element**.

The **Service Request Tracking Text** mail form field is available as a variable for service requests and incident numbers.

If the variable is not available in the mail form, you have to manually add the CRMS_ERMS_SCENARIO_FIELDS structure to the **ERMS** context in Customizing. You do this in Customizing for **Service** by choosing **Basic Functions > Communication Channels > Maintain Attribute Contexts for Mail Forms**.

- You have enabled tracking text insertion by maintaining the **Tracking Text Form** field and selecting the **Track Text on Reply** indicator in Customizing for **Service**. You do this by choosing **Interaction Center WebClient > Basic Functions > Communication Channels > Define E-Mail Profiles**.
- You have defined rules using the **Route to Service Request Responsible** action. For more information, see the actions and parameters table in [ERMS Rule Policies](#).

Features

- E-Mail Tracking Thread

The threading is enabled by the use of tracking text you define. You use a variable that is filled at runtime with the service request number to insert in the tracking text. This variable allows e-mails to be linked to business transactions.

- Tracking Text Insertion

The system inserts the tracking text in the e-mail content when an e-mail is created, either manually or automatically through automated processing, or when the e-mail is replied to. If a service request or incident is automatically created for an incoming e-mail from the auto acknowledgement action, you can link both the incoming e-mail and the outgoing reply to the service request or incident.

- **Thread Detection and Routing**

When customers reply to e-mails, the system determines whether tracking text is inserted. The incoming e-mail can be associated automatically to the related business transaction. Attributes from these transactions, such as the service request number, can be used during rule evaluation. When you create rule policies, you can create rules that route the e-mail to the agent who is currently assigned to the business transaction.

Example

❖ Example

You create a mail form with the following as the tracking text.

```
***** DO NOT DELETE *****
```

```
{SrvReqNo:[8000006814]}
```

```
***** DO NOT DELETE *****
```

You define a rule, "If Service Request Responsible Is Not Equal To "", Then Route to Service Request Responsible to forward the e-mail".

More Information

[E-Mail Threading Process](#)

E-Mail Threading Process

Use

This process describes e-mail threading using service tickets as the relevant business object.

Prerequisites

You have defined [e-mail threading](#).

Process

1. The customer sends an e-mail.
2. Either an agent sends a manual response or the system sends an automatic response.

Manual Response

- a. The agent processes the e-mail and creates a service ticket.
- b. The agent replies to the e-mail.

- c. The system uses the mail form you entered in the agent's e-mail profile. It inserts the mail form text and tracking text into the outgoing e-mail. The system populates the variable with the actual service ticket number that was just created.

Automatic Response

- a. The e-mail response management system processes the e-mail automatically and creates a service ticket, as defined through an [action](#).
 - b. The system sends a reply to the customer as defined in an action, such as **Send Auto Acknowledgement**.
 - c. The system uses the mail form that is defined as a parameter in the **Send Auto Acknowledgement** action. The mail form includes the service ticket tracking variable. When the system sends the reply, the variable is populated with the service ticket number that was just created.
3. The customer replies to the e-mail and sends it back to the interaction center.
 4. The fact gathering services scan the e-mail and pick up the tracking text and store this in the fact base during automated processing.
 5. The e-mail is associated to the service ticket.
 6. The e-mail is routed to the agent who is processing the service ticket through a [rule](#) that uses the **Route to Service Ticket Responsible** action.

E-Mail Escalation

Use

E-mails can be escalated if they are not processed within a specific period of time, for example, e-mails that are not processed within 24 hours after the initial receipt are escalated. You can define the time periods that determine the escalation and you can be notified and view escalated e-mails as soon as the escalation occurs. From reviewing escalated e-mails, you can reroute the e-mail or decide on another action to speed up processing.

Integration

The SAP Business Workflow is used for the escalation process. In [Workflow for ERMS](#), the escalation times you define are used as the **Latest End** date and time.

Prerequisites

You have [defined e-mail escalations](#).

Features

Escalation Definition

Escalated e-mails are determined when defined initial response times are not met. The initial response time is the time between an e-mail is received in the system until the time that an agent processes the e-mail.

Initial response time is calculated by one of the following:

- [Service Level Agreements](#) (SLA)

You can set a deadline for escalation by using a service contract. The system uses the service contract that is assigned to the business partner and reads the duration time from the response profile associated with the service contract.

- [Rules](#) and [actions](#)

In the WebClient user interface, you create a rule using the **Set Attribute Value** action with the value **Response Time Duration** to determine at what point an e-mail is considered escalated. You can also use some fact base attributes as additional criteria, for example, if a customer is a high priority and valued customer with a specific system status, and sends an e-mail with a request, but the e-mail is not processed within a specific number of hours as determined in the rule, the e-mail is escalated.

- E-Mail Addresses

You can determine an escalation time based upon the receiving e-mail address.

Escalation Determination

You can choose which method is used to determine the initial response time. You can either use the:

- Service Level Agreement
- Combination of SLA, rules and actions, and e-mail addresses

The system uses the definition with the shortest initial response time out of all three methods.

Escalation Process

See [E-Mail Escalation Process](#).

Escalation Notification

You are notified of a new escalated e-mail by receiving an e-mail. Mail forms are used to create the notification text.

Escalation Monitoring

After receiving the escalation notification, you use the overview in the [E-Mail Workbench](#) to monitor the number of escalated e-mails per organizational unit and navigate directly to the individual e-mail for further action.

E-Mail Escalation Process

Use

This process describes the e-mail escalation process.

Prerequisites

You have [defined e-mail escalations](#).

Process

1. An e-mail is received in the interaction center (IC).
2. Automated processing begins.
3. The initial response time (IRT) is determined by the escalation definition.

4. If the IRT limit is reached, the e-mail becomes escalated and the workflow escalation process starts.
5. The notification e-mail is sent to the IC manager.
6. The manager accesses the [E-Mail Workbench](#), views the situation and takes further actions with the escalated e-mail.

More Information

[E-Mail Escalation](#)

Defining E-Mail Escalation

Prerequisites

You have authorization to read mail forms.

Context

You use this procedure to define and setup [E-Mail Escalation](#).

Procedure

1. Define the method used for determining initial response times.
 - o Service Level Agreements
 - a. Run transaction CRMD_SERV_SLA.
 - b. Maintain a response profile and assign a response time. For **Name of Duration**, select **Duration Until First Reaction**. The response profile is associated to a service contract. For more information, see [Service Level Agreements \(SLA\)](#).
 - o Rules and Actions

Define a rule in the WebClient user interface and use the **Set Attribute Value** action with the **Response Time Duration** attribute.
 - o E-Mail Addresses

Define the escalation time period for each receiving e-mail address in Customizing for **Service** by choosing **Interaction Center WebClient > Agent Inbox > Settings for Asynchronous Inbound Processing > Define Receiving E-Mail/Fax Settings**.
2. Assign the method used for determining initial response times and assign the fact gathering service.
 - a. In Customizing for **Service** choose **E-Mail Response Management System > Service Manager > Define Service Manager Profiles**.
 - b. Select your service manager profile.
 - c. Under **Directly Called Services**, find the Service ID RE_RULE_EXEC and add the following services underneath to ensure that they are executed in the correct order with the attribute **Invocation Order**. If you do not use Rules and Actions to determine your initial response times, the order of the services could be invoked before RE_RULE_EXEC:
 - i. Under **Directly Called Services**, add the Service ID FG_ESCALTIME if it is missing. You can maintain the property values using the F1 field help.

- ii. Under **Directly Called Services**, add or select Service ID UT_ESCALDETERM and choose **Properties**.
- iii. In the **Property ID** field add or select **METHOD**, and in the **Property Value** field enter the value. Refer to the F1 field help for e-mail escalation values.

i Note

Do not include the service UT_SEND_ESCL as part of your **DEFAULT** service manager profile.

3. Customizing of the notification mail.

- a. In the WebClient UI, define the mail form used for notifying managers of the e-mail escalation. This is done in the interaction center (IC) manager role under **Knowledge Management > Mail Forms**. Insert escalation specific placeholders by clicking **Attributes** in the mail form and adding those specific escalation fields (for example, escalation date, time and type, sender's name and e-mail address and the e-mail subject).

If these escalation objects are not available, ensure that object CRMS_ERMS_ESC_FIELDS has been added and configured to the correct context in the mail form. In Customizing for **Service** choose **Basic Functions > Communication Channels > Maintain Attribute Contexts for Mail Forms**.

- b. Assign the mail form and the receiver's and sender's e-mail information to the service manager profile.

- i. In Customizing choose **Service > E-Mail Response Management System > Service Manager > Define Service Manager Profiles**.
 - ii. Select service manager profile ERMS_ESCALMSG and maintain the property values of the service UT_SEND_ESCL under **Directly Called Services**. Refer to the F1 field help for e-mail escalation values.
- c. In Customizing choose **Service > E-Mail Response Management System > Service Manager > Assign Service Manager Profiles**. Assign the service manager profile ERMS_ESCALMSG to the object **ObjectID** 15107965 without entering the **Address/Number**.
- d. In transaction SWU3 under **Maintain Runtime Environment**, maintain the **Schedule Background Job for Workflow Deadline Monitoring** and specify the interval until next deadline check.

Automated Processes and Architecture

Rule Processing Services

Use

Rule processing, or rule invocation, services is one of the four types of services found in the E-Mail Response Management (ERMS) service manager. There are standard delivered rule processing services available. For more information, see Customizing documentation for **Service > E-Mail Response Management System > Service Manager**.

Features

Rule processing services evaluate a set of predefined rules and determine a set of actions that need to be executed during rule processing.

Rules consist of conditions and actions that are triggered if the conditions evaluate to true. The rule processing services support rules and nested rules. Nested rules are only processed if the condition in the higher-level rule is true.

Rule processing starts with the first rule defined within a given policy and continues sequentially with all subsequent rules in exactly the order defined in the rule modeler. All actions that are triggered during rule evaluation are collected and passed back to the service manager. For each of these actions, the service manager determines the appropriate action handler and calls these handlers accordingly.

Special rule processing

The **Stop Further Rule Processing** action can immediately interrupt rule processing. If this action is triggered, only that actions that have executed to that point are passed to the service manager.

The **Invoke Policy** action is executed by the service manager after the current policy is completely processed. It is not executed immediately.

Content Analysis

Use

This function identifies the categories that best match specific content and makes automatic category suggestions.

Content analysis is used in the E-Mail Response Management System (ERMS) to analyze the content of inbound e-mails. It uses predefined search queries that are assigned to categorization schemas, to determine suitable categories.

Integration

Function	Description
Category Modeler	You define content queries for content analysis in the category modeler. If necessary, you can reset the content analysis in the page area Application Areas for each application area. You do this by choosing Reset Content Analysis.
E-Mail Response Management System (ERMS) - automated processing	ERMS manages a high volume of inbound e-mails. This system partially automates routing and responding to e-mails. The automated processing of e-mails in the ERMS uses content analysis to evaluate conditions assigned to categories. The following ERMS components relate to content analysis: • Fact gathering • Rule operators Matches Category and Rule Attributes
Interaction Center	Content analysis provides suggested categories based on the content of e-mails received in the interaction center and selected for processing in the agent inbox. These categories are then displayed in the relevant fields of the following IC-specific business transactions: • Service order • Service request • Interaction record

Prerequisites

You have made content analysis settings in Customizing for [Service](#) by choosing **Cross-Application Components** > **Multilevel Categorization** > **Define Application Areas for Categorization**.

Features

- Automated categorization

Category suggestions are made automatically using content queries.

- Language detection

Content analysis also supports automatic language detection for textual content.

More Information

[E-Mail Response Management System](#)

Fact Base

Use

The fact base is an E-Mail Response Management System (ERMS) component that serves as a data container for any information that is related to an inbound e-mail and can be relevant during rule evaluation.

Integration

Fact gathering services gather data relevant to the inbound e-mail and add the data to the fact base.

The rule engine runtime retrieves data from the fact base and uses it for rule evaluation and processing.

Features

The fact base stores information that is relevant for exactly one instance of an inbound e-mail. Such information may include e-mail content, business partner information, and the incoming e-mail address to which the e-mail was originally sent.

The fact base is organized as an XML document. Its structure is flexible to include any tree structure and content, depending upon your business requirements.

Updates and read access to the fact base occurs through XSL transformations (XSLT) and XPath queries. These standardized methods are platform- and programming-language independent. This provides maximum flexibility if you need to extend the fact base, for example, for data residing in legacy systems.

Overview Report

This report provides a real-time snapshot of the activity and current status of the [E-Mail Response Management System \(ERMS\)](#). It gives you an overview of the volume of incoming e-mails, which you can display by organizational unit, e-mail address, or Web form ID.

To use this monitoring function, you must be actively using ERMS. To use this function, you must be running Service, and you must activate the necessary BSP service.

Performance Report

You can use this report to monitor the overall effectiveness of the [E-Mail Response Management System](#) (ERMS) in your interaction center. This report provides a real time overview of key performance indicators such as volume, average handling time, average response time, and service level. You can sort this information by organizational unit or by e-mail address.

To use this monitoring function, you must be actively using ERMS.

Workflow for ERMS

Use

This workflow is used for the E-Mail Response Management System (ERMS) in the Interaction Center.

ERMS allows automated processing of inbound e-mails received via SAPconnect. E-mails can be automatically responded to or routed to an agent group for manual processing. An inbound communication triggers the workflow template WS00200001, which controls the work item processing and fulfills several tasks.

The following system tasks are the most important:

- Determining the responsible agent group and routing the work item to that group
- Deleting and stopping further manual processing

For a complete list of tasks, see the workflow template WS00200001 in the workflow definition tools.

Prerequisites

ERMS configuration activities are completed.

Process

1. The incoming e-mail is received.
2. Task TS00207915 **ERMS Rule Execution** invokes the service manager.

The service manager provides capabilities (necessary to process e-mails) such as fact gathering, content analysis, and rule processing.

3. Depending on the outcome of the service manager invocation, the system continues processing the work item by one of the following steps:

- Delete

You can determine when the system deletes an item by using a rule, for example, in cases of junk e-mail (spam). Task TS66107918 **ERMS Delete E-Mail** removes the office document. The workflow is ended.

- Do not process further

You define when processing is stopped in either a rule or in one of the services in the service manager.

- o Route

Task TS00207914 **ERMS Decision** forwards the work item to an agent or agent group as selected by the processor determination.

The processor determination decides which agents are responsible for a work item. Task TS00207914 determines which agents or agent groups are eligible for processing e-mails. The actual determination of which agent or agent group receives the work item is defined in rules.

Technical Execution

The following technical information is useful if you wish to customize your own workflow:

- Workflow template

The actual procedure is implemented in the workflow template WS00200001 (**ERMS1**).

- Triggering events for the workflow template

The event **MailReceived** of Business Object Repository (BOR) object ERMSSUPRT2 is maintained as a triggering event in the template WS00200001.

- Processing methods

The following table explains how methods of BOR object ERMSSUPRT2 are used in the corresponding tasks of the workflow template WS00200001:

Method	Use
RECEIVE	Receives the mail from SAPconnect
RULEEXECUTION	Raises the event MailReceived and triggers workflow

Setting Up ERMS

Use

This procedure describes the basic settings needed for [E-Mail Response Management System](#).

Prerequisites

Your IT administrator has created e-mail addresses for receiving incoming e-mails.

Procedure

Mandatory and Recommended Steps

Step	Description
Create an organizational model.	In Customizing for Service under Master Data > Organizational Management > Organizational Model > Create Organizational Model .
Maintain workflow standard settings.	In Customizing for Service under Basic Functions > SAP Business Workflow > Maintain Standard Settings for SAP Business Workflow .

Step	Description
Define receiving e-mail addresses.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Define Receiving E-Mail Addresses/Fax Numbers .
Assign receiving e-mail addresses to ERMS workflow.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Maintain Recipient Distribution .
Assign agents for e-mail handling.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Assign Agent for E-Mail Handling . Choose Assign Agents . Under workflow ERMS 1 either: • Use a general task for ERMS decision to allow e-mail routing to all agents, depending on the rules, or • Specify individual agents or agent groups to allow e-mail routing only to those agents, depending on the rules.
Activate event linking. Incoming e-mails can use workflow to define how mails are routed and how they are handled. Activating event linking activates the workflow you defined and allows it to be used for ERMS.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Assign Agent for E-Mail Handling . Choose Activate Event Linking . For New Recipient , choose possible entries and select SAP Object Instance . Use business object repository (BOR) Object Type ERMSSUPRT2 (ERMS Support 2) . When this BOR object type is invoked, it triggers workflow WS00200001 (ERMS1) . For more information, see Workflow for ERMS .
Define outgoing e-mail addresses.	In the SAP Menu, choose Interaction Center > E-Mail Response Management System > Settings > Maintain Sender E-Mail Addresses .
Define service manager profiles.	In Customizing for Service under E-Mail Response Management System > Service Manager > Define Service Manager Profiles . Choose Default and double click Directly Called Services to set the DEF_ROUTING for Invocation Order 50 to an organizational unit.
Assign service manager profiles.	In Customizing for Service under E-Mail Response Management System > Service Manager > Assign Service Manager Profiles .
Create ERMS policies and rules	In the Interaction Center, choose Process Modeling > Rule Policies . Choose E-Mail Response Management System as the Context .
Assign standard tasks to communication methods	In Customizing for Service under Interaction Center WebClient > Agent Inbox > Settings for Asynchronous Inbound Processing > Assign Standard Tasks to Communication Methods . Choose INT as the communication method for Object ID 207914 .
Define e-mail templates and e-mail forms.	In the Interaction Center, choose Knowledge Management > Mail Forms .

Optional Steps/Extensions of Default Customizing

Step	Description
Define additional services. You can define services that are available in the service manager. This activity determines how all necessary services for evaluating policies are invoked. You should only	In Customizing for Service under E-Mail Response Management System > Service Manager > Define Services .

Step	Description
execute this activity if you want to change the default settings or create new services.	
Extend the ERMS repository.	In Customizing for Service under E-Mail Response Management System Define Repository .
Define additional e-mail profiles.	In Customizing for Service under Interaction Center WebClient Basic Functions Communication Channels Define E-Mail Profiles .
Define catalogs, codes, and subject profiles. Catalogs are resources which help ensure the uniform use of terms	In Customizing for Service under Basic Functions Catalogs, Codes and Profiles .
Define additional categorization profiles. This influences which categorization schema(s) are displayed in the corresponding business transactions, and the behavior of the auto suggest function.	In Customizing for Service under Interaction Center WebClient Business Transaction Define Categorization Profiles .
Define reporting for e-mail statuses and events.	In Customizing for Service under E-Mail Response Management System Reporting .
Delete reporting data.	In the SAP Menu, choose Interaction Center E-Mail Response Management System Utilities Delete Reporting Data .

Integrated Communication Interface

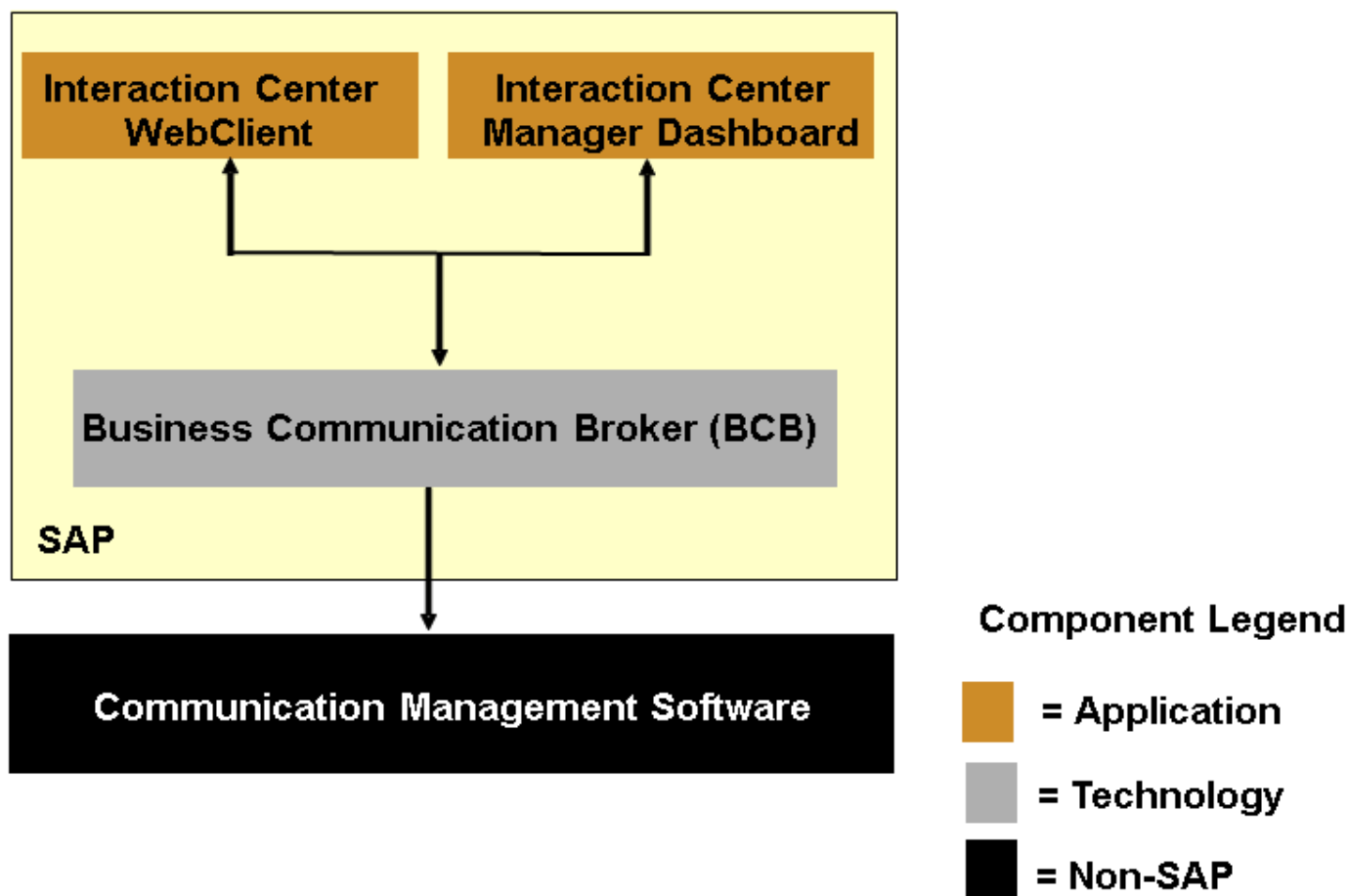
Use

The Integrated Communication Interface (ICI) supports the integration of non-SAP communication products with SAP components that use communication services. The ICI is intended in particular for scenarios involving multiple communication channels such as telephony, e-mail, and chat, but it can also be used in single-channel scenarios. It enables activities such as an agent accepting an inbound phone call or deflecting an inbound chat request, or a supervisor monitoring a queue. The ICI is only one of several components required to implement communication scenarios of this kind.

i Note

An example scenario is the integration of communication management software systems with the interaction center (IC), specifically the IC WebClient and IC manager dashboard. For illustration purposes, this example is used throughout this documentation.

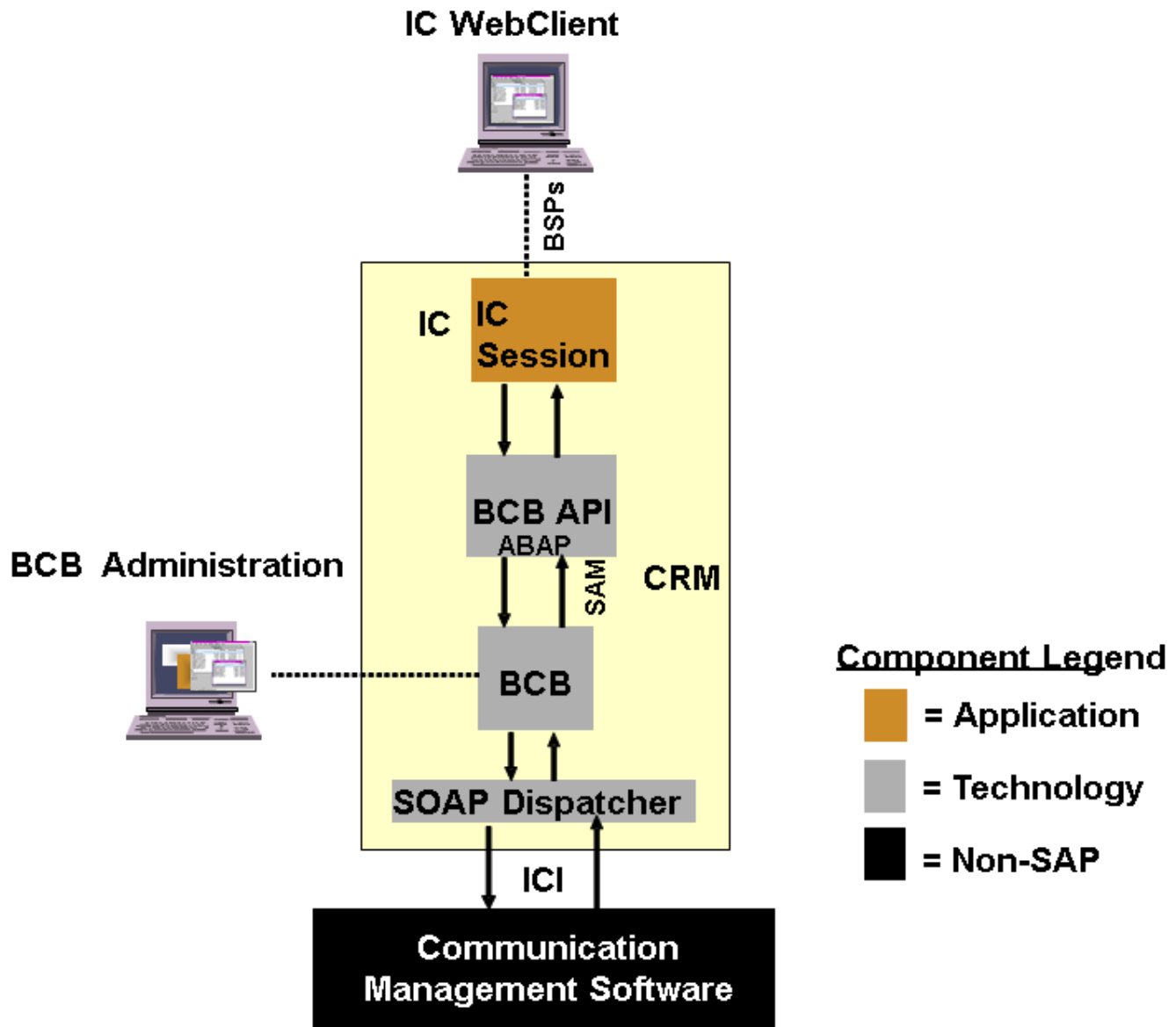
The graphic below represents a simplified view an example scenario without detailing the interfaces. The scenario is made up of functionality from three distinct sources – an application component, a technology component, and an external (non-SAP) component.



Integration

The ICI is based on Simple Object Access Protocol (SOAP) and Extensible Markup Language (XML); they are non-proprietary, Web-oriented, open interface technologies. HTTP is used at the transport level.

The system architecture required to connect the interaction center with external communication management software using the ICI is made up of several components. The system architecture of the interaction center example scenario in a live environment is shown below:



Live System Architecture

BCB Components

The BCB, BCB API, and SOAP dispatcher components enable the CMS. The messages passed between the application component and the external component are passed via the ICI.

i Note

CMS Customizing for IC WebClient is done using transaction `CRMM_BCB_ADM`.

Application-Specific Components

The application-specific components are linked to the ICI-specific components via the Business Communication Broker Application Programming Interface (BCB API). This is the API for application customizing.

In the interaction center example scenario, the application consists of the IC WebClient, which provides an ABAP-based user interface for interaction center agents.

The IC server sends actions to the communication management software via the ICI-specific components and receives events back from the communication management software also via the ICI-specific components. The actions might include agent logon,

initiate outbound phone call, or remove chat line from chat session. The events might include new inbound phone call, or new posting in chat session.

External Components

These are non-SAP communication systems. In this scenario, the external product is the communication management software, which is a multichannel communication product. This is responsible for merging requests from different media, routing requests to specific agents or agent groups, queuing requests, signaling inbound requests to the IC, and so on. The external products communicate with SAP via the ICI using SOAP and XML.

Certification

A certification process is implemented to verify the connectivity of external communication products with SAP using the ICI. An external product does not have to support the ICI in full. Partial certification is possible to enable the connection of single-channel communication products, such as telephony products.

Features

Many application components use external communication services. Integrating an external communication product with an application component using the ICI can provide:

- Integration of multiple communication channels such as phone, e-mail, and chat
- Active reporting of new, changed, or completed phone calls, e-mails, or chat sessions (known generically as items)
- Monitoring and statistics

i Note

Multichannel management (that is, the synchronization and serialization of incoming communications) is not part of this SAP functionality. The ICI is a server-to-server interface. It does not support direct communication between the SAP client and the external communication product.

Signing In and Out

Procedure

Entering your user ID and password in the logon page takes you to the home page, which contains your most important personal information, such as tasks and appointments. From here, you can access the search, quickly create appointments, tasks, and contact persons, and send e-mails.

Click [Sign Out](#) to end your work session.

i Note

Your user ID and password determine the role with which you sign in. You can also choose the language. If you click [Change Password](#), you can enter a new password.

More Information

[Working with Business Roles](#)

Accessibility in WebClient UI

Use

WebClient UI applications provide accessibility options. The applications and controls are accessible by default. You can use settings to personalize keyboard shortcuts, and use the high contrast black or high contrast white theme.

Keyboard Shortcuts

You can access all interactive screen elements, like links, fields, buttons, and so on, using keyboard actions. There are special navigation shortcuts that you can use for setting the cursor to focus on specific screen areas.

Personalizing Shortcuts

1. To open the personalization page, choose the **Personalize** link in the user menu. If you are using the WebClient launchpad integration mode, choose the **My Dashboard** button in the WebClient UI App toolbar to open the dashboard page instead.
2. In the **Shortcuts** block, choose the **Personalize Shortcuts** link.
3. Here, you can define your own shortcuts.

For more information, see

- [Keyboard Shortcuts for Navigation](#)
- [Keyboard Shortcuts for Spreadsheet](#)
- [Keyboard Shortcuts for Standard Buttons](#)

High Contrast Themes

WebClient UI provides a high contrast black and high contrast white theme to support people with visual impairments.

Choosing the High Contrast Black or White Theme

1. To open the personalization page, choose the **Personalize** link in the user menu.
2. In the **Skin** block, choose **SAP High Contrast Black** or **SAP High Contrast White** to switch to the high contrast themes.

In WebClient UI launchpad-integrated mode, use the SAP Fiori launchpad user settings to choose the **SAP High Contrast Black** or **High Contrast White** theme.

Working with Business Roles

Use

You can use predefined business roles to access all the data and information that you require for your daily work, at any time. You sign in with your business role and have access to all applications and data that are assigned to this role.

If you are using SAP Fiori launchpad integrated mode, see [Integration with SAP Fiori Launchpad](#)

i Note

System administration creates your business roles for you, determines the role content, and assigns the required authorizations.

In the organizational information section of the personal data in the central personalization page, you can see with which business role you are signed in to the system.

More Information

[Personalizing the User Interface](#)

What Does the User Interface Look Like?

Use

The SAP Fiori visual theme for the WebClient UI Framework aligns with the look and feel with SAP Fiori 3. Skins from the classical WebClient UI Framework and personalized skins that are not Fiori-compatible will not be available. If you have a Fiori-compatible skin defined and selected in Personalization, that skin will be loaded, otherwise, the system will load the SAP Morning Horizon skin by default.

Features

The table below contains a description of each area:

Area	Use
Header	The header is always visible. Features of the header include: • Menu icon: Allows you to toggle the visibility of the side panel • Breadcrumbs and history navigation • Home button • Logo • Timer • Title • Access to the main menu, if configured • Search launcher • Quick create button • Notification Center • User menu i Note The title and main menu will not be part of the header if you are using the IC WebClient business role.
Side Panel	You can toggle the visibility of the side panel using the menu icon in the header. The side panel contains the logo and logo text, system

Area	Use
	information, and group links. The main menu could also be configured to be visible here instead of in the header.
User Menu	The user menu is accessed by choosing the the user icon in the header. • Your user name • Personal settings • Share • Role Select • System news • Help center • Sign Out
Group Links	This is a set of tabs for navigating. The tabs that appear in your group links depend on your Customizing and personalization settings. Some of the tabs you may see include the main menu (apps) and recent items.
Work Area	The main portion of the screen is the work area. The overview pages as well as the advanced search pages are displayed here. There is a work area header and an anchor bar when applicable. Success messages are toasted in the work area.
Work Area Anchor Bar	The anchor bar is displayed when you have multiple assignment blocks within your work area. You can navigate to these assignment blocks quickly using the anchor bar rather than scrolling vertically.
Footer	The floating footer is displayed when there are messages, or closing or finalizing actions for the work area. On the left side of the footer, you find a message icon, visible when there are messages. On the right side, closing or finalizing actions are displayed, with the left most action emphasized.
Message Area	A message area icon appears in the footer when there are messages. The appearance of this icon changes to match the classification of the most severe message type. By default, in the event of error messages, the message area opens automatically, otherwise, you can open the area by choosing the message button. You can enable and disable the automatic expansion of the message area in your personalization settings. Using the message summary bar, you are able to view all of the messages or filter them based on the message type. If more information is available about a particular message, you can access this information by selecting the chevron to the right of the message. Message types: • Message with an error short text • Message with a warning short text

Area	Use
	• Message with a positive short text • Message with an information short text

More Information

[Personalizing the User Interface](#)

[Keyboard Navigation](#)

What Pages Are There?

Use

On each page, you can execute different tasks. For example, you can search for data in the search page or view business object data in the overview page.

Features

The table below contains a description of the pages in the user interface:

Page	Use
Home Page	This is used as the entry point for your daily work. The page contains your most important personal data and gives you central access to the functions you use most frequently.
Work Center	This is used as the entry point for all role-dependent areas. The page contains data and hyperlinks for pages you access most frequently. Here you can search for business objects or create new business objects.
Search Page	This is used to search for business objects. You can launch the page from the header or the home page.
Overview Page	This contains all the important information on a single business object and the business objects assigned to it. From here, you can navigate to subordinate overview pages. In certain business objects, the details are displayed as a hierarchy. The overview page is used in two modes: • In display mode, you can view all data • In edit mode (editable overview page), you can change business object data as well as the data in most assignment blocks.

Page	Use
Assignment Block	This is in the overview page and contains the most important data for an assigned business object. This data is frequently displayed in a list. You can navigate from here to the edit page, where you can edit the data.
Edit Page	This helps you to edit assigned business objects. The data is displayed in forms or tables.
Report Page	This contains hyperlinks to reports. You can launch the page from the top level in the navigation bar.
Dashboard Page	This contains business intelligence analyses and hyperlinks to more analyses. You can launch the page from the second level in the navigation bar.

More Information

[Navigating Between Pages](#)

Buttons and Icons

Use

You can use buttons and icons to perform your daily tasks, for example, displaying, editing, deleting, or printing data.

Features

The table below contains a description of general icons, buttons, and hyperlinks:

Icons, Buttons, and Hyperlinks	Function
	Creates a new business object
	Saves a new or edited business object You can save a new or edited business object only in the editable overview page. When you save, you can no longer change the data in the overview page.
	Deletes a business object After deleting, you return to the last page you viewed that does not belong to this business object. In assignment blocks, you can use this function to delete the business object assignments.
	Edits a business object

Icons, Buttons, and Hyperlinks	Function
	In assignment blocks, you can use this function to edit the detail data of a business object or an assigned business object.  Example You can edit the business appointments of the account, or the organizational data of the sales order.
✓	Returns to the page that you visited previously You use this function to return from the edit page or editable overview page of an assigned business object to the overview page of the business page; you can save the changed data here.
	Prints the current page
	Opens the online help The online help opens for the page you are currently viewing.
<	Navigates back to the page you viewed previously You use this function to navigate from the page that you are currently on to the page you viewed previously.
>	Navigates to the next page You use this function to navigate to the page that you viewed after a page that you navigated back to.
▼	Opens the dropdown box under < and > In the respective dropdown boxes, you can choose the pages to which you want to navigate.
	Copies a business object
×	Cancels and returns to the previous page Your latest changes are not saved, and you see the data that you saved last.
▼	Additional functions that are grouped in a dropdown box under the pushbutton
	Refreshes the display
	Exports data to spreadsheets, for example, to Microsoft Excel
	Personalizes the display in pages, tabular assignment blocks, and tables
	Opens and closes a table shown as a chart, in result lists and assignment blocks You can choose between pie charts and bar charts.
	Opens the input help or search help for the relevant field
	Opens the calendar

Icons, Buttons, and Hyperlinks	Function
	Adds an existing business object, in assignment blocks, for example
+ and 	Adds and removes additional search lines in the search
	Selects and deselects all search results in the result list and in other lists
▼ and >	Shows and hides assignment blocks on the overview page
	Goes to the top of an overview page This icon is on the right of the title bar in all assignment blocks. In long overview pages with a large number of assignment blocks, you can use this icon to return to the top of the page instead of scrolling vertically to it.
	Starts the configuration of the corresponding page
	Shows and hides the configurable areas on a page When you click the selected area, you start the configuration of the corresponding area.
Button Search	Launches the search on the search page You can also launch the search by pressing the Enter key.
Button Clear	Resets the search fields and the result list in the search
Button Reset to Default	Resets to the default settings, in the personalization of the user interface layout, for example
Button Insert	Inserts an assigned business objects In an assignment block, click Edit List to add more business objects to the list.
Button Mass Change	Applies changes that were made to one business object, to multiple business objects
Button Follow-Up	Creates a follow-up transaction, such as a task, interaction log, business activity, appointment, or e-mail
Button Output	Prints or sends a business object by fax or e-mail
Hyperlinks Show Search Fields and Hide Search Fields	Shows and hides search fields on the search page
Hyperlinks Expand and Collapse	Expands and collapses additional lines in tables You can use these hyperlinks to expand and collapse additional lines in tables, when a specific number of lines is exceeded.
Hyperlink Edit List	Navigates to the list of assigned business objects

Icons, Buttons, and Hyperlinks	Function
	When you click this hyperlink in assignment blocks, you can directly edit the list of assigned business objects.

i Note

In business roles that use communication management software, you require special functions to make and accept calls.

More Information

[Personalizing Pages](#)

[Exporting Data](#)

[Performing a Search](#)

[Personalizing the Search Result](#)

Personalizing

Use

You can personalize the user interface, for example, by selecting various themes. You can personalize individual pages, and adjust the search result to your personal requirements.

More Information

[Personalizing the User Interface](#)

[Personalizing Pages](#)

[Personalizing the Search Result](#)

Personalizing the User Interface

Procedure

When you sign on, the user interface has a particular default layout. You can personalize the user interface, however. That is, you can adjust the display to meet your own requirements.

In the user menu, which is found in the header, you can navigate to the central personalization page using the **Personalize** link.

i Note

If you want to personalize the page currently displayed, click the **Personalize** icon on the top right of the page. In the **Content** dropdown box, you can select the information block that you want to see in this location on the page. If you choose the blank line, the information block remains empty. Click **Save** to save your personalization. If you click **Reset to Default**, the standard settings are restored.

Changing Personal Data

You can change your personal data, such as your first name or last name. You can change communication information such as your telephone number, fax number, or e-mail address. You can view organizational information, such as in which organizational unit or position you are employed. In the **Business Role** field, you can see with which role you are signed into the system.

Changing Your Password

If you click **Change Password**, you can change your current password by entering a new one.

Personalizing General Settings

You can make general settings, for example time zone, date format, or time format. You can activate the configuration mode, that is, activate the configuration of pages and views directly from the user interface. You can activate the tab order mode, that is, you can define your own tab order on the user interface. You can change table settings to enable filter saving, sort saving, and automatic text wrapping. In addition, you can activate favorites and tag clouds.

Defining Groupware Integration

You can set up your e-mail transfer in Microsoft Outlook. You can edit your settings for synchronizing your personal calendar, tasks, and contacts in the system. You can install the new version of Groupware synchronization software on your local computer.

Defining the Layout

You can define the layout of the user interface.

- You can choose between different skins, and view each one in a thumbnail.
- You can choose to the option **Optimize for Touch Input** to increase the size and spacing of controls. This allows you to interact with them more easily using your fingertip. This is useful for hybrid devices that combine touch and mouse events.

You can set up the user interface so that you waive some functions and visual effects, in favor of better performance. You can also disable suggestions for entries in input helps.

Personalizing the Navigation Links

You can personalize the navigation links, by defining which of the links available for your role are displayed. This includes recent items, quick creates and other link groups and apps. You can also determine the sequence of the displayed links.

Personalizing Recent Items

You can define whether the **Recent Items** block is displayed in your business role. In this block, you see the titles of the items that you have recently used, as hyperlinks in the side panel. In addition, you can determine the number of titles displayed, up to a maximum of 25 entries.

Setting Up Keyboard Shortcuts

You can set up personal keyboard shortcuts. The system makes a distinction between those shortcuts with which you focus on certain areas in the user interface, and those with which you control navigation to other pages or business objects.

Example

You can set up shortcuts that focus on the header area, the side panel, or the work area. You can define shortcuts that control navigation to the home page, calendar, or activities.

Add Application Links, Web Links, and Widgets

You can add application links, Web links, and widgets to the home page of your role. You can choose which of the links that are available in your role should be displayed on the home page. You can also determine the sequence of the displayed links.

You can also make these settings directly on the home page of your role, if you click the **Personalize** icon in the menu of the respective information block.

Subscribing to a Feed

You can subscribe to your personal feed. To do so, copy the link to your feed reader.

More Information

[What Does the User Interface Look Like?](#)

[Keyboard Navigation](#)

[Navigating](#)

[Defining Tab Order](#)

[Using Favorites](#)

[Using Tag Clouds](#)

Personalizing Pages

Use

When you sign in, all pages have a particular default layout. However, you can personalize your pages. That is, you can adjust the way that pages are displayed, to meet your own requirements.

Procedure

Personalizing the Home Page, Work Center Pages and Report Pages

On the home page, work center pages, and report pages, you can define which entries are displayed in information blocks. The personalization is the same for all three pages.

When you sign in, the home page is the first thing you see. To navigate to a work center, you choose a role-dependent area in the apps. The report pages are accessed from **Reports** in the side panel.

At the top right of the page, click the **Personalize** icon to personalize the home page.

Using drag and drop, you can change the sequence of the information blocks on the page by moving them from left to right, or up and down.

You can define which entries are displayed in information blocks. Click the **Personalize** icon from the menu in the information block, and select the entries that you would like displayed. In the list of displayed entries, you can define the sequence in which they are displayed in the information block.

Personalizing the Overview Page

You have searched for a business object, for example a service order, and are in the overview page for that service order. Click the **Personalize** icon to display the personalization dialog.

Just like in the home page, work center pages and report pages, you can use drag and drop to position the assignment blocks in the overview page.

In the personalization dialog for the overview page, you can define which assignment blocks are always directly visible and open. To do so, activate the checkbox **Display Expanded** for the relevant assignment block, under **Displayed Assignment Blocks**.

i Note

If you click **Reset to Default**, the standard settings are restored.

Personalizing Columns in Assignment Blocks

In the personalization dialog for assignment blocks, you can determine which columns should be displayed. You can use **Up** and **Down** or drag and drop, to define the sequence of the displayed columns. You can specify the absolute (pixel) column width, and the relative (percentage) column width in relation to the total width. You can also enter the maximum number of rows that should be displayed in this assignment block.

i Note

With the runtime merge of the configuration and personalization, the newly added or removed assignment block or column will be automatically added to the existing personalization. The changes will be reflected in the UI for the user without deleting the existing user personalization.

More Information

[What Pages Are There?](#)

[Personalizing the Search Result](#)

Personalizing Tables

Use

You can personalize tables using the **Personalization** dialog, which you open from the **Personalization** icon on the top right in the title bar of the results list.

Procedure

When you perform a search, your results are displayed in a results list or table.

The result list contains at least 10 lines per page. No blank lines are displayed. That is, if the search finds fewer than 10 results, you see only the number of results found.

The minimum number of lines displayed, and the appearance of columns depend on the business object for which you are searching.

Specifying the Maximum Number of Search Results

Under the search fields, you can define the maximum number of rows that should be displayed in the result list.

i Note

If the search result exceeds this value, a corresponding message appears in the message bar in the header area. The value you set is not exceeded in the result list.

Defining and Fixing Columns

You can define the column width of individual columns

- In relation to the total width
- In absolute width in pixels
- In relative width in percent

You can fix columns on the left and/or right of the table. In this case, you must use column width in pixels.

Showing and Hiding Search Fields

You can use the [Show Search Fields](#) and [Hide Search Fields](#) hyperlinks to show or hide the search fields. This lets you make the result list larger or smaller, for example if you need more space for displaying search results.

If you hide the search fields, the result list contains at least 20 rows by default, unless you enter a different value in the [Maximum Number of Results](#) field.

Sorting Search Results

If you click a column title, you can sort all search results in the corresponding column. A dropdown box appears in which you can choose between the following options:

- Sort ascending

All search results are sorted alphabetically or numerically in ascending order. The column title is marked with a small triangle pointing upwards.

- Sort descending

All search results are sorted alphabetically or numerically in descending order. The column title is marked with a small triangle pointing downwards.

i Note

The sort functions in the dropdown box are separated from the filter functions, by a dashed line.

Filtering the Search Result

If you click a column title, you can filter the corresponding column. A dropdown box appears in which you can choose between the following options:

- (All)

All highlighting and filter values in the corresponding column are reset. Any sorting of the column is kept.

- (Custom...)

If you click this entry, you can enter a filter value according to which the column is filtered. You can use the wildcard characters ' * ' and ' + ' here. The filter icon after the column title shows that this column was filtered.

❖ Example

The wildcard character '*' replaces multiple unknown letters, for example `Wal*` or `Wal*dorf`. The wildcard character '+' replaces one unknown letter, for example `Wal+dorf`.

- (Too Many Entries...)

This appears in the dropdown box if more than 100 different data records were found in the search. If you click this entry, you can enter a filter value according to which this column should be filtered. You can use the wildcard character '*' here, which replaces any number of characters. The filter icon after the column title shows that this column was filtered.

- Data records displayed in the column

The data records displayed are alphabetically or numerically sorted. If there are more than eight data records, you can scroll up and down in the dropdown box. A maximum of 100 data records is displayed in the dropdown box.

If you click a particular data record in the dropdown box, the column is filtered according to this data record. The filter icon after the column title shows that this column was filtered.

i Note

The filter icon appears after the column title even if its text is truncated. If you hover over the column title with the mouse, you see the complete column title in a quick info text.

More Information

[Editing Data](#)

[Creating Data](#)

[Personalizing the User Interface](#)

[Using Charts](#)

[Exporting Data](#)

Defining Tab Order

Prerequisites

Before you define the tab order, you must activate the tab order definition mode in the central personalization page, in the general settings.

Context

You can define your own tab order, that is, you can define the sequence of fields that you edit on a page yourself.

Procedure

1. Select the required business object.

You do this on the overview page of the business object.

2. Click **Edit**.

You are then in the edit page of the business object.

3. Click the icon **Define Tab Order** at the top right of the page.

The page segment now has a blue frame, and is shaded.

4. Click the first field of your tab order.

The number “1” appears in the clicked field.

5. Click the subsequent fields of your tab order.

The respective numbers appear in the clicked fields. To change the sequence, click the numbered fields again. The sequence is changed accordingly.

6. Save your new tab order.

You are then in the edit page of the business object.

Example

In the advanced search page of service orders, for example, enter your search criteria and select a service order. In the overview page of the service order, click **Edit**. On the edit page, click the icon **Define Tab Order**. You can now define your personal tab order.

Next Steps

[Personalizing the User Interface](#)

Searches

Use

You can search for business objects such as accounts, products, opportunities, or activities. The result of your search is displayed in the result list.

A series of search operators, described in one of the following documents, is available for the search. Input help or search help in certain fields also assists you in your search.

More Information

[Using the Central Search](#)

[Performing a Search](#)

[Performing an Archive Search](#)

[Search Operators](#)

[Using Saved Searches](#)

[Personalizing the Search Result](#)

[Using Input Help and Search Help](#)

[Using Charts](#)

Using the Central Search

Use

Searching for data is one of your most important tasks. The central search is a fast and flexible tool for this. In Customizing, you define that the central search is located in the header area of the user interface, and is therefore visible and accessible at all times, regardless of the application you are in.

Procedure

Opening the Central Search

When you open the search menu of the central search in the header area, you can choose between the following options:

- Saved searches
- Search all objects
- Search individual applications

i Note

You can use the wildcard character ' * ' if you are not sure of a spelling, *a* for example.

If you want to further refine your search criteria, execute a search to navigate to the search page of the application and choose [Show Search Fields](#).

Performing Saved Searches

In the search menu, choose [Saved Searches](#). The field label takes on the name [Saved Searches](#). Now you can select and perform a saved search. Selecting the saved search name from the dropdown directly executes the saved search. You can change the search criteria of your saved search and save it again.

Searching All Objects

In the search menu, select [All Objects](#). The field label takes on the name [Search for All Objects](#). You can enter key words, IDs, or free text in the search field. The search is performed in all applications that belong to your role.

If you choose an individual application and the attribute [Free Text](#) in the search menu, you can also enter free text in the search field.

i Note

If you are searching for an ID, you must enter the wildcard character ' * ' in front of the ID, in the search field.

Searching Individual Applications

When you open the search menu, you see an alphabetical list with all applications that belong to your role. Choose an application. You automatically see all attributes that belong to this application. As soon as you choose an application and an attribute, both are highlighted in color. Choose your selection. The field label takes on the name of the search that you want to perform, for example

Service Order/Description. Enter a search term in the search field, and choose the **Search** icon or **[ENTER]** to perform your search.

Result

The result of your search by individual application is displayed in the result list of the respective application. If you search in all the applications of your role, you see the result in the search result page. You see the search result of each individual application in a separate result list on this page.

More Information

[Performing a Search](#)

[Using Saved Searches](#)

[Personalizing the Search Result](#)

Performing a Search

In the search, you have a variety of search criteria with which you can refine your search to find precisely the data you are looking for.

❖ Example

For example, in the service role, you want to find the Service Order business object. The business object Service Order belongs to the Service area.

1. In the Apps, choose the area to which the business object you are looking for belongs.
2. In this area, choose the required business object.

Using the example above, you see the search page for service orders.

i Note

You can also launch the search page for service orders by choosing the business object you want in the corresponding work center or home page under **Search**.

3. Refine your search criteria.

- a. Use the default search criteria or select other criteria from the dropdown list.

Search criteria describe characteristics of the business object that you are looking for.

Example search criteria for service orders are **Search Order ID**, **Description**, **Created By**, or **Status**.

- b. Select the search operators.

Search operators define how data is filtered. Example search operators are **is** or **starts with**.

- c. If necessary, enter data to refine your search.

Use the wildcard character asterisk (*) to replace one or more characters. Use wildcards asterisks when you are not sure how to spell the search term you want, for example, **T*st*** or ***est**.

4. In the field **Maximum Number of Results**, you can enter the maximum number of business objects that should be in the result list for this search.

5. Choose **Search** or to confirm.

In the result list you can see the data that matches your search criteria. If the result list is empty, a message appears in the message bar.

i Note

Click **Clear** to clear all search fields and the result list for the current search. However, this does not reset the search criteria and search operators for the current search. Search lines you may have added in the current search are not removed. The field **Save Search As** is also not cleared.

Additional Search Criteria

To copy or delete a search line, click the icons **Copy Line** and **Remove Line** next to the search line. You can use the copied search lines without making any changes to them, or select an additional search criterion from the dropdown list.

Performing an Archive Search

Procedure

In the WebClient UI, you can search for archived business objects. If you searched for a business object but did not find it, you can perform an archive search.

❖ Example

You can search for archived service orders in the service role. The **Service Order ID** is available as a search criteria.

1. Click **Search Archive** in the search page for the business object under the title of the page.

You see the search page for the archive search. The result list in the search page is empty.

2. Enter your search criteria.

3. Click **Search**.

In the result list you can see the data that matches your search criteria. If the result list is empty, a message appears in the message bar.

i Note

To return to the standard search, click **Back to Search** under the title of the search page.

More Information

[Performing a Search](#)

Search Operators

You can use search operators to refine your search to meet your specific requirements.

The table below lists the search operators you can use in the search:

Search Operator	Description	Use
is	Identical	In text- and date fields, in numeric fields, in ID fields, as well as in search fields with dropdown lists
is not	Not identical	In text- and date fields, in numeric fields, in ID fields, as well as in search fields with dropdown lists
starts with	Texts that start with a particular letter of the alphabet	In text fields
contains	Includes	In text fields, numeric fields, and ID fields
is greater than	Range after a certain value	In numeric fields and ID fields
is less than	Range before a certain value	In numeric fields and ID fields
is empty	No value	In text fields, in numeric fields, in ID fields, as well as in search fields with dropdown lists
is between	Range between two values	In numeric fields, ID fields, and date fields
is earlier than	Range before a certain date	In date fields
is later than	Range after a certain date	In date fields

Using Saved Searches

Procedure

You can save your search criteria for further use. Your saved searches are displayed in the search menu of the central search, in the header area. You can also display your searches in a tree structure where saved searches are grouped by categories.

If you are using the tree structure, you can organize the saved searches by defining the order and using folders as categories. Defining your own order can be done using drag and drop. To create new categories, use the main folder's context menu.

Saving Searches

1. Enter your search criteria in the search fields on the search page.
2. In the **Save Search As** field, enter a name for the search.
3. Click **Save**.

The search criteria are saved. In the message bar in the header area, you get a message stating that your search was saved. Your saved search is displayed:

- In the search menu under **Saved Searches** in the central search in the header area
- On the home page under **My Saved Searches**, if this information block is displayed on the home page
- In the tree structure, if configured.

i Note

If you leave a search field empty, this search criterion is not saved in the search. For example, if you leave the **Country** search field empty, this search criterion is not saved.

If you leave all search fields empty, you cannot save the search. In the message bar in the header area, you get a message informing you to fill in at least one search field.

Performing Saved Searches

1. In the search menu, choose **Saved Searches**.

Select the saved search that you want to perform either from the central search or the saved search name from the tree structure.

The search with your saved search criteria is performed. You see the search results in the result list.

Deleting Saved Searches

1. In the search menu in the central search **Saved Searches**, select the search that you want to delete.

If you are using the tree structure, you can delete a saved search using the context menu of the selected saved search.

2. Click **Delete**.

The saved search is deleted and you receive a corresponding message.

Fiori Layouts

For more information about advanced search pages with Fiori layouts, see [Advanced Search Pages with Fiori Layout](#).

More Information

[Using the Central Search](#)

Using Input Help and Search Help

Procedure

You can use input help and search help to search for values for individual fields. The respective field determines whether the input help or the search help launches.

Example

In the service role, when creating an order, you want to search for a particular currency or contact. You also want to determine the planned start and end date.

Using Input Help to Search

1. Click the icon at the end of the field.

You see the input help.

2. Select the value that you want.

The selected value is copied to the corresponding field.

i Note

Using the SVH_DISABLED parameter in the [Define Parameters](#) Customizing activity for [UI Framework](#) [Technical Role Definition](#) you can set various options for the input help. For example, you could set the input help for to show the last values. In this case, if you double-click in fields with input help, or use the **BACKSPACE KEY** in an empty field with input help, you can see your last five entries in this field. Your last entry appears first, in the list that you see underneath the input field.

Using Search Help to Search

1. Click the icon at the end of the field.

You see the search help.

2. Enter your search criteria in the search fields.
3. Click [Search](#).

You see all the values that match these search criteria.

4. Select the value that you want.

The selected value is copied to the field.

Searching for a Date

1. Click the date icon at the end of the corresponding field.

You see the current month in the calendar.

Use the arrow icon on the left to select the previous month, and the arrow icon on the right to select the following month.

2. Click the date that you want.

The date you want is copied to the field.

More Information

[Performing a Search](#)

Navigating

Use

Navigation follows standard navigation paradigms of various web applications. The WebClient UI Framework combines the hub-and-spoke navigation model with an application network model.

Using the Header to Navigate

You can use the header to navigate to the home page and to recently viewed pages using breadcrumbs and history. The main menu, if configured to be in the header, can be used to navigate to other pages.

Using the Side Panel to Navigate

You can toggle the visibility of the side panel using the menu icon. When visible, the side panel can be used to see system information and system links. If the main menu is configured, you can navigate to other pages from here. You can also navigate to

other pages using configured group links and recent items. Recent items are displayed in the sequence in which you used the items, with the most recent at the top of the list.

Page Navigation

- The home page is your starting point when you sign in. It contains personal information and navigation to your most frequently used tools.
- The work center pages summarize all the business content related to that work center. It contains shortcuts and provides access to related components, including reports.
- The reports page is a central page where you can access analytical information and reports.
- Overview pages provide an overview of the information regarding a single instance of an object such as an account. Assignment blocks show related information. This page contains shortcuts to related information.
- The search page allows you to find and navigate to instances of any type of object that is available for your role from the results list or table.

More Information

[What Pages Are There?](#)

[Keyboard Navigation](#)

[Navigating Between Pages](#)

Navigating Between Pages

Procedure

You can navigate between different pages in the user interface. Once you have signed in, the home page appears, where you find your tasks and appointments.

Navigating to the Work Center

If you select a role-dependent area from the apps, the work center is displayed. Here you have the following options:

- Under **Search**, you can launch the respective search page.
- Under **Create**, you can select the new business objects you want to create.
- Under **Reports**, you can find links to the most important reports in this area.

Example

In the service role, you open the **Service** work center. Under **Search**, you can search for service orders, service requests or service confirmations. Under **Create**, you can create service orders, service quotations, or service requests. Under **Reports**, you find reports as customized.

Navigating to the Search Page

If you select a role-dependent business object, the search page of the corresponding business object is launched.

Example

In the service role, you want to search for service orders in the [Service](#) area.

Navigating to the Overview Page

If you click a hyperlink in the result list on the search page, the overview page of the selected business object is displayed. Here you find the most important information about the selected business object.

❖ Example

In the service order search, if you click a hyperlink in the [Service Order ID](#) column, you see the overview page for the selected service order.

If you click a hyperlink in the assignment block of an assigned business object, the overview page of the corresponding business object is displayed.

❖ Example

If you click a hyperlink in the [Involved Parties](#) assignment block in a service order, the overview page of the selected person or organization is displayed.

Navigating to Lower-Level Overview Pages

If you click a hyperlink in an assignment block on the overview page of a selected business object, you navigate to a lower-level overview page.

❖ Example

In the service role, if you click a selected item in the [Item No.](#) column in the [Items](#) assignment block in a service order, you navigate to the overview page for this item.

More Information

[What Pages Are There?](#)

[Performing a Search](#)

[Editing Data](#)

Keyboard Shortcuts for Navigation

Use

You can use the keyboard to control interactive elements (for example, hyperlinks or pushbuttons) on the user interface (UI).

Features

Standard Keyboard Navigation

Action	Keyboard Command
Activate selected pushbutton	ENTER

Action	Keyboard Command
Open menu of a menu button	ENTER
Go to next entry in menu of a menu button	TAB or DOWN ARROW
Go to previous entry in menu of a menu button	SHIFT + TAB or UP ARROW
Choose entry in menu of a menu button	ENTER
Activate navigation link	ENTER
Select or deselect checkbox	SPACEBAR
Select or deselect radio button	SPACEBAR
Select line of a table	ENTER
Go to next entry in column of editable table, if entry is not in list box	DOWN ARROW
Go to previous entry in column of editable table, if entry is not in list box	UP ARROW
Expand or collapse hierarchy level in hierarchy tree	ENTER
Open list box	DOWN ARROW
Go to next entry in list box	DOWN ARROW
Go to previous entry in list box	UP ARROW
Select entry in list box	ENTER or TAB

Side Panel

The side panel contains menus with two levels.

Action	Keyboard Command
Go to next entry on same level.	DOWN ARROW or TAB
Go to previous entry on same level.	UP ARROW or SHIFT + TAB
Activate entry (applies to first and second level of the menu).	ENTER
Open second level of menu when cursor focus is on an entry in first level of menu.	RIGHT ARROW
Close second level of menu and place cursor focus on corresponding entry in first level of menu.	LEFT ARROW

Standard Shortcuts

Action	Keyboard Command
Place the cursor focus on the first hyperlink in the system links.	ALT + 1

Action	Keyboard Command
Place the cursor focus on the first hyperlink in the navigation bar.	ALT + 2
Place the cursor focus on the first interactive element in the create area of the navigation bar.	ALT + 3
Place the cursor focus on the first interactive element in the work area.	ALT + 4
Place the cursor focus on the first pushbutton of the work-area toolbar. When this shortcut is pressed for a page that has no work-area toolbar, such as the home page, this shortcut does not move the cursor position.	ALT + 5
Place the cursor focus on the first pushbutton of the table toolbar when the actual focus is already inside the table.	ALT + 6
Place the cursor focus on the first interactive element in the page navigation area of a table when the focus is already inside the table.	ALT + 7
Place the cursor focus on the first interactive element of the table when the cursor focus is already in the table.	ALT + 8
Place the cursor focus on the first interactive element in the previous group (an assignment block on an overview page or a group on a home or work-center page). If this group does not contain an interactive element, the cursor focus is placed on the next previous group that contains an interactive element.	ALT + PAGE UP
Place the cursor focus on the first interactive element in the next group (an assignment block on an overview page or a group on a home or work-center page). If this group does not contain an interactive element, the cursor focus is placed on the next group that contains an interactive element.	ALT + PAGE DOWN

If you want to use different key combinations for the shortcuts, click [Personalize](#) in the system links. For each of the shortcuts, you can select your own combination of

ALT

,

CTRL

,

SHIFT

, or

ESC

, with another key from the dropdown box.

More Information

[What Does the User Interface Look Like?](#)

Keyboard Shortcuts for Spreadsheet

Use

You can use the following keyboard navigation and mouse navigation in the spreadsheet.

Features

Keyboard Navigation

Keyboard Command	Usage
ENTER	Triggers a submission of all the modified cells, unless application disables that functionality by setting the <code>commitOnEnter</code> tag attribute to FALSE
TAB	Navigates in top left table cell to bottom right table cell direction Focus only stops on active cells, that is, when clicking Expand/Collapse buttons or with editable input fields • Tabbing through table header and lead column cells highlights the corresponding column or row. • Tabbing through table data cells highlights the corresponding column and row Table will scroll if needed
SHIFT + TAB	Navigates in bottom right table cell to top left table cell direction stopping only onto the active cells, and scrolling if needed
ARROW UP	Navigates in bottom column cell to top column cell direction stopping on every data and header cell Table will scroll if needed
ARROW DOWN	Navigates in top column cell to bottom column cell direction stopping on every header and data cell Table will scroll if needed
ARROW LEFT	Navigates in right row cell to left row cell direction stopping on every data and header cell Table will scroll if needed
ARROW RIGHT	Navigates in left row cell to right row cell direction stopping on every header and data cell Table will scroll if needed
CTRL + S	If the save button exists on the page, it will be triggered.
CTRL + X	In an editable field, cuts the selected characters and saves them to clipboard
CTRL + C	In an editable field, copies the selected characters and saves them to clipboard
CTRL + V	In an editable field, pastes the content of the clipboard to the spreadsheet

Navigation with the Mouse

Action	Usage
Clicking the left mouse button	• Executes a button • Selects a cell
Double-clicking the left mouse button	Positions the cursor on an input field

Action	Usage
Spinning the mouse-wheel	Scrolls the spreadsheet vertically
SHIFT + Spinning the mouse-wheel	Scrolls the spreadsheet horizontally

Keyboard Shortcuts for Standard Buttons

The standard buttons are assigned default shortcuts for basic operations.

Action	Shortcut (Microsoft Windows)	Shortcut (Mac OS)	Relevant Page	Prerequisite
Create or new	CTRL + ENTER	CMD + ENTER	Advance search Object page	
Delete table entry	CTRL + D	CMD + D	Advance search Object page	The row is selected.
Edit page level	CTRL + E	CMD + E	Object page	The focus is set anywhere on the object page.
Save page level	CTRL + S	CMD + S	Object page	The page is in edit mode. The focus is set anywhere on the object page.
Cancel	CTRL + Q	CTRL + Q	Object page	The page is in edit mode. The focus is set anywhere on the object page.
Print	CTRL + P	CMD + P	Advance search Object page	
Go or search	ENTER or RETURN	ENTER or RETURN	Advance search Object page	The focus is in the search area.
Select row in tables	SHIFT + SPACEBAR	SHIFT + SPACEBAR	Advance search Object page	The focus is on a table cell.
Table settings	CTRL + ,	CTRL + ,	Advance search Object page	The focus is set on the table.
Export as	SHIFT + CTRL + E	SHIFT + CMD + E	Advance search Object page	The focus is set on the table.

The development of the application does not necessarily take advantage of these shortcuts. Non-standard buttons may be developed to look like the standard ones. These non-standard buttons are not automatically assigned the standard button shortcuts. The above standard buttons could also be customized to use different shortcuts.

You can confirm button shortcuts from the UI using the tooltip. The tooltip displays the shortcut if there is one associated with the button.

Editing

Use

You can create new business objects or edit, assign, or delete existing ones. You can also create new business objects quickly.

Regardless of the page you are currently in, you can create new tasks, business activities, appointments, and e-mails at any time. You do this using the **Quick Create** button in the header.

More Information

[Creating Data](#)

Creating Data

You can create new business objects and enter their details. The details you enter depend on the type of the respective business object. For example, you can create new business objects in the following pages:

- In the Work Center
- In the overview page for existing business objects
- In the result list in search pages

❖ Example

If you create a new corporate account, enter general data such as name and ID, address- and communication data, and the employee responsible. Then assign planned activities, contact persons, or opportunities to the corporate account.

Creating New Business Objects

In the Work Center under **Create**, if you click the business object that you want to create, you see the edit page in which you can enter the details of the business object.

❖ Example

In the **Accounts and Products** work center under **Create**, click **Corporate Account** to display the edit page for the account. Here you can enter the account details.

Creating New Business Objects Quickly

In the header, click the quick creates icon to display a list of business objects that can be created quickly. Click a business object to display the edit page in which you can enter business object details.

i Note

The business objects you can select using quick creates depend on your role. In the sales pro role, for example, you can create service orders, service quotations, or services requests.

Editing Data

You can edit business object data. The data you can edit depends on the type of the respective business object.

❖ Example

You want to edit the rating of a corporate account. The communication data for an assigned contact has changed. You also want to edit the list of assigned products.

Editing Business Object Data

In the overview page of a business object, if you click **Edit** in the details title bar, the overview page appears in edit mode (editable overview page). Here, you can edit the details of the selected business object. Click **Save** to save your changes. You return to the display mode of the business object's overview page.

❖ Example

In the overview page of the corporate account, click **Edit** in the details title bar. In edit mode of the overview page (editable overview page), change the rating of the account. Click **Save** to save the changed rating and return to the display mode of the account's overview page.

Editing Assigned Business Object Data

In the assignment block on the left next to an assigned business object, if you click **Edit**, the edit page with details of the selected business object appears. Edit your data and click **Back** to return to the overview page of the business object. Click **Save** to save your changes.

❖ Example

In the overview page of the corporate account, in the **Contacts** assignment block, click the **Edit** icon by the contact you want. In the edit page, change the contact's communication data. To return to the overview page of the corporate account, click **Back**. Click **Save** to save the changed communication data.

Editing Lists with Assigned Business Objects

In the assignment block, if you click **Edit List**, a list of all assigned business objects is displayed. In this list, you can add or remove business objects. Click **Save** to save your changes.

❖ Example

In the **Products** assignment block in the overview page of the corporate account, click **Edit List**. Click **Insert** and select a product in the product search. Select a product in the list and click **Delete** to remove the product from the list. Click **Save** to save your changes.

Editing Multiple Data Records Simultaneously

You can use the “Mass Change” function to make the changes made to one business object apply to multiple business objects simultaneously. Select the lines of the business objects you want to edit. Click [Mass Change](#). Edit your data and save your changes.

i Note

The mass change is only possible with business objects which have been configured by applications.

Adding Data

Procedure

You can add more business objects to a list of assigned business objects. To do this, you can either select business objects that are already available or create new ones.

❖ Example

In the [Contacts](#) assignment block in the overview page for the business account, click [New](#) to display the edit page for contacts. In the [Contact ID](#) field in the search help, you can select an existing contact.

Adding an Existing Business Object

1. Click the search help icon at the end of the corresponding field.

You see the search page.

2. Enter your search criteria in the search fields.
3. Click [Search](#).

The result list displays all the search results.

4. Select the value you want.

The value is copied to the corresponding field.

Adding a New Business Object

1. Click the search help icon at the end of the corresponding field.

You see the search page.

2. Click [New](#).
3. Enter the data you want in the fields.
4. Click [Back](#) to return to the edit page.

The value is copied to the corresponding field.

5. Click [Back](#) to return to the overview page.
6. Click [Save](#).

i Note

In both cases, the business object is added to the list of assigned business objects in the assignment block.

More Information

[Using Input- and Search Help](#)

[Performing a Search](#)

Deleting Data

Procedure

You can delete certain business objects. There are other business objects, such as accounts or employees, that you cannot delete. These business objects are part of the master data, that is, they remain unchanged for a long period and contain information that you require again and again.

i Note

The authorization concept for the deletion of business objects is dependent on the respective company. Authorization for deletion can therefore be adapted to the requirements of the respective company. The system administrator sets up and assigns the authorization.

Caution

When you delete a business object, you cannot undo the transaction. The business object itself is deleted, as well as assignments to other business objects.

Deleting Business Objects in the Result List

1. Search for the business object you want to delete.
2. In the result list, choose the required business object.

i Note

You can also select and delete multiple business objects in the result list simultaneously. However, this is not applicable to all business objects.

3. Click the icon **Delete** in the result list.

Deleting Business Objects on the Overview Page

1. Search for the business object you want to delete.
2. In the result list, choose the required business object.
3. Click the hyperlink in the result list.

You navigate to the business object's overview page.

4. Click the icon **Delete** on the overview page.

Deleting Assignments

1. In the assignment block of a business object, click the **Delete** icon.

i Note

The business object is removed from the assignment block. However, the business object itself is not deleted, just the assignment to another business object.

2. In the overview page, click one of the following icons:

- **Save**, if you want to delete the assignment
- **Cancel**, if you do not want to delete the assignment

More Information

[Performing a Search](#)

Exporting Data

Procedure

You can create export files, and reuse data that has already been entered. For example, you can create an export file from the result list of a search. This data is exported to a table processing program, for example Microsoft Excel.

1. Perform the search.

The search result is displayed in the result list.

2. Click the icon **Export to Spreadsheet** on the top right of the result list.

The export file with the search results is created in a table processing program, for example Microsoft Excel, and it then opens.

3. Use the standard processing of the table processing program to edit and save the export file.

More Information

[Performing a Search](#)

Printing Data

Procedure

You can print all business object data that you can access. You can choose between two options.

Printing the Browser View

You are in the overview page of a business object, and you click the **Print** icon. The overview page is launched in the browser view. You can use your browser's print function to print the overview page.

Displaying and Outputting PDF-Based Documents

You are in the overview page of a business object, such as a sales contract. Click **Trigger Output**. A message in the message bar informs you that the output has been triggered. Click **Preview Output** to trigger a PDF view of the overview page. You can print the PDF file by using the PDF print function.

More Information

[Performing a Search](#)

Adding Attachments

You can add attachments, such as documents or presentations in various formats, to most business objects. Most business objects have an **Attachments** assignment block in their overview pages, where you can upload and add attachments. You can add documents or Web addresses as attachments.

Adding Documents

Click **Attachment** to add a document to a business object as an attachment. You can either add a document from Content Management and attach it as a link, or upload a document from your local hard disk. Click **Browse** to search for a document on your hard disk. Enter the name and a description of the document and click **Attach** to add the document as an attachment.

Adding Web Addresses

Click **URL** to add a Web address to a business object as an attachment. Enter the Web address, a name and a description. Click **Attach** to add the URL as an attachment.

Adding Document Templates

Click **With Template** to add a document template to a business object as an attachment. Select the required document template. You receive notification that the selected document template was successfully saved. When you close the document template selection and return to the overview page of the business object, the selected document template appears in the list of attachments.

Editing Attachments

Click **Advanced** to edit attachments. You can choose between the following functions:

Function	Function	Description
Folder		Creates a new folder in the assignment block
Delete		Deletes an attachment from the list of attachments The assignment is deleted, not the attachment itself.
Copy		Copies an existing attachment The copied attachment can be saved under a different name.
Cut		Cuts the content of an existing attachment
Paste		Copies the content of an existing attachment into another one
More		

Function	Function	Description
	Edit	Edits an attachment
	Check Out	Downloads and saves an attachment to a local hard disk or to another server
	Check In	Uploads an attachment from a local hard disk
	Paste Link	Pastes a link to another attachment
	Remove Check Out Lock	Removes the lock for downloading an attachment
	Refresh	Updates the attachment data

Using Charts

Procedure

In result lists and assignment blocks, you can display the information from tables in the form of a chart. You see the chart directly under the result list or assignment block.

Display Chart

The display of tables as charts is identical in both result lists and assignment blocks. The following description applies to result lists, as well as to assignment blocks:

1. Click the icon **Open Chart** in the title bar of the table.

The area in which you see the chart opens, underneath the table.

i Note

If you do not see the chart, click **Play** in the context menu.

2. Choose the column whose data should be shown in the chart.
3. Choose between pie chart and bar chart.
4. Use the zoom function to zoom in or out on the chart.

Update Chart

In the table, if you click the title of the column that you selected, you can filter this column. The chart is updated, and the corresponding bar or segment is highlighted in the chart. If you click another column title in the table and filter it, the table and the chart are both updated again.

Select Bar or Segment

If you select a bar or a segment in the chart, it is highlighted, and marked with a frame in the legend. The table is automatically filtered according to the selected bar or segment.

More Information

Using Favorites

Use

You can create favorites for business objects. Favorites are displayed on the home page of your role.

Prerequisites

Before you can use favorites, you must activate them in the central personalization page, in the general settings.

Procedure

Creating Favorites

1. Select the required business object.
2. Click the icon **Add to Favorites** at the top of the business object's overview page.
3. Enter a name for the favorite, and choose the folder in which it should be created.

You can also create a new folder, which is then added to the existing folder structure.

i Note

If a favorite has already been created for this business object, the path is displayed. If you want to create a new favorite for the selected business object, delete the existing favorite.

4. Select the checkbox **Share**, if you want to share the favorite with all users who are signed in to the system with the same role as you.
5. Save your entries.

Maintaining Favorites

You can see your personal favorites within the folder structure **My Favorites**, in the information block **Favorites** on the home page. You can see the favorites that you share with other users, under **Community Favorites**. All favorites are displayed as hyperlinks. When you click a hyperlink, you navigate to the corresponding business object.

1. Choose **My Favorites** in the dropdown box.

The folder structure with your personal favorites is displayed.

2. Select folders or favorites, and move them one after the other to the required location in the folder structure, using drag and drop.

You can move favorites to other folders, or move whole folders with favorites within the folder structure.

Filtering Favorites

You can filter your personal favorites, as well as the community favorites, according to various criteria. If you choose **All items** in the filter dropdown box, all favorites are displayed. If you select a business object in the filter dropdown box, only favorites of this

type are displayed.

Personalizing Favorites

On the home page, you can click the icon **Personalize** at the top right of the information block **Favorites**, to personalize your favorites using drag and drop. You can create new folders in the folder structure, rename, or delete existing folders and favorites, and copy or insert folders and favorites. You can also move them up and down within the folder structure. When you are finished, save your changes.

More Information

[Personalizing the User Interface](#)

Using Tags

Use

You can add tags to business objects.

Prerequisites

Before you use tags, you must activate the function in the central personalization page, in the general settings.

Procedure

Creating Tags

1. Select the required business object.
2. Click the icon **Tag Current Item** at the top of the business object's overview page.
3. Enter a term for the tag.

In the search help, you can search for existing tags to add to this business object. You can also use tags that other users have assigned to this business object. The system suggests these tags as terms that you can use.

4. Click **Add**, to add the tag to the business object.

Newly added but unsaved tags are shown in italics. You can delete your tags for a business object at any time.

5. Confirm your entries by clicking **Save** or cancel all newly added tags by clicking **Cancel**.

Displaying Tags

You can see tags in tag clouds on the home page of your role, in the information block **Tags**. You can see your tags in your personal tag cloud. In the community cloud, you see the tags from all users who are signed in to the system with the same role as you.

All tags are displayed as hyperlinks. When you click a hyperlink, you navigate to the corresponding business object. If a tag is assigned to multiple business objects, you navigate to a page on which you see the assigned objects.

The bigger a tag is in the tag cloud, the more business objects are assigned to this tag.

Sorting Tags

You can sort tags in your tag cloud and in the community cloud alphabetically, and in ascending or descending order of occurrence.

Displaying Popular Tags

If you select **Popular tags** in the dropdown box, you see the community tags.

Recent Tags

Select **My recent tags** in the dropdown box. You see a maximum of five tags that you have used recently. If you select a tag, you navigate to the corresponding business object or to a page with multiple objects.

Searching for Tags

If you select **Search** in the dropdown box, you can search for objects that belong to your tags, and to the community tags. In the search help, select one of your available tags or a community tag. When you choose **Search**, you navigate to the corresponding business object or to a page with multiple objects.

Deleting Tags

Choose **Clear all my tags** in the dropdown box, to delete your tags.

Personalizing Tags

You can specify how many popular tags you want to display in the tag cloud. Click the icon **Personalize** at the top right of the information block, and choose the number.

More Information

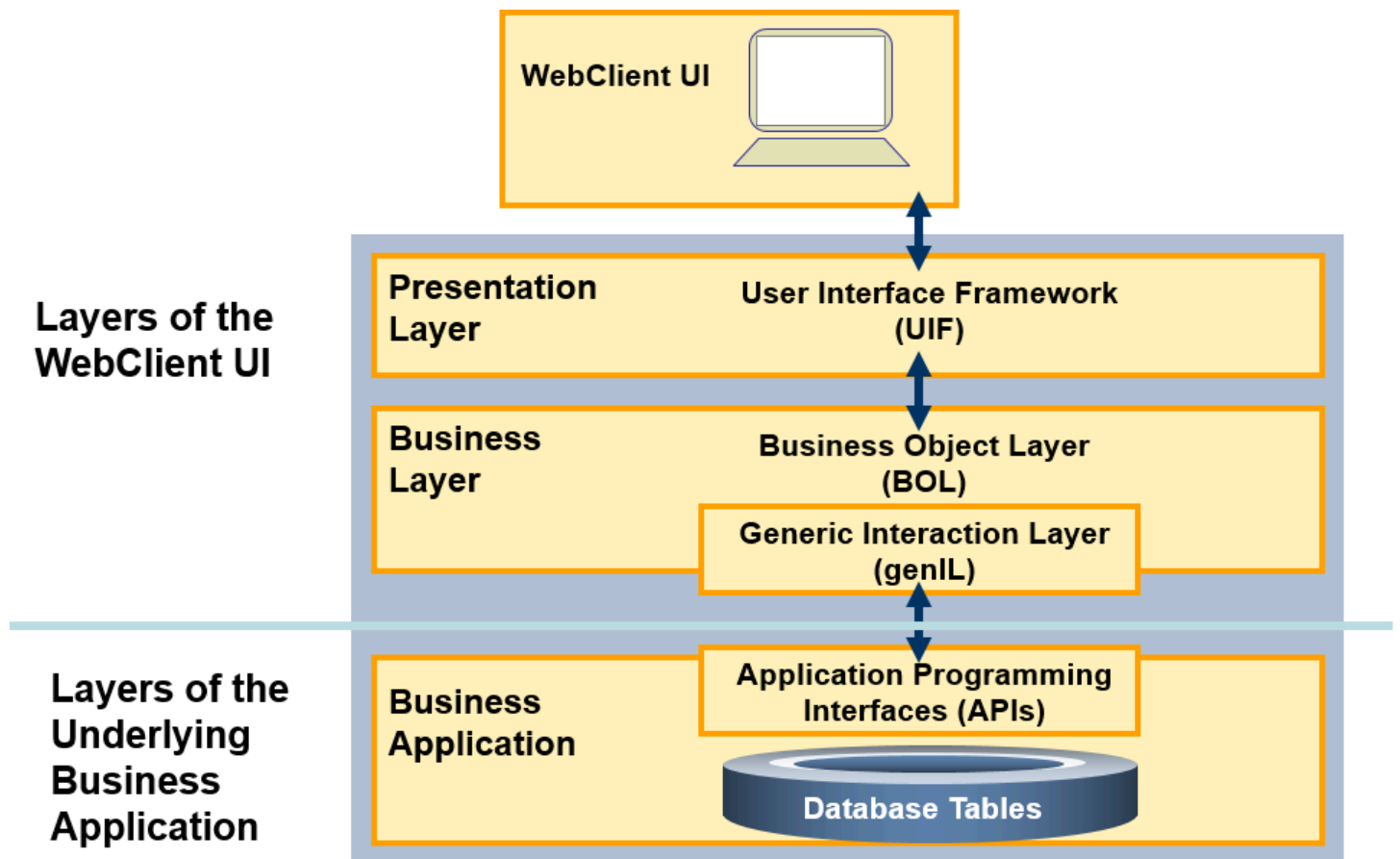
[Personalizing the User Interface](#)

WebClient UI Framework

The WebClient UI framework defines the software architecture of the WebClient UI. With the WebClient UI, SAP delivers a harmonized online user interface. The WebClient UI is designed for the business user and presents a role-based workspace that provides an easy-to-use navigation and user interface.

Framework Architecture

The framework architecture of the WebClient UI is based on different software layers. The figure below illustrates the framework architecture and its different software layers:



The different software layers are completely separated from each other. This separation allows the connection of any business application, via business object layer (BOL) and generic interaction layer (genIL) to the presentation layer.

Presentation Layer

The presentation layer of the WebClient UI is based on the User Interface Framework (UIF), which is the basis for the HTML pages running in the Web browser.

Business Layer

The business layer of the WebClient UI consists of the following software layers:

- Business object layer (BOL)

The business object layer saves the business object data, for example of sales orders, at runtime of the session. This layer guarantees the separation of the WebClient UI and the underlying business logic.

- Generic interaction layer (genIL)

The generic interaction layer handles the data transfer from the business object layer to the application programming interfaces (APIs) of the underlying business engine. It is the connection between the business object layer and the underlying business engine.

Business Application

The business application contains the business logic and the database tables.

Framework Tools

Use

To adapt the WebClient UI to your business needs and to your corporate identity, a bundle of framework tools is available.

The table below shows which task you can perform with which tool:

Task	Tool
Configure views	UI Configuration Tool
Use parameters to save your configuration	UI Configuration Tool
Use customer-specific fields in view configuration	UI Configuration Tool
Implement the same configuration content for several views	Design Layer Customizing
Define input help for customer-specific fields	BSP Component Workbench Design Layer Customizing
Create new UI components and views	BSP Component Workbench
Add customer-specific fields to existing business objects	Application Enhancement Tool

More Information

[UI Configuration Tool](#)

[Application Enhancement Tool](#)

[Design Layer](#)

UI Component Workbench

Use

The UI component workbench is the central framework modeling tool, and provides access to a variety of functions. It offers the following main access points:

- Component structure browser

The component structure browser provides access to the objects of a UI component.

- BOL model browser

The BOL model browser shows the business object layer (BOL) model of a UI component. This BOL model is assigned to the UI component via the runtime repository. The BOL model browser is available only if this assignment exists.

- Runtime repository editor

The runtime repository editor shows the runtime repository of a UI component. In the runtime repository, you can see and edit the view set composition, that is, which views are assigned to a view area and which view areas belong to a view set. In this runtime repository, you can also see and edit the navigational links between the views of a UI component. You can see the component interface description and the component usages.

To start the UI component workbench, you can use transaction BSP_WD_CMPWB.

UI Configuration Tool

You can use the UI configuration tool to adapt the WebClient UI Framework to your company's specific requirements. You can access the UI configuration tool in the UI component workbench under **Component Structure Browser**. You select a view under **Views** and select the **Configuration** tab. This tab is activated only if a view can be configured.

Component Enhancements

You can use the framework enhancement concept to enhance the function of standard components, standard views, and standard controllers that are delivered by SAP. You can perform the enhancement in the UI component workbench by clicking **Enhance Component**.

More Information

[UI Configuration Tool](#)

[Framework Enhancement](#)

Debug Tools

Debugging tools such as **Field Information**, **Conditional Breakpoints**, **Performance Tracker**, **System Information**, and **Hierarchy Information** are available from the Access Debug Tool dropdown list or using keyboard shortcuts.

Use

There are various debugging tools that are available to you. These tools can be accessed using keyboard shortcuts or through a dropdown list. Since the keyboard shortcuts may perform other functions if you are using a remote connection, you may choose to activate their availability in the **Access Debug Tools** dropdown list. To activate this dropdown list in the work area header, add the user parameter BSP_DLC_DEBUG_PARAM with the value WCF_HEADER_T00L to your user profile with debugging rights.

Debug Tool	Keyboard Shortcut	Description	Access Debug Tool Dropdown List Option	Access Debug Tool Function
Field Info	F2	Select an input field and choose F2 to see the Technical Data dialog and to find detailed technical information.	Field Information	Once you select Field Information , the mouse cursor changes into a crosshair cursor. Select the input field where you would like to see the detailed technical information.
Conditional Breakpoints	Alt + F2	Shows the Technical Data (Conditional Breakpoints) dialog where breakpoints can be activated or modified.	Conditional Breakpoints	Same behavior as the keyboard shortcut.
Performance Tracker	Ctrl + Shift + F7	Shows the frontend and backend time and memory performance.	Performance Tracker	Same behavior as the keyboard shortcut.
System Info	Ctrl + Alt + F2 or Ctrl + Shift + F2	Shows the system information.	System Information	Same behavior as the keyboard shortcut.

Debug Tool	Keyboard Shortcut	Description	Access Debug Tool Dropdown List Option	Access Debug Tool Function
View Hierarchy Info	Shift + F2	Select an input field and choose Shift + F2 to see the View Hierarchy dialog and to find detailed information on the backend hierarchy.	Hierarchy Information	Once you select Hierarchy Information , the mouse cursor changes into a crosshair cursor. Select the input field where you would like to view the backend hierarchy information.

More Information

For more information about Field Information, see [Displaying Technical Information](#).

Displaying Technical Information

Use

You are located in an application and want to find out the UI component name and view name. Select an input field, and press F2 to see the [Technical Data](#) dialog. In this dialog, you find the following detailed technical information:

- Selected UI component
- Enhanced view

This information is displayed only if an enhancement set is active.

- Selected field
- Configuration at runtime
- Design layer

If you click [Configure](#) at the bottom of the dialog, you see the view configuration of the selected UI component and view. This button is available only if you have activated the configuration mode on the central personalization page of WebClient UI.

If you click [Show View Location](#), you see the technical structure in which the selected view is embedded. This button is available only if you have debugging rights.

❖ Example

You are in the Account search page and want to configure the search criteria. To find out the application component name and view name, you click the empty entry field in the first row of the search criteria. This is the third field in the row in which you search for [Name 1/Last Name](#). You press F2 and navigate to the [Technical Data](#) dialog, which says that the application component is BP_HEAD_SEARCH and the view is [MainSearch](#).

More Information

[Enhancement Set](#)

User Interface Personalization

You can use a bundle of tools and options to set up and personalize the WebClient UI to your individual preferences.

Home Page Links

Use

You can add the following links to the home page of a business role:

- Application links, which are URLs that are used to navigate inside SAP systems; they are opened in the same window in which the SAP system is running
- Web links, which are simple URLs that are opened in a separate window
- Widgets, which are graphical objects that show dedicated information, typically from an external source

Prerequisites

You can define Web links and widgets that are to be displayed on the home page of a business role, in Customizing for **UI Framework** under **UI Framework Definition > Define Web Links**. These Web links and widgets are assigned to Web link groups and widget groups in the same Customizing activity.

You need to assign Web link groups and widget groups to the function profile of a business role in Customizing for **UI Framework** under **Business Roles > Define Business Role**. The function profiles to be used for Web links and widgets are, respectively, **EXT_WEB_LINKS** and **WIDGET_LINKS**.

The user can personalize Web links, application links, and widgets, in the **Favorites** settings on the central personalization page of the WebClient UI. However, these links are only available if the necessary Customizing settings at business role level have been performed beforehand.

Central Personalization

The following functions for personalizing the WebClient UI are conveniently located on the central personalization page. You can start the central personalization page by clicking **Personalize** in the user menu. The user menu is available using the user icon in the header.

My Data

In this block, you can enter your personal data and your communication data. You can see organizational information, such as the business role with which you are logged on to the WebClient UI. Additionally, you can change your current password.

Settings

In this block, you can

- Make general settings
- Enable tab order definition
- Configure dropdown lists

- Enable UI configuration
- Enable tag clouds and favorites
- Enable message preview
- Enable table personalization
- Enable session timeout parameters

General Settings

You can enter general settings such as time zone, decimal notation, date format, time format and language.

Tab Order Definition Mode

If you want to define your own tab order, you can enable the tab order definition.

Dropdown Lists

You can define the sorting order for items in dropdown lists and you can enable keys to be displayed in dropdown lists as technical information.

Configuration Mode

You can enable the UI configuration, which means that you can start the view configuration and the fact sheet configuration directly from the WebClient UI. For more information, see [UI Configuration Tool](#).

Tag Clouds and Favorites

You can enable tag clouds and favorites that are displayed on the home page of a business role in the WebClient UI. For more information, see [Tag Clouds](#) and [Favorites](#).

Messages

The message icon is located at the left hand side of the footer. A message bar header shows the icon of the corresponding message type and the number of messages that are related to this message type. You can see the short text of new messages in the message popover. If there is a long text, it will be available using a chevron next to the message short text. You can personalize the preview of messages on the central personalization page. For more information, see [Message Area](#)

Table Personalization

You can enable automatic saving of the personalization changes you do directly on any table. If you enable filter saving, the filters that you apply to that table is automatically saved to your personalization. If you enable sort saving, the column sorting that you apply to that table is automatically saved to your personalization.

Session Timeout

You can disable the session timeout clock from being displayed in the header.

Groupware Integration

Under this heading, you can define the settings for the groupware integration in the WebClient UI.

Layout Settings

Personalize Layout**Skin**

You can choose between different skins, and the text size on the user interface. For more information, see [Skins](#).

Performance Settings

- Enable Fast Performance Mode with Fewer UI Features

You can waive some functions and visual effects in favor of better performance. Here you can enable fast performance with fewer UI features. If you select this checkbox, the following functions are not available:

- Highlighting of UI elements

If you place the mouse pointer over an element, it is not highlighted; for example, an entry in the navigation bar, a colored entry in a table, or an underscore under a hyperlink.

- Drag and drop in tables, on home pages, work center pages, and overview pages.

You cannot use drag and drop to change the column width of tables, or the sequence of columns.

- Fixed header row in tables

You cannot fix the column header at the top of the table when you use the vertical scrollbar in tables.

- Exact sizing of tables with vertical scrollbar

The table size cannot be exactly determined, which means that table rows can be cut off.

- Disable the Suggestion of Last Typed Entries for Input Helps

You can disable the suggestion of entries in input helps for performance reasons. If you disable this function, all entries saved for the current user are deleted. To make this checkbox available in the performance settings, you must have performed the necessary settings in Customizing for **UI Framework**. For more information, see [Smart Input Help](#).

- Disable the Suggestion of Background Searched Entries for Input Helps

You can disable the suggestion of entries in input helps for performance reasons. If you disable this function, suggested entries from the backend do not populate in the input field. To make this checkbox available in the performance settings, you must have performed the necessary settings in Customizing for **UI Framework**. For more information, see [Smart Input Help](#)

Skin Color

You can choose the skin color by clicking first the color bar on the left and then the color field on the right. This option is available for SAP standard skins, not for custom or copied skins

Personalize Navigation Links**Navigation Links**

You can choose which logical links are contained in which direct link groups. The quick creates icon is found on the right side of the header.

Which direct link groups and logical links are available in your business role depends on Customizing. You define direct link groups and the corresponding logical links in Customizing for **UI Framework** under **► Technical Role Definition ► Define Navigation Bar Profile ►**. For more information, see [Business Roles](#).

Defining Recent Items

You can use the recent items to navigate to the items that you have visited most recently in your business role.

To display the **Recent Items** block in the navigation bar of your business role, and to define the number of links to be displayed in the personalization, you must have performed the necessary Customizing settings at business role level beforehand. For more information, see [Recent Items](#).

Defining the Location of the Main Menu

You can choose to display the main menu in the header area or the side panel.

In the navigation links, if you select **Apps** for the **Link Type**, you can choose to display the main menu in the side panel when the **Display Apps in Side Panel** is selected otherwise, it will be in the header.

Shortcuts

In this block, you can define your personal keyboard shortcuts. You can define keyboard shortcuts for focus setting and for navigation.

Preferences

Personalize Application Links, Web Links, and Widgets

You can add application links, web links, and widgets to the home page of your business role. To make these links available on the central personalization page of the WebClient UI, you must have performed the necessary Customizing settings at business role level beforehand. You can also personalize these links directly on the home page. For more information, see [Content Block Personalization](#). For more information about home page links, see [Home Page Links](#).

Personalize Categories for Saved Searches

You can create categories for saved searches. The categories are sorted in the saved search dropdown according to the order defined here.

Feed

In this block, you can copy the feed URL and subscribe to this feed in your feed reader. To make the feed settings available on the central personalization page of the WebClient UI, you have performed the necessary Customizing at business role level beforehand. For more information, see [Feed Subscription](#).

Advanced Search Personalization

Use

You can define default searches for search pages in the personalization dialog.

The personalization icon is available in the view of the search page next to the **Hide Search Fields** link. In the personalization dialog, you can select a saved search or the current search criteria as a default. You can also specify whether the default search is executed automatically when you access the search page. This allows you to start working directly with the result list.

Prerequisites

To enable the personalization for advanced search pages, you have to make the following code change and Customizing settings:

i Note

For this function, you must carry out the following additional development activities on a project basis.

If your search page was created using the Search Page Wizard, no effort is required. If you have created a search page manually, you need to enhance an inbound plug and a view set layout. Proceed as follows:

1. Enhance the inbound plug for handling the **Advanced** pushbutton on the header of the navigation frame of the WebClient UI. The inbound plug is typically called `IP_SAVEDSEARCHEDIT` and resides in the view controller of the search criteria.

You can see the full code in your system in method `IP_SAVEDSEARCHEDIT` of UI component `UIF_FLIGHT_S` in view `AdvSearchFlight`.

- a. Add the public method `INVOKE_DEFAULT_OR_AUTO_SEARCH` in class `CL_WCF_BSP_DSP_F4` to the method `IP_SAVEDSEARCHEDIT`.

The public method `INVOKE_DEFAULT_OR_AUTO_SEARCH` uses cross-component navigation. The method can only be called if no search criteria are passed through the collection.

- b. Make sure that event `QUERY_LOADED` is raised afterwards.

2. Change the BSP page of the view set.

You can see the full code in your system in file `AdvSearchFlightVS.htm` of UI component `UIF_FLIGHT_S` in view set `AdvSearchFlightVS`.

- a. Add public method `SEARCH_CRIT_PERS` in class `CL_WCF_BSP_DSP_F4`. The method determines whether the personalization icon is to be shown by checking the Customizing settings.
- b. Pass the result to parameter `personalizable` of tag `searchFrame`.

Customizing

You enable default searches in Customizing for **UI Framework** under **Technical Role Definition** > **Define Parameters**.

To enable default searches for your business role, you need to assign the parameter `DEFAULT_SEARCH` with profile value `TRUE` via function profile `PARAMETERS` to your business role. You do this in Customizing for **UI Framework** under **Business Roles** > **Define Business Role**.

Depending on these Customizing settings, the personalization icon is shown or hidden for each business role.

Table Personalization

Use

You personalize tables in the **Personalization** dialog, which you open by using the **Personalize** icon, for example, on the right in the title bar of the result list or in the toolbar of the overview page.

Features

Selecting and Ordering Columns

You can select which columns you would like displayed from the list of available columns using the checkboxes. You can use the [Up](#), [Down](#), [Top](#) and [Bottom](#) buttons or drag and drop to order the columns.

i Note

You can also position the columns quickly and easily by using drag and drop. This function is not active if you have specified on the central personalization page that you want to waive some user interface functions, as well as some visual effects, in favor of better performance.

Specifying the Maximum Number of Rows per Page

You can define the maximum number of rows that should be displayed on each page of a table.

Defining the Column Width

You can define the column width of the individual columns, in relation to the total width. You can specify the absolute width in pixels, and the relative width in percent. If the entry in pixels exceeds the width of the window, a horizontal scroll bar is automatically displayed.

If you enter a number without a unit of measurement, the system automatically calculates the column width in pixels. If you want relative column widths, enter the unit of measurement %.

i Note

You can also define the column width in tables, for example in the result list, quickly and easily by using drag and drop. This function is not active if you have specified on the central personalization page that you want to waive some user interface functions, as well as some visual effects, in favor of better performance.

Fixing Columns

You can fix columns on the left and/or right of the table. The column width must be defined in pixels (px) for columns that you want to fix. Using the [Fix Left](#) and [Fix Right](#) buttons, you can fix columns (turn the icon to lock). You can [Remove Fix](#) columns one at a time (turn the icon to open lock) or you can remove the fix from all or the columns using the [Remove Fixed Columns](#) button at the top of the [Displayed Columns](#) section.

Filtering Columns

You can filter columns in the table. To save these filters, ensure that you have enabled the [Enable filter saving](#) option in the [Table Personalization](#) section of [Personalize Settings](#). Then, when you open the [Personalization](#) dialog, you can hover over the [Remove Filter](#) button to see what filter is set and remove the filter if desired. To remove all filters, use the [Remove Filters](#) button at the top of the [Displayed Columns](#) section.

Saving Table Views

You can save several personalized views for tables. You can then select the table view using a variant menu in the table toolbar. If you move columns using drag and drop or if you resize columns, your changes are automatically saved to the current view.

Defining Scrolling and Paging

You can define whether you want to use scrolling or paging in tables, or both.

More Information

[Central Personalization](#)

Personalization of Overview Pages with Tiles

Use

You can personalize overview pages with tile layout in the WebClient UI.

Features

You can personalize overview pages with tile layout directly in the WebClient UI, or in the personalization dialog, which is started by clicking the personalization icon in the toolbar of an overview page. You can use drag and drop for the personalization in the WebClient UI, and also in the personalization dialog.

You can change the order of assignment blocks that are represented by tab links on each tile. You can move tab links from one tile to another. The first assignment block on a tile is visible by default when you access the tile layout.

In the personalization dialog, you can add assignment blocks to a tile and choose which assignment blocks are to be displayed on each tile. You can also change the order of the assignment blocks on a tile or move assignment blocks between tiles. You can remove assignment blocks from the list of displayed assignment blocks to the list of available assignment blocks (assignment block repository).

Content Block Personalization

Use

You can personalize the content of content blocks on home pages, work center pages, and report pages.

Features

In content blocks where personalization is available in the content block header, for example, [Web Links](#), [Navigation Links](#), or [Widget Links](#) content blocks on the home page, you can start the personalization dialog by clicking the personalization icon. You can personalize available links and decide whether they are displayed or not, and in which order. In some content blocks, you can additionally create your own links and delete existing links. You can modify the description and the URL of existing links by using the edit icon.

Logo Personalization

Use

You can replace standard the SAP logo with your own logo.

Prerequisites

To change the logo in the WebClient UI, you need to make the necessary settings in Customizing for [UI Framework](#) under [UI Framework Definition](#) > [Skins and Icons](#) > [Define Icons](#) >

Features

You can change the SAP logo, by using your own logo or overwriting the SAP logo. The SAP logo is replaced in the entire WebClient UI, including the header area and the screen in which you select a business role for logon. Since logos are skin-dependent, you also need to ensure that the logo is available for all skins.

More Information

[Skins](#)

Tab Order Definition

Use

You can define your own tab order in the WebClient UI.

Prerequisites

Before users can define their own tab order, they need to enable the tab order definition mode in the [Settings](#) block, on the central personalization page of the WebClient UI.

Features

You can use this function to define the tab order for particular fields or all fields in a flexible way, without having to tab through a default or predefined tab order.

The tab order definition is available in configurable form views; it supports input fields, dropdown list boxes, checkboxes, radio buttons, and buttons.

To define the tab order, you need to select edit mode and click the tab order icon in the toolbar of the form view. The form view is now surrounded by a blue frame. When you choose a field or press **ENTER**, this field is activated in the selected order, and the order number is displayed in this field. Clicking a field twice means that this field is removed from the tab order definition. When you save your tab order definition, you exit the tab order definition mode and return to the form view in edit mode.

Example

You select an account on the account search page. In the [Account Details](#), you click [Edit](#) and then the tab order icon in the toolbar. The account details are surrounded by a blue frame in which you can now define the tab order.

CSS Files and JS Files

Use

You can add your own Cascading Style Sheet (CSS) files and JavaScript (JS) files on top of the CSS and JS files that are delivered by SAP.

Prerequisites

To add a CSS file, you make the necessary settings in Customizing for **UI Framework** under ► **UI Framework Definition** ► **Skins and Icons** ► **Define Path for CSS Files** ►.

To add a JS file, you make the necessary settings in Customizing for **UI Framework** under ► **UI Framework Definition** ► **Define Path for JS Files** ►.

Features

Instead of copying and pasting an existing skin and then adapting the copied skin to your requirements, you can simply create a CSS file. You can change, for example, the styles that are used to change the background color of an input field. You can also change the image at the top of the navigation bar, or the highlighting of navigation bar links.

CSS files are skin-dependent, which means that you can add multiple CSS files, depending on the skin. Since they are also browser-dependent, you can use multiple CSS files for Microsoft Internet Explorer and Mozilla Firefox browsers, depending on the browser version.

Instead of changing backend files, you can simply create a JS file and, for example, change quick info texts or create your own mini-calendar, which is started when you click the date selection.

You can add several JS files and define the priority that determines the ascending order in which the files are loaded.

More Information

[Skins](#)

Generic User Interface Functions

You can use a variety of generic functions on the WebClient UI.

Central Sharing Tool

Use

You can use the Central Sharing Tool to share items with other users, organizational units, positions or business roles. You can share the following items:

- Tags

For information, see [Tag Clouds](#).

- Favorites

For information, see [Favorites](#).

- Saved searches

- Favorite reports

When sharing an item, you can enter a message for the recipient. The recipient can view the shared items in their **Inbox**.

For the description of the saved searches, translation is not available.

Integration

You can access the Central Sharing Tool using a new system link ([Share](#)) that is available in the header area. You can also make the function available in the user menu by selecting the user icon in the header. On the home page of your business role, you can display the [My Sharebox](#) content block, which displays the folders that contain the shared items: [Inbox](#) and [Sent](#).

You can make the Central Sharing Tool available on your home page as an application link using the central personalization.

Prerequisites

Customizing

You have enabled the Central Sharing Tool for business roles in Customizing for [UI Framework](#) under [►Technical Role Definition ► Define Function Profile ►](#) and you have defined the level of authorization for the business role:

- Standard sharer
 - Can share items. The authorization to receive shared items is included in a sharer's authorizations.
- Supersharer
 - Can copy shared saved searches to the recipient's saved searches.
- Recipient with limited authorizations
 - Can receive shared items, but cannot delete shared items from the [Inbox](#).
- Recipient
 - Can receive shared items.

To enable the usage of the Central Sharing Tool in a business role, you have assigned the function profile CENTRAL_SHARING_TOOL to your business role in Customizing for [UI Framework](#) under [►Business Roles ► Define Business Role ►](#).

UI Component Workbench

If the [My Sharebox](#) content block is not available in the home page personalization of your business role, you can make it available as follows:

1. Open the UI Component Workbench by using the transaction code BSP_WD_CMPWB.
2. Open the UI component where your home page resides, and open the [Runtime Repository Editor](#).
3. Add UI component WCF_UST_SH_BOX in your component usages with outbound plug UST.
4. Delegate the outbound plug to your window. Then, in the mapped outbound plug, add the following code:

```
me->fire_outbound_plug( iv_outbound_plug = 'UST'
iv_data_collection = iv_data_collection ).
```

5. Add <usageName>.WCF_UST_SH_BOX/MainWindow as a view in your home page view set.

To display the [My Sharebox](#) content block on the home page, you have to add the content block to the displayed content blocks in the page configuration.

Activities

Sharing and Receiving Items

You can:

- Share items with users, organizational units, positions or business roles
- View the messages that the sharer has attached to the item
- Search shared items and filter the search results, and you can also forward the search results
- Forward items you sent or received to other users by choosing the relevant icon left of the shared item
- Make received saved searches available in the central search by choosing the relevant icon left of the shared item
- Make received saved searches available in the **My Saved Searches** tray by clicking on **Add to My Saved Searches** option in the menu icon on left of the shared item

A saved search that was received as a shared item keeps a dependency with its original saved search. That is, if the sharer changes the original saved search, the recipient's saved search is also updated. For saved searches, you share only the search criteria, but not the search result.

- When sharing a saved search, a supersharer can make this saved search available in the central search of the recipient. The users who receive the shared items can remove the received saved search from their central search.

Personalizing the My Sharebox Content Block

In the personalization of the **My Sharebox** content block, you can:

- Delete shared items. The item is also deleted in the recipient's **Inbox**
- Create new folders below the **Inbox** and **Sent** root folders
- Rename and delete folders that you have created
- Move items up and down within a root folder
- Select and forward multiple shared items
- Cut and paste items and folders that you have created

Smart Input Help

Use

If you double-click fields with an input help, or use the backspace key in an empty field with input help, you can see your last five entries in this field. Your most recent entry appears at the top of the list that is displayed below the input field.

When you make a new entry in this field, and when you then take further action, such as navigating to another page, the new entry becomes the topmost entry in the dropdown list for this input field.

Prerequisites

You can customize the smart input help in Customizing for **UI Framework** under **Technical Role Definition** **Define Parameters**.

To disable the smart input help on the central personalization page, the smart input help must be enabled in Customizing, otherwise the indicator is not available in the layout settings.

Features

In addition to the five entries in fields with a simple input help, a background search is performed; it is a simple input help, where the available entries are immediately displayed if the input help was clicked by the user. The background search starts as soon as you enter the first character in the field. The data records found are displayed in alphabetical order below the five entries, from which they are separated by a dashed line. If more than 50 records are found in the background search, [\(Too Many Records ...\)](#) appears at the end of the list with the data records found.

You can use report `SVH_DELETE_LAST_ENTRIES` to delete the entries of any user in the smart input help.

You can also use smart input help and value help together or separately. You can use the type-ahead features of the smart input help with a value help option on an input field. The value help can be set up to use an F4 description or a navigation descriptor. This navigation descriptor could be for advanced search.

Activities

Users can disable the smart input help for performance reasons, for example, or they can delete all entries that have been saved for the user. They can do so on the central personalization page under the layout settings.

More Information

For more information on disabling smart input help or deleting saved entries, see [Central Personalization](#).

Input Field With CDS View Based Value Help

Use

Input fields with CDS views based value help can be integrated within WebClient UI applications. This allows you to leverage predefined CDS view metadata, enabling you to make selections as you type text and receive real-time suggestions. When you focus on the input field, the dialog displays recently used items for quick access. Additionally, the F4 value help dialog is available for extended search and selection.

This feature can only be used in object pages.

Prerequisites

The CDS view or CDS view entity must be exposed by the OData service.

i Note

Only OData version 2.0 is supported.

Features

Input fields with CDS view based value help have the following features:

- When you focus on the input field, you will see your last entries for this field in a dialog, with the most recent entry at the top of the list.
- When you make a new valid entry in this field and perform another action, like navigating to another page, the new entry becomes the topmost entry in the recently used list.

- In addition to the most recent entries, a background search is performed on the CDS view, starting as soon as you enter a character in the field. The search finds matches to your input in the different searchable columns, highlighting the matches in the results list. If there are more than 10 results, you will get a link to [Show all items](#). If you use this link, you will see the F4 help dialog with options to filter for your desired entry.
- You can use the F4 value help dialog to search for and select your entry.

Procedure

1. Indicate the OData Source

To enable the value help based on CDS entities, the URL of the OData service must be specified in the [DataSources](#) section of the runtime repository of the UI component.

i Note

The names of the nodes and attributes are case sensitive.

The `<DataSources>` node is opened right after the closure of the `<Models>` node. Inside the `<DataSources>` node, add the `<DataSource>` with the following attributes:

Attribute	Mandatory	Description
Id	No	Data source identifier. i Note Currently, not used but may be required in future.
Uri	Yes	OData relative path.

Example

Sample code for DataSource in RT repository XML:

Sample Code

```
<Models>
  <!-- Reference to BOL Component Set -->
  <Model id="ONEORDER"/>
</Models>
<DataSources>
  <DataSource Id="mainService" Uri="/sap/opu/odata/sap/C_BUSSOLUTIONORDER/" />
</DataSources>
```

`<DataSource>` can be added locally within the same component in which the field is being used (for example: BT112H_SC) or can be added to the parent component in the view hierarchy (BT112M_SC). In the case of the `<DataSource>` being defined in multiple places in the hierarchy, the closest component will take precedence.

2. Add Vocabulary Annotations (optional)

You can also add vocabulary annotations that help with how data should be interpreted, displayed, or processed by the view. To enable this, you must open the [DataSource](#) node, append the child node [Annotations](#) where you create another

child node **Annotation** with the following attributes:

Attribute	Mandatory	Description
Uri	Yes	Vocabulary annotation relative path

❖ Example

Sample code for vocabulary annotation:

☰ Sample Code

```
<DataSources>
  <DataSource Id="mainService" Uri="/sap/opu/odata/UA_S4CRM_REUSE_VH/">
    <Annotations>
      <Annotation Uri="/sap/opu/odata/IWFND/CATALOGSERVICE;v=2/Annotations(TechnicalName='l
    </Annotation>
  </DataSource>
</DataSources>
```

3. Redefine the V-Getter

In the v-getter method, return the value help descriptor object of class `cl_bsp_wd_vh_cds_entity_descr`.

In the constructor, use the following parameters:

Parameter	Mandatory	Value	Default
IV_HELP_ID	Yes	<String value>	
IV_HELP_ID_KIND	Yes	IF_BSP_WD_VALUEHELP_F4DESCR=>HELP_ID_KIND_CDS_ENT	
IV_INPUT_MAPPING	Yes	<Internal table>	
IV_OUTPUT_MAPPING	Yes	<Internal table>	
IV_OBJECT_REF	No	<Object>	<Initia

Parameter	Mandatory	Value	Default
IV_TRIGGER_SUBMIT	No	<ABAP boolean value>	<ABAP False>
IV_F4TITLE	No	<String value>	<Empty>
IV_OUTPUT_MAPPING_FORMAT	No	<Internal table>	<Initia

Additional optional properties can be set via the method `set_additional_property` in the v-getter method. The following attributes are supported:

Property	Value	Description
<code>cl_bsp_wd_vh_cds_entity_descr=>co_p_columns_to_display</code>	String table	Define the columns of the table in the F4 popup and the selection box. If not defined, the displayed columns will be as default in the value help CDS entity metadata.

❖ Example

Sample code for redefine v-getter:

☰ Sample Code

```
METHOD get_v_soldto_name.
```

```
DATA lr_cds_vh_descr TYPE REF TO if_bsp_wd_valuehelp_f4descr.
DATA lv_in_mapping_tab TYPE if_bsp_wd_valuehelp_f4descr=>gtype_param_mapping_tab.
DATA lv_in_mapping TYPE if_bsp_wd_valuehelp_f4descr=>gtype_param_mapping.
DATA lv_out_mapping_tab TYPE if_bsp_wd_valuehelp_f4descr=>gtype_param_mapping_tab.
```

```

DATA lv_out_mapping TYPE if_bsp_wd_valuehelp_f4descr=>gtype_param_mapping.

lv_in_mapping-context_attr = 'SOLDTO_NAME'.
lv_in_mapping-f4_attr = 'BusinessPartner'.
APPEND lv_in_mapping TO lv_in_mapping_tab.

lv_out_mapping-context_attr = 'SOLDTO_NAME'.
lv_out_mapping-f4_attr = 'BusinessPartner'.
APPEND lv_out_mapping TO lv_out_mapping_tab.

CREATE OBJECT lr_cds_vh_descr TYPE cl_bsp_wd_vh_cds_entity_descr
EXPORTING
  iv_help_id          = 'C_SrvcOrderSoldToPartyVH'
  iv_help_id_kind     = if_bsp_wd_valuehelp_f4descr=>help_id_kind_cds_ent
  iv_input_mapping    = lv_in_mapping_tab
  iv_output_mapping   = lv_out_mapping_tab.

DATA lt_cols TYPE string_table.
lt_cols = VALUE #( ( `BusinessPartnerName` ) ( `PostalCode` ) ( `CityName` ) ( `Country` ) ).

lr_cds_vh_descr->set_additional_property(
  iv_attr_name = cl_bsp_wd_vh_cds_entity_descr=>co_p_columns_to_display
  iv_value     = lt_cols
).
rv_valuehelp_descriptor ?= lr_cds_vh_descr.

ENDMETHOD.

```

4. Additional Field Properties in P-Getter

In the p-getter method, some additional properties can be set as follows:

Property	Value	De
FP_TEXT	<String>	
FP_TEXT_IN_EDIT_MODE_SOURCE	if_bsp_wd_model_setter_getter=>FC_TXT_EDIT_SRC_VALUELIST if_bsp_wd_model_setter_getter=>FC_TXT_EDIT_SRC_NONE if_bsp_wd_model_setter_getter=>FC_TXT_EDIT_SRC_VALUELIST_NVAL if_bsp_wd_model_setter_getter=>FC_TXT_EDIT_SRC_VALUELIST_WARN	if
FP_DISPLAY_BEHAVIOUR	if_bsp_wd_model_setter_getter=>FC_DISP_BEHAVIOR_DESC_AND_ID if_bsp_wd_model_setter_getter=>FC_DISP_BEHAVIOR_DESC_ONLY if_bsp_wd_model_setter_getter=>FC_DISP_BEHAVIOR_ID_AND_DESC if_bsp_wd_model_setter_getter=>FC_DISP_BEHAVIOR_ID_ONLY	if

Property	Value	De
	if_bsp_wd_model_setter_getter=>FC_DISP_BEHAVIOR_AUTO	

❖ Example

Sample code for p-getter:

⌵ Sample Code

```
METHOD get_p_soldto_name.  
  
CASE iv_property.  
  WHEN if_bsp_wd_model_setter_getter=>fp_text.           "#EC NOTEXT  
    rv_value = 'SOLDTO_NAME'.  
  
  WHEN if_bsp_wd_model_setter_getter=>fp_text_in_edit_mode_source.  
    rv_value      = if_bsp_wd_model_setter_getter=>fc_txt_edit_src_valuelist_nval.  
  
  WHEN if_bsp_wd_model_setter_getter=>fp_display_behaviour.  
    rv_value      = if_bsp_wd_model_setter_getter=>fc_disp_behavior_id_only.  
ENDCASE.  
ENDMETHOD.
```

Recent Items

Use

You can navigate to the items that you have visited most recently in your business role, by using the recent items.

Prerequisites

To display the **Recent Items** block in the navigation bar, you need to define a direct link group with type **Recent Items** in Customizing for **UI Framework** under **Technical Role Definition > Define Navigation Bar Profile**.

This direct link group needs to be enabled in your business role, in Customizing for **UI Framework** under **Business Roles > Define Business Role**.

Features

The **Recent Items** block is located in the side panel. It contains links with the item titles, in the order you have visited them. The most recently visited item title appears at the top. If you click a link in the **Recent Items** block, the item title is emphasized until you navigate to another item. The item title is also emphasized if you have searched for the item and opened it.

You can also hover over items in the **Recent Items** block to see a preview overlay with additional information for the item. You can set the visible attributes for the information that is displayed in the preview overlay for each object. This is based on [Tag Clouds](#).

You can decide if you want to display the **Recent Items** block in your business role. You can make the necessary entries in the **Personalize Navigation Bar** dialog. You can start this dialog in the **Layout** block on the central personalization page by clicking the link **Personalize Navigation Links**. Here you can also select the number of links to be displayed, by choosing a maximum of 25 entries.

Feed Subscription

Use

You are provided with a feed that is related to your daily work. Every user is provided with his or her own feed.

Prerequisites

Browser Settings

If you are using the Mozilla Firefox browser, you need to adapt your browser settings to use the **Copy Link** button in the feed personalization dialog, which you can access at the central personalization page of the WebClient UI. You need to configure the browser settings, so that the user can use this button. In this case, he or she needs to select the link and copy it manually to the feed reader. For more information about how to adapt the browser settings, see

<http://www.mozilla.org/editor/midasdemo/securityprefs.html> .

To adapt the URL generation to your own requirements, you can adapt the Web dispatcher settings and the table HTTPURLLOC. For more information, see <http://help.sap.com>  **SAP S/4HANA**  and search for **Web Dynpro ABAP**  **URL Generation in AS-ABAP – Web Dispatcher Configuration**  **Configuration Table HTTPURLLOC** .

Customizing Settings

You can define feeds and feed groups in Customizing for **UI Framework** under **UI Framework Definition**  **Feeds** .

You need to assign a feed group with its defined value to the function profile CRM_FEEDS of a business role in Customizing for **UI Framework** under **Business Roles**  **Define Business Role** .

To enable a feed in a business role, you need to assign the function profile CRM_FEEDS with value DEFAULT to this business role.

Features

The feed currently includes worklist items, such as alerts, workflow tasks, and business object inbox items.

You can subscribe to a feed by using the RSS feed feature of the browser, for example by clicking the RSS icon in the address bar of the Mozilla Firefox browser. Or you subscribe to a feed by clicking the feed icon in the feed personalization dialog. You can start this dialog if you click the **Personalize Feed Settings** link in the feed block on the central personalization page of the WebClient UI. In this dialog, you can use the **Copy Link** button to copy the feed link to the feed reader.

However, these settings are only available if the necessary Customizing settings at business role level have been performed beforehand.

Message Area

Use

You can use the message icon to display messages in the message area.

Features

The message icon is located at the left hand side of the footer. You can move the message bar to any part of the browser window below the branding area by using drag and drop. When new messages are available, the message bar is displayed in its original place.

A message bar header shows the icon of the corresponding message type and the number of messages that are related to this message type. The following message types are available: warnings, alerts, error messages, and information messages.

Messages Popover

You can see the short text of new messages in the message popover. If there is a long text, it will be available using a chevron next to the message short text.

If there are more than five messages, a scroll bar appears.

You can personalize the preview of new messages on the central personalization page by clicking **Personalize** in the header area of the WebClient UI. For more information, see [Central Personalization](#).

Navigation to Field That Issued a Message

By clicking a link, you can navigate directly from the message to the field that issued the message.

This function is automatically available for the following fields:

- Fields that are defined as mandatory in the UI Configuration Tool
- Fields that are defined as mandatory in the business object layer (BOL)

For more information, see [Mandatory Fields](#).

For all other cases where messages are related to fields, you can enable the navigation links from the message to the field that issued the message.

i Note

For this function, you must carry out additional development activities on a project basis.

For information about enabling the navigation links to the fields that issued the message, see [Enabling Links in Messages](#).

Coloring of Field That Issued a Message

You can define the color of input fields and dropdown lists to reflect their status.

i Note

For this function, you must carry out the following additional development activities on a project basis.

You can do this by using the method `SET_FIELD_STATE`. Possible values for parameter `IV_STATE` are **W** (warning) for yellow coloring and **E** (error) for red coloring. The field status is reset at every roundtrip. So there is no need to delete the status that is set. If, after revalidation, the status of the field is still problematic, method `SET_FIELD_STATE` must be called again.

Enabling Links in Messages

Enabling Links to Fields in Forms

Proceed as follows:

1. Start transaction WCF_MSG_FLD_MAPPING.

You can see all mappings of messages and fields that are defined.

2. Choose **New Entries**.

3. Define the mapping and fill all required fields.

4. You can edit the message text and message variable fields. This restricts the mapping to the relevant message.

5. Choose **Save**.

The mapping takes effect at the next logon to the WebCient UI.

As an alternative, for fields in forms, you can also follow the procedure below.

Enabling Links to Fields in Tables

i Note

To enable links in fields of tables, you must carry out the following development activities on a project basis.

Proceed as follows:

1. Provide the mapping between the message and a field.

```
lr_msg_service = view_manager->get_message_service( ).
lr_msg_service->add_message_mapping(iv_message_class = 'CRM_BRC_MESSAGES'
iv_message_number = '013'
iv_binding_string = '//BRCBUSINESSROLE/STRUCT.CRMUI_NAVBAR'
iv_page_id = component_id ).
```

2. You can modify the mapping using the following methods:

- REMOVE_MESSAGE_MAPPING

This removes the mapping for a specified message (message class, message ID) and for a specified view.

- REMOVE_ALL_MESSAGE_MAPPING

This removes all mappings for a specified view.

The mapping is kept and reused for the entire session.

If a message is mapped to several fields and you click the link in the message, you navigate to the first field on the screen.

If a field is not found on the page, the navigation does not happen. However, you can specify that a server event is to be triggered. Provide the optional parameter IV_SERVER_FALLBACK if you add the mapping using method ADD_MESSAGE_MAPPING with value ABAP_TRUE. This triggers the server event focusField. You need to define an event handler.

For the binding string that is provided to method ADD_MESSAGE_MAPPING, you can use a single attribute or table attribute.

♣ Example

Type	Example
Single attribute	//BRCBUSINESSROLE/STRUCT.CRMUI_NAVBAR
Table attribute	//BRCKEYBOARDSHORTCUTS[x]/LINK_ID

X corresponds to the position of the corresponding object in the collection wrapper. For a binding string that uses a table attribute, make sure that the index (specified in angular brackets) is the index of the targeted entity in the collection wrapper. That is, if sorting happens, the binding string of the message attribute mapping needs to be updated with the index.

Copy and Paste in Tables

Use

You can copy text from Microsoft Excel and paste it into tables in the WebClient UI.

Features

You can paste any text into the WebClient UI. This includes normal text (numeric and alphanumeric text, for example), dropdown list boxes, checkboxes, and so on.

To paste text into the WebClient UI, you press **CTRL** + **V** in the edit mode. In the display mode, pasting is not possible.

To identify a table in the WebClient UI, you need to place the cursor on the table cell into which the first table cell from the Excel list is to be pasted. Read-only cells and cells with dropdown list boxes, however, cannot be selected as the first cell.

Pasting is supported only for adjacent table cells in the WebClient UI. If pasted text exceeds the visible lines in the WebClient UI due to scrolling, this text is pasted into the table. It is only visible when the user scrolls down the table, however.

i Note

If pasting exceeds the number of rows and columns in the WebClient UI, the copied text from the Excel list is discarded. Pasting to value-dependent fields is not supported.

You cannot paste text into read-only cells. In this case, the copied text from the corresponding cell in the Excel list is discarded, and the content of the read-only cell in the WebClient UI remains unchanged.

If you paste values into dropdown list boxes in the WebClient UI, the system tries to find matching key/value pairs. If a matching key/value pair is found, the correct value is pasted into the dropdown list box. If no matching key/value pair is found, the values in the dropdown list box in the WebClient UI remain unchanged.

If you paste values for checkboxes into the WebClient UI, **1** or **X** is used for a selected checkbox, and **0** or an empty space is used for an unselected checkbox.

i Note

If either selecting a value in the dropdown list box or selecting a checkbox requires a roundtrip, for example, a communication step with the backend system, pasting into dropdown list boxes and checkboxes in the WebClient UI is not executed.

If you cannot paste values for a dropdown list box or a checkbox into the WebClient UI (because the value provided is incorrect), the cell is highlighted with red color.

In hierarchies in the WebClient UI, you can only paste text into visible cells, not to lower levels that are currently collapsed and so not visible.

Toggle Buttons

Use

You can use toggle buttons in the WebClient UI.

Features

Toggle buttons have two states: They are on when they are clicked, and off when they are not clicked. When toggle buttons are not clicked, they look like pushbuttons. When toggle buttons are clicked, they have a different background color. The same applies to pushbuttons: They also have a different background color, when they are clicked.

Buttons with a mouseover effect now have a more distinctive look, when users move the mouse over the button. This applies to all button types.

Since buttons are skin-dependent, the new look and feel of buttons is adapted to the color and design of the standard skins that are delivered by SAP.

More Information

[Skins](#)

Tag Clouds

Use

You can use tag clouds to display your tags, on the home page of a business role in the WebClient UI.

Prerequisites

Enable Tag Clouds

To enable tag clouds, you need to select checkbox **Enable tag clouds** in the **Settings** block on the central personalization page of the WebClient UI. If tag clouds are enabled, you can tag business objects by clicking the corresponding icon in the toolbar.

To enable tag clouds in your business role, you need to assign parameter TAG_CLOUDS_ALLOWED with profile value TRUE via the function profile PARAMETERS to your business role. You do this in Customizing for **UI Framework** under **Business Roles** > **Define Business Role**.

You can, however, disable tag clouds for individual UI object types in Customizing for **UI Framework** under **UI Framework Definition** > **Tag Clouds** > **Define Settings for Tag Clouds**. These settings apply to all business roles.

Enable Community Sharing

To enable community sharing in your business role, you need to assign parameter TAG_CLOUDS_COMMUNITY with profile value TRUE via the function profile PARAMETERS to your business role. You do this in Customizing for [UI Framework](#) under [Business Roles](#) [» Define Business Role](#).

You can, however, disable the community sharing for your business role, if you assign parameter TAG_CLOUDS_COMMUNITY with profile value OFF via the function profile PARAMETERS to your business role in the above-mentioned Customizing activity.

Define Tag Result Lists

You can define the columns that are to be displayed in every tag result list in Customizing for [UI Framework](#) under [UI Framework Definition](#) [» Tag Clouds](#) [» Define Settings for Tag Result Lists](#).

Enable Dynamic Navigation Support

To enable navigation support, you need to define the generic outbound settings in Customizing for [UI Framework](#) under [Technical Role Definition](#) [» Define Navigation Bar Profile](#).

Features

You can add tags to any business object in the WebClient UI. If you click the tag cloud icon in the toolbar on the overview page of a business object, you can add a tag to the current business object. You can also search for existing tags to add them to this business object, and easily delete your own tags assigned to this business object. If other users have added tags to a business object, you can also add these suggested tags to the same business object.

Your tags are organized in tag clouds that are displayed on the home page of your business role. You can see your own tags in your personal tag cloud, and those of all users who are logged on to the system with the same business role, in a community tag cloud. The text sizing of tags is used to represent the most commonly used tags, which means that the font size of tags in a tag cloud is based on the number of objects that are assigned to this tag.

In the tag cloud, you can do the following:

- Organize tags in your personal tag cloud and in the community tag cloud alphabetically and according to occurrence, in ascending and descending order
- Search for your own tags or for all community tags. The search result is displayed in a tag cloud.
- Search for business objects that are related to your own tags, or for business objects that are related to all community tags. The search result is displayed in tag result lists on a tag result page.
- Display popular tags, which are community tags that are listed in a table. You can personalize the number of popular tags that are displayed in the tag cloud.
- Display your five most recent tags. If you choose one of the most recent tags, you navigate either to the tag result page or to the overview page of a single business object.
- Clear your personal tags

If you click a tag that is assigned to several business objects, a result page is started where the business objects are grouped in tag result lists. Every tag result list is based on an individual UI object type.

Activities

To make tag clouds available on the home page of your business role, proceed as follows:

1. Add the content block for tag clouds to the home page

You need to add UI component WCF_TAG_CLOUDS to the component usages in the runtime repository of the UI component workbench. You need to do this in the UI component that contains the home page of your business role.

2. Configure the tag result page

You define the columns that are to be displayed on the tag result page. You can change the order of tag result lists that are displayed on the tag result page, by using business add-in WCF_MIXED_LIST.

After you have defined the columns of the tag result lists, you need to adapt the UI configurations of the corresponding UI object type. These configurations are available via UI component WCF_MIXED_LIST in the UI configuration tool.

3. Enable tag clouds in your business role

4. Enable dynamic navigation support

You need to make sure that the navigation bar profile of your business role supports dynamic navigation, which is necessary either when you navigate to the overview page of a business object or when tag result lists are displayed.

Share Link

Use the [Share Link](#) button to display the URL of a persistent business object so that you can copy the link to your clipboard. You can then share it with others so that they can access the business object quickly and without searching for it or you can save it as a bookmark.

You can use the [Share Link](#) button to manually copy the URL of a persistent business object to your clipboard.

Prerequisites

1. To enable the [Share Link](#) button, you need to set up the business role using the [Define Parameters](#) Customizing activity available in the [UI Framework](#) > [Technical Role Definition](#) .
 - o Determine whether SHARE_LINK_ALLOW is listed in the [Define Parameters](#). If not, you can add it using the [New Entries](#) button.
 - o Go to the [Profile Definition](#) and select or add the profile where you would like to add the Share Link capabilities.
 - o Select this profile and open the [Profile Assignment](#) so that you can add the SHARE_LINK_ALLOW parameter and set it to TRUE.
2. Assign the parameter profile to the corresponding business role using the [Define Business Role](#) Customizing activity available in [UI Framework](#) > [Business Roles](#) .

Features

When you use the [Share Link](#) button, the displayed URL will open the business object. You can copy this link and share it with others so that they can access a business object quickly, rather than searching for it or bookmark it for later use.

Dashboards

Use

You can display dashboards in a content block on the home page of a business role. You can display XCelsius dashboards and BW reports in this content block. By clicking a link, you can display a report in full-screen mode. In the personalization of this content block, you can select a default dashboard to be displayed. You can also select the dashboards to be available in a dropdown list.

You can choose the report to be displayed on your home page in the personalization of the content block. By default, the first report in your personalized list of reports is displayed. If a user does not have reports in their personalized list of reports, then the report configured by the superuser is displayed.

You can display multiple content blocks with different reports. The order of the reports in the personalized list determines the order of the reports in the content blocks.

Prerequisites

To make dashboards available in the home page of a business role, you have made the necessary settings in Customizing for **UI Framework** under **►►Technical Role Definition► Define Function Profile ►**. To enable a default report to be displayed on the home page of a business role, enter the logical link of the report that you want to display by default as a value for the **DASHBOARD_ON_HOMEPAGE** function profile. Then, you need to add the **DASHBOARD_ON_HOMEPAGE** function profile to your business role.

To make the **Dashboards** content block available in the home page personalization, you have added the UI component **WCF_DASHBOARD** to the view set of the home page using the UI Component Workbench. To do this, proceed as follows:

1. Open the UI Component Workbench by using the transaction code **BSP_WD_CMPWB**.
2. Open the UI component where your home page resides, and open the **Runtime Repository Editor**.
3. Add **<usage.Name>.WCF_DASHBOARD/MainWindow** as a view in your home page view set.

If you wish to add more than one content block, you have to repeat these steps and use a different title for the content block.

To display the **Dashboard** content block on the home page, you have to add the content block to the displayed content blocks in the page configuration.

Favorites

Use

You can add business objects to your favorites, which are displayed on the home page of your business role.

Prerequisites

Enable Favorites

To enable favorites, you need to select the **Enable favorites** checkbox in the **Settings** content block on the central personalization page of the WebClient UI. If favorites are enabled, you can add business objects to the favorites list by clicking the corresponding icon in the toolbar.

To enable favorites in your business role, you need to assign parameter **FAVORITES_ALLOWED** with profile value **TRUE** via the function profile **PARAMETERS** to your business role. You do this in Customizing for **UI Framework** under **►►Business Roles► Define Business Role ►**.

You can, however, disable favorites for individual UI object types in Customizing for **UI Framework** under **►►UI Framework Definition► Define Settings for Favorites ►**. These settings apply to all business roles.

Enable Community Sharing

To enable community sharing in your business role, you need to assign parameter FAVORITES_COMMUNITY with profile value TRUE via the function profile PARAMETERS to your business role. You do this in Customizing for [UI Framework](#) under [Business Roles](#) > [Define Business Role](#).

You can, however, disable the community sharing for your business role, if you assign parameter FAVORITES_COMMUNITY with profile value OFF via the function profile PARAMETERS to your business role in the above-mentioned Customizing activity.

Enable Dynamic Navigation Support

To enable navigation support, you need to define the generic outbound settings in Customizing for [UI Framework](#) under [Technical Role Definition](#) > [Define Navigation Bar Profile](#).

Features

If you click the favorites icon in the toolbar on the overview page of a business object, you can add this business object to your favorites list. You can choose to which folder you want to add this favorite. You can also create a new folder, which is then added to the existing folder structure.

If you want to add a business object to your favorites list that has already been added, the path where the favorite is located is displayed under [Existing Favorites](#). To add the business object to your favorites, you can delete the existing favorite.

You can decide whether you want to share your favorites with all users who are logged on to the system with the same business role. In that case, your favorites are displayed in the community favorites list on the home page of your business role.

Your favorites are displayed in your personal favorites list and in the community favorites list, on the home page of your business role. You can see the folder structure of your personal favorites. You can use drag and drop to move your favorites. The community favorites are displayed in a flat list. If you click the favorites that appear as hyperlinks, you navigate to the corresponding item.

You can filter your personal favorites and the community favorites by UI object type. In the standard system, the criterion [All Items](#) is displayed in the filter. If you select a criterion in the filter dropdown list box, the favorites are automatically filtered with this criterion.

You can organize your personal favorites by clicking the [Personalize](#) icon in the favorites content block header. You can create new folders in the folder structure, rename existing folders and favorites, delete folders and favorites, cut and paste folders and favorites, and move folders and favorites up and down in the folder structure. You can use drag and drop to personalize your favorites.

Activities

To make favorites available on the home page of your business role, proceed as follows:

1. Add the content block for favorites to the home page

You need to add UI component WCF_FAVORITES to the component usages in the runtime repository of the UI component workbench. You need to do this in the UI component that contains the home page of your business role.

2. Enable favorites in your business role.

3. Enable dynamic navigation support

You need to make sure that the navigation bar profile of your business role supports dynamic navigation, which is necessary when you click a favorite and directly navigate to the overview page of the business object.

Action Menu

Use

You can use the action menu to perform, for example, actions on fields in form views and tables in the WebClient UI. These actions enable the use of various communication channels. You can perform actions such as showing if a person is online or offline, sending e-mails and SMS, sending instant messages, and making phone calls by using SAP Business Communications Management and Skype technologies.

If you click the icon next to the field, the action menu opens, in which you can choose an action. The action starts immediately, either in the same window or in a new window, by clicking the entry in the action menu.

Prerequisites

To define action providers, you need to make the necessary settings in Customizing for **UI Framework** under **UI Framework Definition** > **Design Layer** > **Define Action Providers**.

Features

You can use one of the following options to assign an action menu to a field:

- Assign an action menu tag to a field
- Assign an action menu from a configurable view to a field
- Assign an action menu to a particular field in multiple views by using the design layer

Assign an Action Menu Tag to a Field

To assign actions to a specific field, this field needs to be inserted into the action menu tag. The attribute of the action menu tag corresponds to the ID of the specific field.

❖ Example

```
<thtmlb:actionsMenu ... for="BP_RESP" ...>
```

```
<thtmlb:inputField id="BSP_RESP" ...>
```

```
</thtmlb:actionsMenu>
```

The following table explains which attributes can be assigned to a specific field:

Name	Mandatory	Description	Possible Values
id	X	Specifies the ID for the action menu instance; unique name that identifies the BSP element; the attribute is used in event handling and data handling.	Character values, for example, <code>actionMenuforBP1</code>

Name	Mandatory	Description	Possible Values
actionItems		Defines a table that describes the menu items. Every row specifies a menu item, for example, ID, text, icon, and so on.	Table type WCFT_THTMLB_ACTION_MENU_ITEMS
actionProvider		Retrieves the action items. This is an alternative to actionItems. If this attribute is specified, the actionItems attribute is skipped.	Class that inherits the abstract class CL_BSP_WD_ACTION_PROVIDER and implements only the abstract method IF_BSP_WD_ACTION_DESCR_EXT~BUILD_ACTIONS
actionMenu		Specifies icon source and tooltip for the action menu	Structure WCFS_THTMLB_ACTION_MENU
for	X	Explicitly associates the actionMenu with another BSP element, for example, textView. The value of this attribute must be the same as the value for the id attribute of the element wrapped by this actionMenu.	
loadOnDemand		Specifies that the HTML for the menu is retrieved from the backend only on demand, which means only when the menu is opened. This attribute is provided for performance reasons: reducing HTTP response size and rendering time.	TRUE/FALSE
iconSrc		Defines the URL of the icon. The WebClient UI framework provides a standard URL.	
switchId		Used to switch the tag on and off that uses the switch framework transaction. If the switch is active, the tag is rendered.	
tooltip		Defines the tooltip of the action menu button	

Assign an Action Menu from a Configurable View to a Field

To assign an action menu from a configurable view to a field, you need to implement a special getter method. The getter returns all the information that is required to build the action menu. The framework runtime engine generates the action menu, which is assigned to the corresponding field.

 Example

Field **City** (CITYTO) is part of the configurable view. You need to provide the getter method `get_ac_City` (GET_AC_CITYTO) to assign the action menu to that field.

The following table explains the signature of the getter method:

Parameter Name	Type	Typing Method	Associated Type	Details
IV_COMPONENT	Importing	Type	STRING	
IV_INDEX	Importing	Type	I	Only if it is used in table view context node
IV_MODE	Importing	Type	CHAR1	
IV_DISPLAY_MODE	Importing	Type	BOOLEAN	
RV_ACTION_DESCRIPTOR	Returning	Type Ref To	CL_BSP_WD_ACTION_PROVIDER	

Use the Design Layer for the Action Menu

You can assign the action menu to a field by using the design layer. You can assign an action provider to one attribute of a design object. If you change the attribute, for example, the implementation class, this change is displayed in all views in which this design object is used.

You can define action providers in Customizing for **UI Framework** under  **UI Framework Definition**  **Design Layer**  **Define Action Providers** 

More Information

[Design Layer](#)

Time Slider

Use

You can use the time slider in the WebClient UI to select a time frame, by using drag and drop.

Features

You can select a time frame on three levels of granularity: year, month, and day. You can drill down for more granularity and expand the time frame to the left and to the right. The time slider offers the following levels to display years, months, and days:

- On the first level, you can see the entire year and a small representation of every month. If you click a year, you can see the levels below in more detail. If you click a month, you can see the entire month.
- On the second level, you can see a detailed representation of every month.

- On the third level, you can see the days of a month.

The selected time frame can be used by applications, for example, to display documents for which the selected time frame is applicable.

To access predefined time frames with one click, the time slider supports buttons, which can be used by applications to save their own time selections. It also offers input fields with date selection to enter exact dates. Buttons and date fields are optional and can be hidden.

You can use the time slider to display items, indicating special incidents, which can be defined by the applications. These incidents are displayed even if no time frame has been selected. You can start a specific action if you click the incident, which is represented by an icon, in the time slider. These actions can be defined by the applications.

To predefine a time frame in the time slider, you can use specific parameters. First, you define the start date and end date of your own selection. Second, you define the level where the selection starts and where it ends. Since the time slider supports three different levels, you need to define both levels.

Mandatory Fields

Use

You can set a field as mandatory in one of the following ways according to your business requirements:

- In the business object layer (BOL)

You can use the GenIL Model Editor to define a mandatory field in the object model. The property from the BOL is made available on the user interface (UI).

- In the application by implementing a property getter (P-getter) method

Implement the following P-getter that evaluates the property IF_BSP_WD_MODEL_SETTER~FP_INPUT_MANDATORY and return ABAP_TRUE if the field is to be mandatory:

```
METHOD GET_P_EMPLFULLNAME.
CASE iv_property.
WHEN if_bsp_wd_model_setter_getter~fp_input_mandatory.
IF gv_display_mode = abap_true.
rv_value = ABAP_FALSE.
ELSE.
rv_value = ABAP_TRUE.
ENDIF.
ENDCASE.
ENDMETHOD.
```

- In the view configuration

In the UI Configuration Tool, you can define a field as mandatory. If you make this setting in a layer, you cannot override it in the next layer above it. For this reason, mandatory fields are marked with an asterisk (*) as part of the label for editable input fields. If the field is left empty, a red frame is displayed around the field and you cannot save or process your data.

Prerequisites

You have made the settings in Customizing for **UI Framework** under **Generic Interaction Layer/Object Layer** **Basic Settings**

Show More in Object Page

Use

You can display more or less information in the object page assignment blocks with responsive layouts.

Usability is improved by showing concise overview of complex objects with the following:

- The most important information displayed first.
- The option to show the hidden fields when needed.

Features

- In the display mode, only the first row of groups is displayed by default.
- You can select the **Show More** button at the end of the groups to display the hidden groups of fields.
- When all the fields are displayed, you can use the **Show Less** button to hide the groups.
- In the edit mode, all the groups are displayed by default. You can select the **Show Less** button to show less fields.
- If you select the button to show more or show less fields, your setting will be persisted.

Cancel Confirmation Popover

Use

You can add **Cancel Confirmation** popover in the overview pages. If you make changes and select the **Cancel** button, a popover is displayed to confirm that the changes are discarded. You must select outside the popover to close it.

Procedure

The popover is not added by default to all the object pages.

- To add the confirmation popover, you must enable the feature in the button attributes.
- Locate the overview page and method where the **Cancel** button is added in the toolbar and add the button attribute.

❖ Example

☰, Code Syntax

```
ls_button-btn_usage = 'Confirm'.
```

Tables

You can use the following table functions:

You can define whether and how many columns are fixed when horizontal scrolling is used in tables and trees.

Navigation Using Cursor Keys

You can use the cursor keys to navigate from one field to another field in a table. If this navigation is done in navigation mode, the text in input fields is highlighted with a dotted line. If you start typing in a table field, you automatically switch to input mode and the cursor moves within the input field. Alternatively, you can press **F12** to switch from navigation mode to input mode and back to navigation mode again.

Selection of Multiple Rows

You can choose multiple rows by selecting the checkboxes

Optimized Scrolling

You can use optimized scrolling in tables for which vertical scrolling is enabled. With optimized scrolling, only visible rows are initially loaded. The remaining rows are loaded during scrolling and are displayed either when you slow down or stop scrolling. Once rows have been displayed, they are kept on the frontend, which accelerates the scrolling even further.

Additional Functions that You Can Enable

i Note

For the functions described below, you must carry out additional development activities on a project basis.

In addition to the functions described above, you can enable the following functions on a project basis:

- Fast row selection

You can enable fast selection of table rows, that is, the selection of rows does not trigger any roundtrip unless explicitly triggered by a button or any other action on the screen programmed to do so. You cannot use fast row selection in tables that use pagination and tables that use optimized scrolling.

For more information, see [Enabling Table Functions](#).

- Fast Row Creation

You can enable fast row creation. In tables for which edit mode is enabled, fast row creation enables you to quickly create new table rows with an accelerated roundtrip.

For more information, see [Enabling Table Functions](#).

If a table is in edit mode, you can add new lines at the end of the table by pressing the tab key in the last table cell.

Using fast row creation, you can insert one or multiple rows into a table without a roundtrip.

When pasting data into a table, for example, from Excel, new rows are automatically created. For more information, see [Copy and Paste in Tables](#).

Enabling Table Functions

Use

You can enable the following table functions:

- Fast row selection
- Fast row creation

i Note

For these functions, you must carry out the following additional development activities on a project basis.

Procedure

Fast Row Creation

To append rows at the end of the table, proceed as follows:

1. Add attribute `row_template` to the context node of the table view.
2. If you are modifying an existing button, remove attribute `on_click` from the button definition and add attribute `on_client_click`. If you wish to append more than one row when the button is chosen, change the call to the `cl_thtmlb_util=>get_table_row_creation_event` by passing parameter `iv_number_of_rows`.
3. Set the `create_status` of the newly appended rows in your append event handler to `true`.
4. Remove any code that is related to maintaining cellerator tag attribute `visibleFirstRow`.

To insert rows above the last selected row, proceed as follows:

1. Add attribute `row_template` to the context node of the table view and pass parameter `iv_mode` as `INSERT`.
2. If you are modifying an existing button, remove attribute `on_click` from the button definition and add attribute `on_client_click`. If you wish to insert more than one row when the button is chosen, change the call to the `cl_thtmlb_util=>get_table_row_creation_event` by passing parameter `iv_number_of_rows`.
3. Retrieve the index in which rows should be inserted and set the `create_status` of the newly inserted rows in your insert event handler to `true`.
4. Remove any code that is related to maintaining cellerator tag attribute `visibleFirstRow`.

Fast Row Selection

To enable fast row selection, proceed as follows:

1. In file `.htm`, remove attribute `onRowSelection` from the table or tree tag.
2. Specify for the tag which toolbar button is enabled when 0, 1 or n rows are selected. Toolbar buttons can be enabled when 0, 1 or n rows are selected. Enhance the code where the button table is built to be passed to attribute `actions` of `configTable` or `configTree`. You can use the constants that are predefined in class `CL_THTMLB_BUTTON` and assign them to attribute `frontend_states`.
3. In method `do_handle_data` of the view controller, call method `mark_selected_rows` for the relevant context node.

i Note

`iv_id`: The table ID passed to tag `configTable` or `configTree` must be identical with the table ID in file `.htm`.

Enable Row Navigation and Display Chevron

Use

You can enable the row navigation and display a chevron to navigate in WebClient UI tables.

Features

You can select and press **Enter** anywhere on the row or chevron to navigate to the corresponding page.

You can enable navigation indicator to display the last opened item.

The row navigation and chevron are available in display and edit mode.

The key object identifier is displayed in bold to indicate the navigation trigger.

If the key object identifier is removed via personalization, the navigation is still enabled.

If the object identifier is removed via configuration, the feature is disabled and navigation is no longer possible.

The feature is automatically disabled in popups.

If implementing the feature in the advanced search result list, application developers must make sure that the navigation is leading to the same page as the navigation when there is only one result object (if applicable), or else navigation will be inconsistent between these two functionalities.

Procedure

Application developers can enable the feature by calling the public method `SET_ROW_CLICK_NAV` of their table's context node instance that derives from class `CL_BSP_WD_CONTEXT_NODE_TV`, and pass the following parameters:

- `IV_LEADING_NAV_ATTR`: Mandatory parameter for the name of the attribute in the context node that leads to navigation when selected
- `IV_DISPLAY_NAV_EVENT`: Optional parameter for the name of the server event that performs navigation in display mode
- `IV_EDIT_NAV_EVENT`: Optional parameter for the name of the server event that performs navigation in edit mode.

If only the parameter `IV_LEADING_NAV_ATTR` is provided, the server event will be fetched from the p-getter of that attribute for both display and edit mode. Application developers must ensure that a p-getter exists (either from the application controller class or from design layer) and implementing onclick and linkevent property, for the leading navigation attribute, and that it returns the server event for the onclick action if they want to omit the optional attributes of method `SET_ROW_CLICK_NAV`.

If `IV_DISPLAY_NAV_EVENT` is provided, but `IV_EDIT_NAV_EVENT` is omitted, the event used in edit mode will be the same as in the display mode. The same happens when `IV_EDIT_NAV_EVENT` is provided but `IV_DISPLAY_NAV_EVENT` is omitted.

There is also a possibility to concatenate the `EVENT_DEFINED` string to the server event name for parameters `IV_DISPLAY_NAV_EVENT` and `IV_EDIT_NAV_EVENT`, which is commonly used in one click action event handlers as a way to differentiate between actions. These will allow application to use this row click navigation functionality for one click action server events, without changing their server event code line. The syntax is the following:

Code Syntax

```
IV_DISPLAY_NAV_EVENT = '<SERVER_EVENT_NAME>#<EVENT_DEFINED>'
```

❖ Example

```
IV_EDIT_NAV_EVENT = 'ONE_CLICK_ACTION#EDIT'
```

In this example, when the user selects the table row while in edit mode, the server event method `EH_ONONE_CLICK_ACTION` will be executed and the string `EDIT.X` (where `X` is the number of the table row selected by the user) will be supplied in property `EVENT_DEFINED` of the `HTMLB_EVENT_EX` object parameter of that method.

≡ Sample Code

```
METHOD GET_TABLE_LINE_SAMPLE.
```

```
* Application code...
```

```
set_row_click_nav(
    iv_leading_nav_attr = 'PO_ITEM_POS'
    iv_display_nav_event = 'TO_ITEM_DETAILS'
    iv_edit_nav_event   = 'ONE_CLICK_ACTION#EDIT'
).
```

```
ENDMETHOD.
```

The method `SET_ROW_CLICK_NAV` can be called from anywhere as it is public, and it can be called as soon as the context node is instantiated. It can be called from a specific application class to target a specific table, or it can be called from a generic class inherited by many application classes and in that way enabling the feature for multiple tables at once. The values are kept during the life of context node class. Some placement suggestions are listed below:

- `GET_TABLE_LINE_SAMPLE` method of table's context node class deriving from `CL_BSP_WD_CONTEXT_NODE_TV`
- `WD_CREATE_CONTEXT` method of application controller class

i Note

If the table tag has row selection event defined with property `onRowSelection`, application developers must make sure that `selectionColumn` property is set to `TRUE` or else the user will not be able to select the row. This happens because selecting the row will navigate and will not fire a row select event when the row click navigation feature is enabled. After implementing this feature, the only way users can select the row is by clicking on the select column (controlled by property `selectionColumn`).

Export to Spreadsheet

Use

You can export data that has already been entered in the WebClient UI to a table processing program. When you export to excel .xlsx format, exported hyperlinks allow you to return to the referenced data object directly.

Prerequisites

To disable the authorization for the export to spreadsheet on business role level, you need to make the necessary settings in Customizing for [UI Framework](#) under [▶▶Technical Role Definition ▶ Define Parameters ▶](#). With parameter `EXPORT_DISABLE` and parameter value `TRUE` you can disable the export.

To disable the export to spreadsheet in your business role, you need to assign function profile `PARAMETERS` to this business role. You can do so in Customizing for [UI Framework](#) under [▶▶Business Roles ▶ Define Business Role ▶](#).

To set the preferred file format on a role level, set the parameter `TABLE_EXPORT_FORMAT` to `XLSX` or `CSV`. Default is `XLSX`.

To set the preferred file format on a user level, set the user parameter `CRM_CSV_EXPORT_ONLY` to `TRUE`, `FALSE`, `XLSX` or `XML`. Default is `XLSX`.

For exporting hyperlinks via `XLSX`, you must be in standalone mode and select `XLSX` file format.

Features

You can perform the export to spreadsheet by clicking the icon [Export to Spreadsheet](#), for example, on the top right of the search result list. The export file with the search results is created in a table processing program, for example, Microsoft Excel, and it then opens. Hyperlinks that connect items in the results list to the WebClient UI will be included if you fulfill the requirements above. You can use the standard processing of the table processing program to edit and save the export file.

You can export result lists (for example, to review the exported data) of up to hundreds of result lines. However, if you need to export larger amounts of data, you should use BI-focused extraction functions.

Authorization for Export to Spreadsheet

To authorize the export to spreadsheet, you can use authorization object `S_GUI` in SAP GUI. This authorization object is assigned to the authorization role `SAP_CRM_UIU_FRAMEWORK`, which is delivered in the SAP standard system.

You can disable the export to spreadsheet for individual users by removing the authorization object. You can also disable the authorization for the export to spreadsheet on business role level in Customizing.

Export to Google Workspace

Use

You can now export spreadsheets in the WebClient UI applications to a drive in the cloud, for example, Google Workspace. In the [Export As](#) dialog box you choose the file name and destination to export the file.

Prerequisites

- You have enabled the export to spreadsheet functionality on the selected table. For more information, see [Export to Spreadsheet](#).
- You have obtained a license for the SAP Document Management Service. For more information, please contact your SAP Account Executive.
- You have configured integration with Google Workspace in your system.

Features

You can perform the export to Google Workspace by selecting the down arrow icon next to **Export to Spreadsheet**, and select the **Export As** button. This opens the **Export As** popup, where the user can modify the following:

- File name
- File format
- Destination: Local or Cloud
 - Local - to your local machine drive
 - Cloud - to a remote vendor drive, for example, Google Drive

When you select the **Export To** button, the dialog provides a file browser to select the cloud folder to export data to the cloud.

In the file browser, you can create a new folder and export to that folder. You can sort the columns **Name** and **Last Changed On** in the file browser.

Once the user selects the **Export** button, the table opens up as a spreadsheet file in a new browser tab page within Google sheets or workspace.

More Information

For more information on how to set up and configure your system, see SAP Help Portal at <http://help.sap.com> → **SAP Documentation Management Service** → **Integration** → **Integrating SAP S/4HANA with Google Workspace Using SAP Document Management Service**.

Table Charts

Use

You can use bar charts, column charts, pie charts, donut charts, line charts, scatter plot charts, and combined charts to view table-based result lists and assignment blocks in the WebClient UI.

Integration

In the standard delivery, charts are enabled in result lists and in assignment blocks.

Prerequisites

You have installed version 9 or higher of the Adobe Flash Player on your computer.

To enable or disable charts in tables, you make the necessary settings in Customizing for **UI Framework** under **Technical Role Definition** > **Define Parameters**.

You need to assign the PARAMETERS function profile to your business role in Customizing for **UI Framework** under **Business Roles** > **Define Business Role**.

Features

You can open the chart by clicking the **Open Chart** icon, if the function is enabled, in the toolbar of assignment blocks and result lists. You can select the column that you want to view in the chart and choose between a pie chart or a bar chart. You can use the

zoom function, including zoom in and zoom out.

If you click the title of the column that you have selected, in the table, you can filter this column. The chart is updated, and the corresponding bar or pie section is highlighted in the chart. If you click and filter another column, the table and the chart are updated again.

If you click a pie section in the pie chart or a bar in the bar chart, this section is highlighted and the corresponding entry in the legend is highlighted by a frame. In both cases, the table is filtered according to the pie section or the bar that you have selected in the chart.

i Note

If no chart appears when you click the **Open Chart** icon, you can open the context menu and choose **Play**.

You can create charts from one data source or two data sources in a table.

One Data Source in a Table

When you use one data source in a table, use the following chart types:

- Pie charts
- Donut charts
- Column charts
- Bar charts

Use the first dropdown list for selections as the source of data. The second dropdown list for selections is used only for charts with two data sources.

Two Different Data Sources in a Table

When you use two data sources in a table, use the following chart types:

- Line charts
- Scatter plot charts
- Combined charts

The order in the table is reflected in the charts. When chronology is expected, for dates for example, the column from the table must be ordered before rendering the chart.

Select the chart type to match the data to be displayed. For example, scatter plot chart can only be displayed with a source of data that are of numerical values.

The order of the axis also is a factor to visualize the data. Consider which is relevant for axis X versus axis Y (the axis can be interchanged accordingly).

You must select data in both DDLBs and data must be from two different data sources.

More Information

For more information, see SAP Note [3027226](#) .

Transactional Inbox

Use

For a quick overview of the day-to-day business a transactional inbox on work center level is available in your business role.

Features

The transactional inbox is flexible, and role-based Customizing of search queries for business transactions is possible. The following features are available:

- Predefined filters plus counter provide a summary of current transactions at a prominent place in the work center.
- Convenient navigation to the corresponding search result pages with executed search
- Filters and underlying search queries can be added, configured, or removed on role level by the system administrator.

Context Menu

Use

You can open the context menu by right-clicking an element on the user interface (UI) or by pressing **SHIFT** + **F10**.

Prerequisites

i Note

For this function, you must carry out the following additional development activities on a project basis.

You have assigned a context menu to UI elements, for example, input fields, images, views. To do that, use the `contextMenu` tag and the ID of the UI element. In addition, pass the name of the tag and the ID of the area inside that UI element, for example, for the cellerator, the column ID or row ID.

Example: Creating a Home Page for the Administrator Business Role

Use

The following example describes the creation of the home page for the **Administrator** business role, which is provided by SAP. You can use this procedure as a reference to create a home page for your custom business roles. You have to replace the technical names with your custom technical names.

The home page is the starting point for every user in their daily work. It contains the most important personal information and provides access to the most frequently used tools. In addition to links, reports can also be included directly on this page.

Procedure

To create the home page for the **Administrator** business role, you proceed as follows:

1. In the UI Component Workbench, create a UI component called WCF_ADMIN_HOME.
2. Open the Component Structure Browser and create an overview page called HomePage.
3. Implement interface IF_BSP_WD_HISTORY_STATE_DESCR~GET_STATE_DESCRIPTION.
4. Create an inbound plug called DEFAULT.
5. Open the Runtime Repository Editor (edit mode).
6. Add inbound plug DEFAULT to window WCF_ADMIN_HOME/MainWindow.

This is necessary to be able to navigate the UI component from the navigation bar.

7. Add view WCF_ADMIN_HOME/HomePage to the WCF_ADMIN_HOME/MainWindow. This is necessary to see the home page in the browser and to add the assigned view to the home page by using the UI configuration.
8. Add component usage WebLinkFavorites as follows:
 - a. Use component UICMP_WC_FAVORI.
 - b. Add interface view UICMP_WC_FAVORI/IFVFavoriteWebLinks.
9. Open view area OverviewPage under view set WCF_ADMIN_HOME/HomePage. Add view WebLinkFavorites. UICMP_WC_FAVORI/IFVFavoriteWebLinks.
10. Add WCF_ADMIN_HOME/MainWindow as an interface view under the component interface.

This is necessary to use the UI component in the navigation bar.

11. Open the Component Structure Browser.
12. Open the view WCF_ADMIN_HOME/HomePage and choose the **Configuration** tab (edit mode).
13. Create a configuration for the home page.
14. Move view WebLinkFavorites. UICMP_WC_FAVORI/IFVFavoriteWebLinks to one of the visible columns.
15. Test the UI component by pressing F8. A home page appears with one empty content block with the title **Web Links**.
16. To add new links into this block, you can open the personalization of the block.
17. In Customizing for **UI Framework** under **►Technical Role Definition ► Define Work Area Component Repository ►**, create a new entry with the following settings:
 - Component name: WCF_ADMIN_HOME
 - Window name: WCF_ADMIN_HOME/MainWindow
 - Description: Administration Homepage
18. Double-click the entry, choose **Inbound Plug Definition**, and create a new entry for the inbound plug definition with the following settings:
 - TARGET_ID: WCFADMHOME

The target ID must be unique.

 - Inbound plug: DEFAULT
 - Description: Administration Homepage

Skins

Use

You can use skins to adapt the layout of the user interface to your corporate identity. All steps in changing existing SAP skins need to be done manually. You can also exclude existing skins that you do not want to delete. Excluded skins are no longer available for users on the central personalization page.

The end user can choose between different skins on the central personalization page of the WebClient UI.

Features

All available skins are saved on the Web Application Server, which hosts the WebClient UI.

You can find all available skins in Customizing for **UI Framework** under **UI Framework Definition > Skins and Icons > Define Skins**. To make a new or adjusted SAP skin available again, you need to make a new entry in this table.

Making New Skins Available on the UI

Context

You can make new skins available on the central personalization page of the WebClient UI. You can decide if you want to make the new skin automatically available on the central personalization page of the WebClient UI. In this case, you do not need to enter the source path in Customizing. If you have decided not to make the new skin automatically available on the central personalization page, you can do this later by following these steps.

Procedure

1. In Customizing, choose **UI Framework > UI Framework Definition > Skins and Icons > Define Skins**.
2. Choose **New Entries**.
3. Enter the skin name, a description, and the source path.
 - The source path of SAP standard entries refers to the style folder on the SAP Web Application Server.
 - You can use a source path on your local computer or on a server.

Caution

Make sure that you are using the correct slash type in your source path entry. For a source path, you need a forward slash: /.

Example

- Example for a source path on your local computer: `c:/zthtmlb_styles/belize/`.
- Example for a source path on a server: `/sap/bc/bsp_dev/sap/zthtmlb_styles/belize`

4. Select a version.

The version depends on the `thtmlb_core_stand.css`, which is the generic CSS file for all skins. You can use your own `thtmlb_core_stand.css`, the `thtmlb_core_stand.css` of the SAP default skin, or no `thtmlb_core_stand.css` at all, for example, if you have made a lot of CSS changes in a previous release.

Results

The new skin is available in the dropdown list box on the central personalization page of the WebClient UI. To check the result in the WebClient UI, save the CSS file and delete the temporary Internet files in your browser before you start the WebClient UI.

Defining the Default Skin

Context

You can define a default skin and assign this skin to all business roles in which you want to use it.

Procedure

1. In Customizing for **UI Framework**, choose ► **Technical Role Definition** ► **Define Technical Profile** ►.
2. Select a technical profile and enter the skin name.
3. In Customizing for **UI Framework**, choose ► **Business Roles** ► **Define Business Role** ►.
4. Select a business role and assign the technical profile.

Results

You have now assigned a default skin to a business role. However, you can select another skin on the central personalization page of WebClient UI.

Excluding Skins

Context

You can exclude skins that you do not need anymore. Instead of deleting these skins, you can deactivate these skins, for example, if you want to use season layouts.

Procedure

1. Start the system.
2. Enter the transaction code SM30.
The page **Maintain Table Views: Initial Page** appears.
3. Choose the table CRMC_THTMLB_SKNE.
4. Choose **New Entries**.
5. Choose the skin(s) that you want to exclude.

Results

Excluded skins are no longer available in the dropdown list box on the central personalization page.

Adjusting Cascading Style Sheets

Use

You can adapt cascading style sheets (CSS) that you have created by copying an SAP skin using the Skin Workbench to your needs. You can change the colors and the size of the different elements.

i Note

Instead of copying and pasting an existing skin and then adapting the copied skin to your requirements, you can create your own CSS file on top of the CSS files provided in the SAP system. You do this in Customizing for **UI Framework** under **UI Framework Definition** > **Skins and Icons** > **Define Path for CSS Files**.

Integration

You can find the Skin Workbench in Customizing for **UI Framework** under **UI Framework Definition** > **Skins and Icons** > **Access Skin Workbench**.

Features

All general format styles for fonts and color definitions of the WebClient UI are saved in cascading style sheets (CSS).

The main layout file of the WebClient UI is called `thtmlb_stand.css`. It is structured by comment entries that look like this: `/* ... */`. Certain sections start with this comment, which describes the following style tags and elements that are influenced by it.

The following CSS files are available among others:

- `thtmlb_core_stand.css`: Generic CSS file for all skins
- `thtmlb_stand.css`: Main CSS file that controls browser-independent colors and the background of a skin
- Browser-specific CSS files
- `thtmlb_rtl_stand.css`: Right-to-left (rtl) support; in general, rtl is automatically managed by the browser, but in some cases the flow needs to be changed manually in the CSS file
- `thtmlb_visuals_stand.css`: Control of visual effects such as quick info texts in the navigation bar
- `thtmlb_flex.css`: Styles for FLEX content
- `printpreview.css`: Printing

Activities

You can adjust skins that you have created by copying an existing SAP skin using the Skin Workbench.

Adjusting Skins in the Skin Workbench

1. Choose the file that you want to modify in the panel below the panel with SAP skins, in the Skin Workbench.
2. Select the file and choose **Document** > **Edit** in the context menu.

The file is opened in edit mode.

3. Change the color and size of the different elements.
4. Save your data and close the editor.

Adjusting Skins Manually

1. Open `thtmlb_stand.css` with an editor.

You can find this file in the root folder of the skin, for example `default`, that you copied earlier to your local computer or to a server.

2. Change the color and size of the different elements.

❖ Example

```
.th-l-title {
color: green;

font-weight: bold;

font-style: italic;

font-size: medium;
}
```

Example: Changing Logo

Use

You can change the logo, to adapt the layout of the user interface to your corporate design.

Prerequisites

You have entered the source path of the Web folder that contains the existing skin, in Customizing for **UI Framework** under **UI Framework Definition > Skins and Icons > Define Skins**.

Integration

You can find the Skin Workbench in Customizing for **UI Framework** under **UI Framework Definition > Skins and Icons > Access Skin Workbench**.

Activities

In Customizing

You can change the logo in Customizing for **UI Framework** under **UI Framework Definition > Skins and Icons > Define Icons**. For more information, see the Customizing documentation.

Using the Skin Workbench

1. Choose the file, for example `logo.png`, in the panel below the panel with SAP skins, in the Skin Workbench.
2. Double-click the file, which is displayed in the right-hand side of the panel.
3. Click the icon **Edit Document** in the toolbar.

An editor is started on your local computer.

4. Modify the picture.
5. Save your changes and close the editor.

Manually Changing Logo

1. Open **default > styling > lshape**.
2. Replace **logo.png** with a new logo.

i Note

When you use other names and file types, you also need to adjust the file `thtmlb_stand.css`.

Integration with WalkMe

Use

WebClient UI applications offer integration with WalkMe when applications are launched within SAP Fiori Launchpad.

Features

In the SAP Fiori Launchpad integrated mode, you can launch WalkMe Plugin and use it with WebClient UI applications.

Prerequisites

The configuration for embedding WalkMe is performed in Customizing. For the WalkMe integration, a script must be loaded from WalkMe. The path to this script is maintained in the backend.

In Customizing, navigate to **ABAP Platform > UI Technologies > SAP Fiori > SAP Fiori Launchpad Settings > Configure Backend URL for WalkMe Integration**.

Here, you can open the customizing activity in the backend by following the path provided. For each sub-node of the customizing activity, you can check the documentation available there and follow the steps described to perform the required activities.

For more information about the WalkMe Integration, see [Integration with WalkMe](#).

Integration with UI Theme Designer

Use

You can use the UI theme designer to create your own themes to adapt the visual appearance of the applications.

In standalone mode, you can use special WebClient UI previews in the theme designer.

If you are using FLP integrated mode, you can create your custom themes in theme designer for Fiori apps. The same themes will apply to the WebClient UI apps in the FLP integrated mode.

Features

In the standalone mode, you can use the following WebClient UI previews in the theme designer.

- [Calendar](#)
- [Calendar Monthly View](#)
- [Calendar Weekly View](#)
- [Guided Activity](#)
- [Home New Design](#)
- [L-shape New Design](#)
- [Overview](#)
- [Overview Tile](#)
- [Search](#)
- [Workcenter](#)

Procedure

1. To create a new theme, you can use transaction [/n/UI5/THEME_DESIGNER](#) to launch the UI theme designer
2. Select the [Create a New Theme](#) button
3. Choose the base theme that you want to use from the list and enter the required information
4. Select [WebClient UI Control Previews](#) from the [Test Suites](#) section

i Note

The [WebClient UI Control Previews](#) is only available for supported themes

5. Make your changes based on the different levels of theming within the tool, then save and build your theme.

In the standalone mode, your custom theme is available in [Personalization](#) > [Personalize Layout](#) in the WebClient UI.

In the FLP integrated mode, your custom theme is available in FLP user settings.

More Information

To transport your custom theme from the test or quality system to the production system, you can use transaction [/n/UI5/THEME_TOOL](#).

For more information about the theme designer, see SAP Note [1852400](#) .

Integration with Microsoft Teams

Use

WebClient UI Applications offer [Share Menu](#) in the object page and advance search page toolbar in Fiori launchpad integrated mode to enable the share features. With this feature, users can share the app with other users.

Features

In the FLP integrated mode, you can share the object page or advance search page with other users. You can share via:

- **E-mail**
- **Microsoft Teams - As Chat, As Tab** (if Microsoft Teams integration is enabled)
- **Save as Tile**

The **Share** functionality is not available for newly created objects which are not yet persisted, so the button is not visible.

Prerequisites

To use **Share** with Microsoft Teams option, you must have completed the set up described here: [Integration with Microsoft Teams](#)

Procedure

The share feature is available in the header toolbar within object pages and advance search pages.

Users can click on the **Share** button and choose an option to share via **E-Mail** or **Microsoft Teams – As Chat, As Tab** as well as **Save as Tile** in the Fiori launchpad.

- **Send Email**

When a user chooses this option, the link to the page opens in the default email client that is configured in the system. When a user chooses the link, the application page opens in the same state in which it was shared.

- **Microsoft Teams**

Users can use Microsoft Teams:

- **As Chat**

Collaborate and quickly resolve issues, if necessary, simply by using Microsoft Teams **As Chat** to share a direct link with co-workers. You can provide them with access to a specific state of an application, for example, so they can easily process any requests you have or tasks that come up.

- **As Tab**

Work efficiently with a group of co-workers on specific content that you share in the form of a Microsoft Teams tab. Use the Microsoft Teams environment to work on the same context of an SAP Fiori application, and use the tab conversation option in parallel to share ideas, discuss, and collaborate.

- **Save as Tile**

When a user chooses this option, the corresponding **Save as Tile** dialog is displayed. When a user chooses the tile, the application page opens with which the tile was created. Application developers can customize the title and the sub-title of the tile.

More Information

For more information, see [Integration with Microsoft Teams](#).

UI Integration with Attachment Service

Use

The Attachment Service is offered as a central unified solution for uploading attachments across applications. This feature allows you to embed reuse components in the application UI.

WebClient UI offers a reusable component that can be consumed by any WebClient UI application in SAP S/4HANA. SAP applications and customer built applications can use this feature to enable attachment capabilities. The data management and context is handled by the application consuming this utility.

Attachments are stored via Document Management (DMS) or Generic Object Services (GOS).

With this feature you can upload, download, rename and delete attachments. You can add URLs as attachments. The check-in check-out features are not supported.

Prerequisites

All of the customizing and backend implementations required in [Customizing](#) of the Attachment Service are completed.

The wrapping object page must have a valid business object layer (BOL) entity that you can get from the parent window through the standard method IF_BSP_WD_HISTORY_STATE_DESCR~GET_MAIN_ENTITY.

Procedure

1. In the component where you've defined your overview page, add the **ComponentUsage** WCF_GC_ATTACH.
 - Provide the usage **ID** (for example ATTACHMENTS)
 - Enter the **Used Component**: WCF_GC_ATTACH
 - Enter the **Interface View**: WCF_GC_ATTACH/MainWindow
 - The **InboundPlugs** and **OutboundPlugs** aren't required.
2. Add the usage that you created in step 1 to the overview page view area (runtime repository) and adjust the page configuration.
3. Locate your overview page and if you haven't already, implement the DO_VIEW_INT_ON_ACTIVATION method. Follow these steps to set the values of the object type and to resolve the object key.
 - You retrieve the attachments node or create it if it doesn't already exist. Use the same usage ID that you used in step 1
 - The node object provides attributes OBJECTTYPE and OBJECTKEY to set the object type and object key respectively. These attributes are required to identify the attachment. In some scenarios the final key isn't available, for example when a business object is being created. Applications can use temporal keys (attribute OBJECTKEYDRAFT) and be updated with the actual key later on. The values of these attributes can be one of the following:
 - a. Constant value
 - b. Binding path (for example: //STRUCT.ATTRIBUTE_NAME)
 - c. BOL path (use prefix bpath for example bpath: ./@FLDATE)

Sample Code

```
DATA(lr_attach_node) = cl_wcf_attachments_service=>get_instance( )->get_node( iv_us
```

```

IF lr_attach_node IS NOT BOUND.
    " Set the properties for service contract attachment navigation
    cl_wcf_attachments_service=>get_instance( )->create_node( EXPORTING iv_usage_name  =
                                                                RECEIVING rv_node      =
                                                                )
ENDIF.

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTKEYDRAFT'
                             iv_value     = 'bpath:./@CRM_GUID' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTKEY'
                             iv_value     = 'bpath:./BTOrderHeader/@OBJECT_ID' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTTYPE'
                             iv_value     = 'bpath:./BTOrderHeader/@OBJECT_TYPE' ).

```

i Note

The `DO_VIEW_INIT_ON_ACTIVATION` method is executed each time you activate the view (for example, the navigation to the view). As long as you used a binding or BOL path, the actual value is resolved automatically in each roundtrip as these are dynamic. If the main entity isn't available, the application must calculate the values. The `DO_PREPARE_OUTPUT` method is more appropriate than `DO_VIEW_INIT_ON_ACTIVATION` method in this case.

4. To activate the document type selection for DMS repository, the application must set the attribute `SETDOCTYPE` in the method `DO_VIEW_INIT_ON_ACTIVATION`.

Sample Code

```

DATA lr_oc_node1 TYPE REF TO CL_WCF_ATTACHMENTS_NODE.

lr_oc_node1 ?= CL_WCF_ATTACHMENTS_SERVICE=>get_instance( )->get_node( iv_usage_name = '
IF lr_oc_node1 IS NOT BOUND.
    lr_oc_node1 ?= cl_wcf_output_control_service=>get_instance( )->create_node(
                                                                iv_usage_name = 'CUBTAtt
                                                                class_name   = 'CL_WCF_

ENDIF.
CHECK lr_oc_node1 IS BOUND.

lr_oc_node1->set_property( EXPORTING
                           iv_attr_name = 'setDocType'
                           iv_value     = 'true' ).

```

If attachments are uploaded from DMS source, you can select the document type from a dropdown list when you are in edit mode. The document type is linked to the uploaded attachment. Attachments that are saved show the [document info record](#).

5. To enable Harmonized Document Management, set additional attributes in step 3 in the overview page.

To use HDM for attachments at header level, the application developer must pass `SAPOBJECTTYPE` corresponding to the object type used in HDM Customizing. The `OBJECTTYPE` should be initial.

Sample Code

```

DATA(lr_attach_node) = cl_wcf_attachments_service=>get_instance( )->get_node( iv_us

IF lr_attach_node IS NOT BOUND.

```

```

" Set the properties for service contract attachment navigation
cl_wcf_attachments_service=>get_instance( )->create_node( EXPORTING iv_usage_name  =
                                                         RECEIVING rv_node      =
ENDIF.

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTKEYDRAFT'
                             iv_value     = 'bpath:./@CRM_GUID' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTKEY'
                             iv_value     = 'bpath:./BTOrderHeader/@OBJECT_ID' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTTYPE'
                             iv_value     = ' ' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'SAPOBJECTTYPE'
                             iv_value     = 'MaintenanceServiceOrder' ).

```

To enable HDM for attachments on item level, the application developer must pass both SAPOBJECTTYPE and SAPOBJECTNODETYPE in the item overview page.

Sample Code

```

DATA(lr_attach_node) = cl_wcf_attachments_service=>get_instance( )->get_node( iv_us

IF lr_attach_node IS NOT BOUND.
  " Set the properties for service contract attachment navigation
  cl_wcf_attachments_service=>get_instance( )->create_node( EXPORTING iv_usage_name  =
                                                           RECEIVING rv_node      =
ENDIF.

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTKEYDRAFT'
                             iv_value     = 'bpath:./@GUID' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTKEY'
                             iv_value     = 'bpath:./@HEADER ' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'OBJECTTYPE'
                             iv_value     = ' ' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'SAPOBJECTTYPE'
                             iv_value     = 'MaintenanceServiceOrder' ).

lr_attach_node->set_property( EXPORTING
                             iv_attr_name = 'SAPOBJECTNODETYPE '
                             iv_value     = 'MaintenanceServiceOrderItem' ).

```

Transactional Handling

The save and cancel are synchronized automatically with the main BOL entity commit and rollback. This is achieved as the WebClient UI attachment component flags the BOL entity as modified by raising event `if_bol_comm~object_modified` when there's an update in the attachments, for example when you upload or delete an attachment. The WebClient UI attachment component delegates the save or cancel to the `CL_ODATA_CV_ATTACHMENT_API` API.

i Note

The automatic synchronization of the save and cancel relies on the entity transaction context. If the commit is executed using the global BOL transaction context, the synchronization doesn't work. This is because the global transaction context doesn't raise event `IF_BOL_TRANSACTION_CONTEXT~ON_TX_ACTION_REQUESTED`, which is required to define the type of action (for example commit or revert). The information provided by the `IF_BOL_TRANSACTION_CONTEXT~ON_TX_ACTION_REQUESTED` event is used in the `IF_BOL_TRANSACTION_CONTEXT~ON_TX_ACTION_FINISHED` event to proceed with the actual saving and canceling of attachments. The application implementation must call the API `CL_ODATA_CV_ATTACHMENT_API` manually at the time of the save or cancel.

Draft Handling

The WebClient UI attachment service component supports the draft feature. The attachments become persistent when saved and they're deleted on cancel.

This functionality is based on reuse libraries in the front-end system. In the hub scenario, where the front-end and the back-end systems are separate, the call to the reuse library from the back-end to the front-end server may be blocked due to **Cross-Origin Resource Sharing (CORS)** settings. If this is the case, you need to maintain the **CORS** setting in the front-end server to allow request from the back-end server in the transaction UCONCOCKPIT.

More Information

- For more information about Attachment Service, see [Attachment Service](#).
- For more information about DMS, go to [SAP Help Portal](#), and see under [SAP S/4HANA](#) > [Product Assistance](#) > [Cross Components](#) > [SAP Enterprise Content Management](#) > [Document Management](#) .
- For more information about GOS, see [Generic Object Services](#).
- For more information about Attachment Service data protection, see the Attachment Service section at [SAP Help Portal](#), and search for the S/4HANA product page. On the **Implement** tab, choose [Security Guide](#) > [SAP S/4HANA Enterprise Business Applications](#) > [R&D / Engineering](#) > [Product Lifecycle Management](#) > [Attachment Service](#) .
- For more information about HDM, see <https://help.sap.com/viewer/DRAFT/36802406aebb4b96b1598246e1d316ee/2021.000/en-US/bc275e43f1b043cc8f97710d5de64e97.html>

UI Integration With Output Control

Use

SAP S/4HANA output control is the central entry point in SAP S/4HANA to output functions such as printing, sending e-mails, or maintaining form templates for the output. WebClient UI offers a reusable component that can be consumed by any WebClient UI application in SAP S/4HANA for UI Integration with Output Control.

SAP and customer-built applications can use this feature to display the output control section in WebClient UI object pages. The data management and data context are handled by the application consuming this component.

Prerequisites

To set up SAP S/4HANA output control, you must complete the tasks listed in [Setup of SAP S/4HANA Output Control](#).

Procedure

1. In the component where you've defined your overview page, create the component usage with the following:

→ Tip

Record the **ID** for the next steps. It will be referred to as Usage ID.

- **ID**: usage ID of your choice.
- **Used Component**: **WCF_GC_OUTPUTC**.
- **Interface View**: **WCF_GC_OUTPUTC/MainWindow**.
- **InboundPlugs/OutboundPlugs**: not required.

2. In the **WD_USAGE_INITIALIZE** method of your component controller, set the binding with the context node from which the values for object type and object key can be resolved.

- **iv_controller_type**: **cl_bsp_wd_controller=>co_type_component** OR **cl_bsp_wd_controller=>co_type_custom**
- **iv_name**: required if you use **cl_bsp_wd_controller=>co_type_custom**
- **iv_target_node_name**: your context node
- **iv_node_2_bind**: **BINDINGCONTEXT**

≡ Code Syntax

```
WHEN 'OutputControl'.
    iv_usage->bind_context_node(
        EXPORTING
            iv_controller_type = cl_bsp_wd_controller=>co_type_component
            iv_target_node_name = 'BTADMINH'
            iv_node_2_bind      = 'BINDINGCONTEXT'
    ).
```

To activate the edit mode, link the Output Control view to the parent view group context. In this way if the overview page is in edit mode the Output Control view will follow: **cl_wcf_output_control_service=>get_instance()->set_usg_view_gp_ctxt_to_parent(iv_usage)**.

≡ Code Syntax

```
WHEN 'CUCRMS40OutputControl'.
    iv_usage->bind_context_node(
        EXPORTING
            iv_controller_type = cl_bsp_wd_controller=>co_type_component
            iv_target_node_name = 'BTADMINH'
            iv_node_2_bind      = 'BINDINGCONTEXT'
    ).
```

```
cl_wcf_output_control_service=>get_instance( )->set_usg_view_gp_ctxt_to_parent( iv_usg
```

i Note

Partial edit for Output Control is not supported.

3. If your bound context nodes are not used to resolve the current entity at runtime, locate your overview page and implement the method `DO_VIEW_INIT_ON_ACTIVATION`. To set the attributes from which values of object type and object key can be resolved, complete the following:

- a. Retrieve the output control node or create it if it does not exist. Use the same usage ID that you used in step 1.
- b. The node object provides attributes `OBJECTTYPE` and `OBJECTKEY` to set the object type and object key respectively. These attributes are required to identify the output documents. The values of these attributes can be a constant value, a binding path (for example: `//STRUCT.ATTRIBUTE_NAME`), or a BOL path. If the value of the attribute is a BOL path, you must use the prefix `bpath` (for example: `bpath:./@FLDATE`)

Code Syntax

```
"output control parameters
DATA lr_oc_node TYPE REF TO cl_wcf_output_control_node.

lr_oc_node ?= cl_wcf_output_control_service=>get_instance( )->get_node( iv_usage_name =
IF lr_oc_node IS NOT BOUND.
    lr_oc_node ?= cl_wcf_output_control_service=>get_instance( )->create_node(
                                iv_usage_name = 'OutputControl'
                                class_name      = 'CL_WCF_OUTPUT_CONTROL_NODE' ).
ENDIF.
CHECK lr_oc_node IS BOUND.

"set object type
lr_oc_node->set_property( iv_attr_name = 'OBJECTTYPE' iv_value = 'SERVICE_CONTRACT' ).

"set object key
lr_oc_node->set_property( iv_attr_name = 'OBJECTKEY' iv_value = 'bpath:./@OBJECT_ID' ).
```

i Note

The WebClient UI uses the bound context node to resolve the current entity at runtime and these paths to resolve the values for the object type and object key. In this case, you do not need to implement the method `DO_VIEW_INIT_ON_ACTIVATION`. You can set the binding parameters in method `WD_USAGE_INITIALIZE`.

4. The applications using the output control can activate the actions **Delete** and **Determine Items**. The application must set the previous actions to **true**.

To activate the actions **Delete** and **Determine Items**, the following properties must be set in the node `lr_oc_node` in the method `DO_VIEW_INIT_ON_ACTIVATION` as described in the step 3.

Code Syntax

```
* Activation of Delete and Determine items Actions
lr_oc_node->set_property( EXPORTING
```

```
iv_attr_name = 'setHideDeleteAction'
iv_value     = 'true' ).
```

```
lr_oc_node->set_property( EXPORTING
    iv_attr_name = 'setDeterminationMergeOption'
    iv_value     = 'true' ).
```

If the business scenario requires to hide the **Send Output** button, for example in edit mode, set the attribute `setHideIssueOutputAction` with value `true` in method `DO_VIEW_INIT_ON_ACTIVATION`.

Code Syntax

```
lr_oc_node->set_property( EXPORTING
    iv_attr_name = 'setHideIssueOutputAction'
    iv_value     = 'true' ).
```

- Next, you add the usage that you created in step 1 to the overview page view area (runtime repository) and adjust the page configuration.

This functionality is based on reuse libraries in the frontend system. In the hub scenario, where the frontend and the backend systems are separate, the call to the reuse library from the backend to the frontend server may be blocked due to **Cross-Origin Resource Sharing (CORS)** settings. If this is the case, you need to maintain the **CORS** setting in the frontend server to allow request from the backend server in the transaction `UCONCOCKPIT`.

More Information

- For more information, see [SAP S/4HANA Output Control](#).
- For more information about Data Protection and Privacy in Output Control, see [Blocking of Receiver Data](#).
- For more information about UCON Cockpit, see [Using the UCON Cockpit](#).

UI Integration With Situation Handling

Use

Situation Handling is a concept for bringing business issues to the attention of specific user groups. It helps the user to recognize, understand, and resolve the situation by gathering all relevant information and recommending solution proposals. This feature enables you to integrate and use Situation Handling service with WebClient UI object pages in SAP S/4HANA. The users can view situation message strip in the object page and take necessary actions such as dismiss the situation. You can use this feature when you're using WebClient UI applications in SAP Fiori launchpad integrated mode and standalone mode.

Prerequisites

You must have completed Situation Handling framework set up and configuration.

Procedure

The situations are linked to your application's object page document through the anchor object. The anchor object can be identified with the following two fields:

- **anchorObjectKey:** Must be bound to the unique ID of the business object displayed by the object page.
- **anchorObjectType:** The name of your **Anchor Object Key** as specified in the situation. The **Anchor Object Key** and type can be found in the Monitor Situations app.

Go to your overview page and call method **SET_ANCHOR_OBJECT** of class **CL_WCF_SITUATION_HANDLING** in the method **DO_PREPARE_OUTPUT**. Pass the parameters **iv_anchor_object_key** and **iv_anchor_object_type**.

Code Syntax

```
*Situation integration
TRY.
    lv_objkey = me->typed_context->btadminh->collection_wrapper->get_current( )->get_property(
        'KEY' ).

    CHECK lv_objkey IS NOT INITIAL.

    cl_wcf_situation_handling=>set_anchor_object(
        EXPORTING
            iv_anchor_object_key = lv_objkey
            iv_anchor_object_type = 'ServiceContractUUID' ).
CATCH cx_root.
ENDTRY.
```

This functionality is based on reuse libraries in the frontend system. In the hub scenario, where the frontend and the backend systems are separate, the call to the reuse library from the backend to the frontend server may be blocked due to **Cross-Origin Resource Sharing (CORS)** settings. If this is the case, you need to maintain the **CORS** setting in the frontend server to allow request from the backend server in the transaction **UCONCOCKPIT**.

In addition, for the hub scenarios you must activate the OData service **C_SITNINSTANCETP_CDS** with transaction **/IWFND/MAINT_SERVICE, Activate and Maintain Services** on the backend server and the frontend server. For more information on OData service activation, see SAP Help Portal at <http://help.sap.com> → **SAP Gateway Foundation (SAP_GWFND)** → **SAP Gateway Foundation Developer Guide** → **OData Channel** → **Basic Features** → **Service Life-Cycle** → **Activate and Maintain Services**.

More Information

- For more information, see [Situation Handling](#).
- For more information about UCON Cockpit, see [Using the UCON Cockpit](#).

Situation Indicator and Popover in Tables

Use

The situation indicator and popover are displayed in the table when there is a situation in an item or object. User can see the details of the situation in popover and can navigate to the object.

Prerequisites

You have a column in the table that contains a number of situations.

Procedure

To display the alert icon, implement the P-getter of the situation attribute in the table context node.

Pass the `iv_property` to static method `cl_wcf_situation_handling=>handle_table_p_getter`.

Code Syntax

```
METHOD get_p_sitn_num_of_instances.

    rv_value = cl_wcf_situation_handling=>handle_table_p_getter( iv_property = iv_property ).

ENDMETHOD.
```

To display the button and popover, you must enhance the above code by passing the anchor object key and anchor object type to get the details of the situation.

Code Syntax

```
METHOD get_p_sitn_num_of_instances.

    rv_value = cl_wcf_situation_handling=>handle_table_p_getter(
        EXPORTING
            iv_anchor_object_key = lv_anchor_object_key "Situation object key
            iv_anchor_object_type = 'ServiceContractItemUUID' "Situation anchor object type
            iv_property = iv_property
            iv_display_mode = iv_display_mode
    ).

ENDMETHOD.
```

With the [Show Details](#) button, the navigation is triggered via SAP Fiori launchpad. In some scenarios, if you want to navigate to a WebClient UI application instead of SAP Fiori launchpad navigation, a mapping can be implemented from intent to outbound plug. This can be done in the `do_prepare_output` method of the implementation class.

Code Syntax

```
cl_wcf_situation_handling=>map_ibn_to_wcf_op(
    EXPORTING
        iv_sitnsemanticobject = 'ServiceContract'
        iv_sitnsemanticobjectaction = 'display'
        iv_outbound_plug = 'TO_ITEM_DISPLAY'
).
```

Search Functions

You can use the central search, the advanced search, and the simple search to search for business objects in the WebClient UI. You can use the search page wizard to create search pages.

Central Search

Use

You can use the central search to search for business objects the WebClient UI. The central search fields are displayed when you choose the search icon. You can always access the search fields with this icon, regardless of the business object that you are processing.

In the search menu you can choose between a search for all business objects, saved searches, or a search for individual business objects.

Features

The option **Saved Searches** is the default search option for the central search. Below it in the menu you see **All Objects** and then follows a hierarchical list of business objects, in alphabetical order. When you have selected a saved search, the search is automatically performed.

Saved Searches

When you select **Saved Searches**, the search menu contains all your saved searches, and the field label changes to **Saved Searches**.

i Note

The WebClient UI Framework displays the data that the application has set in the search functionality. There are no controls in place to ensure that this data does not contain sensitive personal information. You can create, modify, and delete saved searches. Use these functionalities to ensure that the pre-filled search fields in Saved Searches are up-to-date and follow data protection regulations.

All Objects (Cross-Object Search)

When you select **All Objects**, the field label changes to **Search for All Objects**. You can now search across all business objects that belong to your business role. This search function behind this search technique is "simple search" (see [Simple Search](#)).

i Note

The **All Objects** entry only appears if more than one searchable business object has been registered in the central search Customizing.

Business Objects (Object-Specific Search)

When you position the cursor over a business object in the menu, a cascading menu with the search attributes assigned to that business object opens.

Search Using Business Object Attributes

After you have selected a search attribute, you can enter the search value in the input field. The field label changes accordingly, for example, to [Accounts/Category](#).

The search function behind these search techniques is “advanced search” (see [Advanced Search](#)).

Free Text Search

You can perform a free text search for a business object by selecting **Free Text**. If **Free Text** is the only search attribute of a business object, it is omitted.

This search function behind this search technique is “simple search” (see [Simple Search](#)).

i Note

To improve the performance when you start searching for a business object, the table for the search objects displayed in the search menu is cached in a server cookie. This cache depends on the business role and language. Therefore, Customizing changes that have been made for the central search might not be immediately visible in the WebClient UI.

More Information

[Configuring the Central Search](#)

Configuring the Central Search

Making Business Objects Available in the Central Search

For every business object that is to be supported by the central search, make the necessary settings in Customizing for **UI Framework** under **►Technical Role Definition ► Define Central Search ►**.

Making Business Objects Invisible in the Central Search

If you want to make business objects invisible at business role level in the central search, make the necessary settings in Customizing for **UI Framework** under **►Business Roles ► Define Business Role ►**.

Defining Search Criteria

If you want to include or exclude search criteria from the search page of a business object, you can activate or deactivate the corresponding indicator in the UI configuration of the selected application component and view. For more information, see [Search Pages](#).

❖ Example

To configure the Account search criteria in the central search of your business role, start the UI configuration tool and select application component BP_HEAD_SEARCH and view **MainSearch**. Activate the indicator **Use in Central Search** for the search criteria that you want to be displayed in the central search.

Adjusting Entries in the Central Search Menu

You can change entries in the central search menu in the Online Text Repository (OTR). If you want to change these entries, start transaction SOTR_EDIT for the OTR. Choose the correct language, search for the OTR text by entering the OTR ID in the **Alias**

field, and change the SAP standard text. The following entries are available in the central search:

OTR ID	SAP Standard Text	Description
CRM_CENTRAL_SEARCH/SEARCH_FOR_ALL_OBJECTS	Search for All Objects	Field label that appears when the user clicks All Objects in the search menu
CRM_CENTRAL_SEARCH/SEARCH_FOR_OBJECT	Search for &1	Field label that appears when the user selects a business object in the search menu; &1 is replaced by the selected business object.
CRM_CENTRAL_SEARCH/SEARCH_IN_ALL_FIELDS	Free Text	Entry in the search menu that allows the user to enter free text in the search field; this entry is used by the simple search.
CRM_CENTRAL_SEARCH/ALL_OBJECTS	All Objects	Entry in the search menu that starts the search across all business objects
CRM_CENTRAL_SEARCH/SEARCH_FOR_OBJECT_BY	Search for &1 by &2	Field label that appears when the user selects a business object and a search attribute in the search menu; &1 is replaced by the business object and &2 is replaced by the search attribute.

Advanced Search

Use

You can use the advanced search in the WebClient UI to search for business objects.

Features

On the search page, you can define the search criteria and start the search. You have a variety of search criteria and search operators with which you can refine your search to find precisely the data you are looking for. In the result list, you can see the data that matches your search criteria.

You can show and hide the search criteria, by clicking the corresponding hyperlinks. You can add or remove search rows, by choosing the icons next to the search row. You can enter the maximum number of business objects that should be in the result list of this search.

You can save your current search, which can be made available as saved search in the central search.

You can define default search criteria in the personalization of an advanced search page. For more information, see [Advanced Search Personalization](#).

To adapt the result list to your personal requirements, you can make the necessary entries in the personalization dialog box.

Search Criteria Displayed as Read-Only

To indicate that certain search criteria that have an impact on the search result may not be changed, certain search criteria can be displayed as read-only. This cannot be modified by users. Plus or minus buttons to add or delete a row in the search criteria are not available.

i Note

In the SAP standard system, the search criteria are not displayed as read-only. This feature is used only in special cases.

Mandatory Use of the Input Help

To ensure that certain search fields are only filled by using the input help, the use of the input help in these fields is mandatory. In this case, the look and feel of the search field is as if it is read-only, and users cannot manually place the cursor focus on the search field.

If users click the search field or the input help icon, the input help is automatically started. If users press **ENTER** in the search field, the input help is also started, instead of executing the advanced search. When users tab to the search field, the focus lies on the search field and the input help icon, because they are one entity. In this case, the input help can only be started by pressing **ENTER**. If the input help is closed and the focus is still on the input help icon, pressing **ENTER** restarts the input help, instead of executing the advanced search.

i Note

In the SAP standard system, the use of the input help is not mandatory. This feature is used only in special cases.

❖ Example

You are logged on to the system as sales professional: You start the account search page or the contact search page and select search attribute **Marketing Attribute**. You can now use the input help in the search field.

You start the account search page and select search attribute **Account Classification**. You can now use the input help in the search field.

Pasting of Multiple Values as Search Criteria

You can copy a list of values from another application, such as spreadsheet or text processing application, and paste them as search criteria in an advanced search by pressing **CTRL** + **V**. The values are handled as separate search criteria.

In the source application, each value should be in a new row separated by tab or new line. Non-adjacent row selection support depends on the source. After pasting the search criteria into the search, they are shown as separate search criteria in individual rows.

You make the settings for this function in Customizing for **UI Framework** under **Technical Role Definition > Define Parameters**. By default, the system limits the number of rows that can be pasted at once to 200 to avoid accidental pasting of an overly large amount of data. You can override this default maximum number by adding parameter **SEARCH_MAX_ROW_PASTE** in function profile **PARAMETERS** of the relevant business role.

Groups of Rows

For usability reasons, when a large number of search criteria rows have been created by pasting multiple search criteria, successive rows that have the same attribute and the same operator are grouped. These rows are collapsed and appear with an arrow icon pointing to the right of the root row. A row group that results from pasting is by default collapsed and shown in display mode. To edit the rows that were created, you have to expand the row group. With **- (Remove Line)**, you can delete all rows of the group. With **+ (Copy Line)**, you can expand the group and add a new row at the bottom of the expanded group. By default, the rows are grouped if there are more than two rows of search criteria that have the same attribute and operator.

More Information

[Central Search](#)

[Table Personalization](#)

[Advanced Search Pages with Fiori Layout](#)

Relative Dates

A relative date is a date or date range that is relative to the current date such as yesterday, last week, or next month. You can use relative dates for date type criteria on advanced search pages.

Use

On advanced search pages, you can use relative dates for date type criteria. A relative date is a date or date range relative to the current date such as yesterday, last week, last year, next month or during the next three quarters.

Prerequisites

- You or your Administrator has defined relative dates and associated them to [Dynamic Queries](#) using the [Define Relative Dates](#) Customizing activity available from [UI Framework](#) > [Generic Interaction Layer/Object Layer](#) > [Component Specific Settings](#).
- You or your Administrator has created relative dates for the default handler using the Maintain BRFplus Settings for the [Relative Dates Default Handler](#) Customizing available from [UI Framework](#) > [Generic Interaction Layer/Object Layer](#) > [Component Specific Settings](#).
- You or your Administrator may have used the [BAdI : Relative Dates Handler](#) in Customizing available from [UI Framework](#) > [Generic Interaction Layer/Object Layer](#) > [Component Specific Settings](#) > [Business Add-Ins \(BAdIs\)](#) to create a component handler to define relative dates.

Features

If your system is customized to use relative dates, the relative date section of the date picker will only be visible when the search criteria operator on the advanced search page is set to [is](#).

- You can either choose the relative date value from the date picker or enter it in the [Date](#) field.
- Since relative dates supports type ahead functionality, suggestions will be provided while you are typing within the [Date](#) field.
- You can enter the key (for example: WCF_TODAY) or the value (for example: Today) in the [Date](#) field. If you have used the key, the value will be displayed after the round trip is completed. Note that the key and value are not case sensitive.
- You can enter a specific date in the [Date](#) field.

Advanced Search Pages with Fiori Layout

Use

The WebClient UI advanced search pages have a user experience consistent with the Fiori layout. The concept of search variants is used to access saved searches. The search criteria view uses tokens. You can pin and unpin as well as expand and collapse the header content.

Features

- **Search Variants:** Search variants allow the user to select, create, update, and delete variants for filter settings and control parameters such as layout, table column visibility, sorting, and filtering.
- **Search Criteria:** In the search criteria section, you can use the input fields and operators to create tokens as your search criteria. The classic search criteria and maximum number of hits are available under the [Adapt Filters](#) dialog.
- **Pin and Unpin:** The pin and unpin button is directly below the header, which is used to pin and unpin the header content. Using the pin feature keeps the header expanded and pinned to the top while you scroll through the page content. This mode remains fixed until you use the pin button again or trigger a snap on click.
- **Expand and Collapse:** The expand and collapse button is located directly below the header, which is used to expand and collapse the header content. When the pin is active, the expand and collapse header button is still available and overrules the pin. Alternatively, you can also expand or collapse the header by selecting the header title bar.
- **App State:** In the Fiori launchpad integrated mode, the search criteria and values are saved in the Fiori launchpad app state and can be shared with other users using the URL or the [Share Menu](#) in the advanced search toolbar.

Activities

- **Search Variants:** When you activate this feature, it will appear on the advanced search pages. You can use the variant list to swap between your search variants, previously known as saved searches. When you save a new variant, the search fields, values, and result list layout will be saved. When you change your search criteria, an asterisk (*) will appear on your variant name, and you can re-save it, or save it as a new one.
- **Manage Variants:** In this dialog, you can do the following actions:
 - **Favorite**
Favorite variants will appear in the variant list on search pages. If you un-favorite a variant, it will be removed from this list, but it will still be available in other areas, like saved searches. New variants are automatically favorited.
 - **Rename**
You can change the name of the variant in the [View](#) column.
 - **Apply Automatically**
When this is selected, the search will execute automatically when you select the variant.
 - **Default**
You may select a variant to be your default, which will be the one loaded when you open this search page.
 - **Delete**
You can delete variants. The [Standard](#) view cannot be deleted.
- **Search Criteria Input:** In the search criteria, you can select or type values to the search criteria. After you tab out or press enter, your search criteria will be turned into a token. This token will display your search values with the operator. You can easily click on the **X** of the token to remove it from your search. Choose [Adapt Filters](#) to see a more detailed view of the search criteria. Choose [Go](#) to execute your search.
- **Search Criteria Adapt Filters:** In this dialog, you can do the following:

- View the classic search view with the displayed operators.
- Add new search criteria items using the dropdown list box.
- Add multiple rows of the same search criteria with the **Copy Line** button.
- Remove criteria with the **Remove Line** button.
- Clear all the values with the **Clear** button.
- Change the maximum number of results shown in the search results list.
- **Search Criteria Tokens:** You may specify operators in the input fields as follows:

Operator	Pattern
is / equal to	= [value] (default when no operator is typed)
contains	* [value] *
between	[value1] ... [value2]
starts with	[value] *
ends with	* [value]
less than	< [value]
less than or equal to	<= [value]
greater than	> [value]
greater than or equal to	>= [value]

Activation

To enable or disable advanced search pages with Fiori layout, use the function profile parameter `SEARCH_LAYOUT= ' FIORI '` (enabled) or `' CLASSIC '` (disabled).

By default, the advanced search pages with Fiori layout are enabled.

Limitations

The search criteria view is only available in advanced search pages. It is not available in dialog.

Public variants are not supported.

Simple Search

Use

You can use the simple search to search for business objects in the WebClient UI using keywords, IDs, or free text, which you enter in a single search field.

Integration

The simple search can be integrated in the UI as follows:

- In the central search (see [Central Search](#)).
- In the search pages for the business objects.

On search pages, users can switch easily between the simple search and the advanced search.

To use the simple search in the search pages, applications must have the corresponding business function active.

Prerequisites

To use simple search you need to configure Enterprise Search (see [Enterprise Search Integration](#)).

To make the simple search available in the **central search** of the WebClient UI, you have made the following settings in Customizing for **UI Framework**:

- You have registered your business objects for the central search under **Technical Role Definition > Define Central Search**.
- For the relevant roles, you have activated your business objects for the central search under **Business Roles > Define Business Role**.

To make the simple search available in the **master data search pages**, you make settings under **UI Framework > Enterprise Search Integration > Define Availability of Simple Search in Business Object Search**. For information about assigning the settings to a business role, see the documentation for the Customizing activity.

Features

Free Text Search

You can enter keywords, IDs, or free text in the easily accessible search field, and search across all business objects that belong to a business role. The search is executed in all attributes that belong to a business object.

Search for IDs

If you search for an ID, a wildcard character asterisk (*) is automatically added as a prefix to the ID. This applies to the free text search and the custom filter in result lists on the search result page. The asterisk is necessary because all IDs are automatically saved with leading zeros in the enterprise search index. For example, if you enter 2349 in the search field, business objects with ID 00002349 are displayed in the result lists. For more information about use cases for simple search, see your application specific documentation.

Activities

Using Simple Search in the Central Search

To start the simple search, you select **All Objects** in the search menu of the central search. Or you select the search attribute **Free Text** of an individual business object in the search menu and enter free text in the search field. The results list is a simple list without further functions.

Using Simple Search in the Master Data Search Pages

To start the simple search on a master data search page, either the field is visible by default or you toggle from the advanced search, depending on Customizing. The results list offers the same functions as are available as for the advanced search results list for the relevant object.

Enterprise Search Integration

Use

The WebClient UI has been integrated with Enterprise Search (ABAP). The main goal of the Enterprise Search integration is to provide the WebClient UI with a search function based on keywords and free text. This means that you can enter keywords, IDs, or free text in the search field, and search across all business objects that belong to a business role. This search function is named simple search.

Integration

The simple search is based on Enterprise Search (ABAP) or Embedded Search, and uses the complex functions of this search engine, such as full indexing or incremental indexing. Embedded Search is the Enterprise Search component in the Application Server (AS) ABAP.

Features

Enabled Business Objects

You can find the business objects that have been enabled for Enterprise Search (ABAP) in your business role in Customizing for **UI Framework** under **Business Roles > Define Business Role**. Select your business role and choose **Adjust Central Search Objects**. All business objects with object action **Product Enterprise Search** are enabled and delivered in the standard system. They are available in the central search. For more information about the central search, see [Central Search](#).

More Information

For more information about Enterprise Search (ABAP), see SAP Help Portal **S/4HANA > Enterprise Search > Enterprise Search**.

Enterprise Search Modeling Workbench

Use

You can use the ES modeling workbench to connect business objects to Enterprise Search (ABAP). You can use the workbench to model templates for business objects and technical objects. These templates are used by Enterprise Search (ABAP) to search for business objects.

Integration

You can find the ES modeling workbench in SAP Easy Access Menu for **CRM ES Modeling Workbench** at **Service > Architecture and Technology > Enterprise Search Integration > Tools**.

Prerequisites

Before you start working with the ES modeling workbench, you need to make the necessary settings in Customizing for [UI Framework](#) under  [Enterprise Search Integration](#).

Features

Technical objects are mainly templates based on text tables, which are used to extract and index the language-dependent text maintained in these tables. If the user enters, for example, a keyword or free text in the search field, the technical objects are searched for this keyword or the free text.

The structure of the templates in the ES modeling workbench corresponds to a tree, which is based on the Business Object Layer (BOL) Model, associated to the BOL root object as defined in the template header. This tree structure is used to maintain these templates in dedicated detail screens on node, relation, and attribute level.

In templates delivered by SAP, the search fields are a subset of the search attributes that are defined in the advanced search of a business object. You can find all search attributes of a business object, in the corresponding advanced search object in the BOL Model Browser (transaction GENIL_MODEL_BROWSER).

Foreign Relation Support

You can model foreign relations to templates in the ES modeling workbench. For every foreign relation, you can define an individual set of search attributes and search result attributes from the referenced foreign template. Search criteria and search result fields can be adapted to specific search use cases.

Complete Attribute Overview

You can display all selected attributes of a business object template in the ES modeling workbench. This information is helpful when you are modeling a business object template, and want to see a list of all attributes that are relevant for the search. You can export this list using different file formats, for example, a spreadsheet.

Activities

Specifying the Search Model

Before you can start modeling your business objects and technical objects in the ES modeling workbench, you can specify the search model. This means that you can define the attributes that are used in the search request and in the result list of a business object.

Modeling the Search Template

You start the modeling process in the ES modeling workbench by creating a new template set, which corresponds to a component set in the BOL model browser. A template set contains two types of templates, which are business templates and technical templates.

In the modeling process you maintain the BOL nodes, relationships, and the relevant fields of your business object. At attribute level, you can maintain the following information:

- Key indicator
 - Specifies fields that can be used as key fields in the relations of a business object
- Search criteria
 - Specifies attributes that are used when users search for a business object
- Result

Specifies attributes that are available in the search result of a business object

- Select

Specifies fields that are extracted for Enterprise Search

- Text

Specifies fields that contain human-readable text

- Language code

Specifies fields that are language-dependent

- Template ID

Specifies the template ID

Transferring the Search Template to Enterprise Search (ABAP)

After you have modeled your business object templates in this workbench, you need to transfer them to Enterprise Search (ABAP).

Working With Templates

Use

When you are working with the ES Modeling Workbench, you can either copy existing SAP templates to your namespace and then adapt these templates to your needs, or you can create new templates.

Procedure

Copying Existing Templates

If you copy existing SAP templates, proceed as follows:

1. Copy the SAP template to your namespace.
2. Make the necessary changes.
Adjust the required search request attributes and the required result list attributes.
3. Save your template.
4. Check if the template is consistent.
5. Transfer your search template to Enterprise Search (ABAP).

Creating New Templates

If you create new templates, proceed as follows:

1. Create a new template.
Define the required search request attributes and the required result list attributes in the workbench.
2. Maintain your search model.

3. Save your template.
4. Check if the template is consistent.
5. Transfer your search template to Enterprise Search (ABAP).

Configuration and Administration Tools

Use

The following configuration and administration tools are available:

Tools

- Enterprise Search Modeling Workbench
- Enterprise Search Administration Cockpit

Utilities

- Structure Generation for Search Queries
- Client Copy Utility of Templates
- Transport Utility of Templates
- Model Transfer Utility of Templates

Report

- Report to synchronize Enterprise Search templates and templates

You can find the above tools, utilities, and report on the [SAP Easy Access](#) screen, under [Service](#) [Architecture and Technology](#) [Enterprise Search Integration](#).

Prerequisites

If you want to use the Enterprise Search Administration Cockpit, you need the authorization role `SAP_ESH_CR_ADMIN`, which is necessary for administration and monitoring in this tool.

If you want to define technical and application-specific Customizing settings, you need authorization role `SAP_WCF_ES`. This role contains authorization object `S_WCF_ES`, which controls access to Enterprise Search.

For more information about authorization roles, see Customizing for [UI Framework](#) under [Business Roles](#) [Define Authorization Role](#).

More Information

Topic	Source of Information
Tools, utilities, and template synchronization report	Selection screens of each individual tool, utility, and report, under Program Documentation .
Enterprise Search Modeling Workbench	Enterprise Search Modeling Workbench

Topic	Source of Information
Enterprise Search (ABAP)	For more information about Enterprise Search (ABAP), see SAP Help Portal ► S/4HANA ► Enterprise Search ► Enterprise Search ►

Search Page Wizard

Use

You can use the search page wizard to create search pages for the WebClient UI. You can create more than one search page for a UI component.

The search page wizard is integrated into the UI component workbench and guides the user through a series of steps.

After the wizard has finished the generation, you need to configure which fields are to be displayed in the search query views and search result views in the WebClient UI.

Features

The search page wizard generates a search query view, a search result view, and a view set. These views are based on a common Business Server Page (BSP) layout.

The search page wizard offers the following functions:

- View set
 - Hiding of search query options
- Search query view
 - Search button
 - Clear button
 - Saved searches (optional)
- Search result view
 - Selection mode: single selection, and multiselection (optional)
 - Navigation links for fields in the result list (optional)
 - Buttons (optional)

Business Roles

Use

You can use business roles to work with content from different applications. You can combine this content according to your personal requirements.

Features

Business Roles in the WebClient UI

You can perform the following activities in the business role application:

- Search for business roles, create new business roles or copy existing ones, and delete business roles
- Start transaction PFCG to assign authorization roles (PFCG roles) to business roles by clicking [Authorization Roles](#)
- Choose or create a transport request, after you have changed or deleted an existing business role, or created a new business role
- Personalize the overview page, including the assignment blocks of your business role, by clicking the [Personalize](#) icon
- Export the content of the individual assignment blocks to Microsoft Excel
- Translate the business role description, and titles of direct link groups, work centers, and logical links into other languages by clicking [Translate](#)
- Assign icons to work centers and to logical links in direct link groups

Where Do I Customize Business Roles?

Use

You can use the business role Customizing to tailor the navigation frame of your business roles to your personal requirements.

Prerequisites

To start the business role Customizing in the WebClient, you need to include a link in the navigation bar of your business role. You have assigned the logical link CT - BR - SR to the work center link group CT - ADM - SR, in Customizing for [UI Framework](#) under [▶ Technical Role Definition ▶ Define Navigation Bar Profile ▶](#).

Features

You can customize your business roles in SAP GUI and in the WebClient UI.

Customizing in SAP GUI

You access the business role Customizing in SAP GUI, by choosing [▶ UI Framework ▶ Technical Role Definition ▶](#) and [▶ UI Framework ▶ Business Roles ▶](#).

Customizing in the WebClient UI

You access the business role Customizing in the WebClient by clicking a link in the navigation bar of your business role. The business role application consists of a search page and an overview page. The overview page contains the header block [Business Role Details](#), with the [General Data](#) and [Profiles](#) areas. In the [General Data](#) area you find general information, such as business role ID and description, role configuration key, and the assigned authorization role (PFCG role). In the [Profiles](#) area, you see all profiles that are assigned to your business role, such as navigation bar profile, layout profile, and technical profile. Under the header block you can see the following assignment blocks on the overview page:

- Navigation Bar
- Direct Link Groups
- Keyboard Shortcuts

- Central Search
- Function Profiles

In the header block of your business role you can change the business role details. In the assignment block you can customize the side panel, assign function profiles, and adjust central search objects.

More Information

For more information about the central search, see [Central Search](#).

How Do I Customize Business Roles?

Use

The following document describes the business role Customizing in the WebClient.

Prerequisites

Before you start your Customizing activities, you have created a navigation bar profile that you want to use in your business role. The navigation bar profile contains a collection of work centers, direct link groups, logical links, and central search objects. It can be assigned to a business role. You have created the navigation bar profile for your business role in Customizing for **UI Framework**, by choosing ► **Technical Role Definition** ► **Define Navigation Bar Profile** ►.

Activities

You can perform the following Customizing activities in the business role application in the WebClient UI.

Maintaining Business Roles

The following functions are available for maintaining business roles in the WebClient:

Searching for Business Roles

After you have started the business role Customizing you navigate to the business role search page. Here you can enter your search criteria and start the search. All business roles that correspond to the search criteria are displayed in the result list.

Displaying Business Roles

You can select a business role in the result list by clicking the technical name in the **Business Role** column. You navigate to the overview page of your selected business role. Here you can see the most important information about your business role, at a glance.

Changing Business Roles

You can change the business role details in the header block by clicking the **Edit** icon. You can change the information in every assignment block by clicking **Edit List**. You can use the **Clean Up** button to delete rows in which the indicator **Deleted** is set. This means that the corresponding rows are not part of the underlying navigation bar profile any longer, and that these entries are no longer relevant for this business role. You can use the **Translate** button to translate titles into other languages, for example work center titles in the navigation bar. Save your changes after you have finished changing your business role.

Copying Business Roles

You can copy a business role by clicking the **Copy** icon either on the search page at the top of the result list or on the overview page. You are asked to confirm that you want to copy all dependent entries of the current business role to the new business role.

Creating New Business Roles

You can create a new business role by clicking **New** either on the search page at the top of the result list, or on the overview page. You navigate to a page that contains the **Business Role Details** header block in which you need to enter business role ID, authorization role ID, and navigation bar profile, which are mandatory fields. The content in the assignment blocks depends on the navigation bar profile that you have assigned to your business role in the header block. Save your new business role.

Deleting Business Roles

You can delete an existing business role by clicking the **Delete** icon either on the search page at the top of the result list, or on the overview page. You can delete more than one business role at once on the overview page.

Adjusting Work Centers and Logical Links

In the assignment block **Navigation Bar**, you can see the work centers that are displayed in the navigation bar of your business role. The work centers are displayed at the first level and the corresponding logical links that belong to a work center are displayed at the second level, in the assignment block. The work centers are displayed in the order defined by the navigation bar profile.

Adjusting Work Centers

You can set the indicator **Visible in Menu** to active, if a work center is to be displayed in the navigation bar of your business role. Additionally, you can change the work center title in the **Title** field. You can change the order of the work centers by using **Up** and **Down**.

Adjusting Logical Links

You can set the indicator **Visible in Work Center** to active, if a logical link is to be displayed on the corresponding work center page. Additionally, you can change the logical link title and the order of logical links on a work center page.

Defining the Sorting of Logical Links

The sorting of logical links on work center pages is in alphabetical order. If you want to change the sorting, you can do so in Customizing for **UI Framework** under **Technical Role Definition > Define Parameters**.

Enter a profile, for example PARAMETERS, and assign the parameter WC_LINK_ORDER, which defines the sorting of logical links on work center pages, to that profile. Enter a value of your choice for the parameter WC_LINK_ORDER. For more information about the available values, see the above-mentioned Customizing. Finally, you need to assign the profile, for example PARAMETERS, to your business role in the assignment block **Function Profiles**.

Adjusting Direct Link Groups and Logical Links

In the assignment block **Direct Link Groups** you can see the direct link groups that are displayed in the navigation bar of your business role. The direct link groups are displayed at the first level and the logical links that belong to a direct link group are displayed at the second level, in the assignment block. The direct link groups are displayed in the order defined by the navigation bar profile.

Adjusting Direct Link Groups

You can set the indicator **Visible** to active, if a direct link group is to be displayed in the navigation bar of your business role. Additionally, you can change the direct link group title in the **Title** field. You can change the order of direct link groups by using **Up** and **Down**, and assign icons that are displayed in front of the link group title in the navigation bar of your business role.

Adjusting Logical Links

You can set the indicator **Visible** to active, if a logical link is to be displayed in the corresponding direct link group. Additionally, you can change the logical link title and the order of the logical links in a work center. You can assign icons that are displayed in front of the logical link in the navigation bar of your business role.

Defining Keyboard Shortcuts

In the assignment block **Keyboard Shortcuts**, you can define shortcuts for navigation operations and focus operations. Navigation operations navigate to a view or component, using a logical link. Focus operations set the focus to an area on the user interface.

In SAP GUI you define keyboard shortcuts in Customizing for **Customer Relationship Management** under ► **Business Roles** ► **Define Business Role** ► **Define Keyboard Shortcuts** ►.

Making Business Objects Invisible in the Central Search

In the assignment block **Central Search** you can make business objects invisible in the central search.

You can make business objects invisible in the central search in Customizing for **UI Framework** under ► **Business Roles** ► **Define Business Role** ► **Adjust Central Search Objects** ►.

Assigning Function Profiles

In the assignment block **Function Profiles** you can assign function profiles to your business role.

You can assign function profiles to your business role, in Customizing for **UI Framework** under ► **Business Roles** ► **Define Business Role** ► **Assign Function Profiles** ►.

Defining Function Profiles

You can find all available function profiles that you can assign to your business role, in Customizing for **UI Framework** under ► **Technical Role Definition** ► **Define Function Profile** ►.

More Information

For more information about the central search, see [Central Search](#).

Defining the Organizational Assignment

Use

Before you can start working with your business role, you need to assign your business role to a node, either an organizational unit or a position, in the organizational model. All users who are assigned to that node can access the assigned business role.

Activities

You can perform both assignments in the business role Customizing in SAP GUI or in the WebClient UI.

Assigning a Business Role to an Organizational Unit

You assign your business role to an organizational unit or a user in the organizational model, which is available in transaction PPOMA_CRM. The user needs to be assigned to the same organizational unit.

- In the WebClient UI, you can start the organizational model in the navigation bar of your business role via the logical link MD - ORG - SR.
- In SAP GUI, you define the organizational assignment in Customizing for **UI Framework**, by choosing **Business Roles** > **Define Organizational Assignment** .

Assigning Authorization Roles

Use

Before you can start working with your business role, you need to assign an authorization role (PFCG role) to it. If you are using SAP Fiori launchpad integrated mode, see [Integration with SAP Fiori Launchpad](#).

Prerequisites

To maintain authorization roles (PFCG roles) in the WebClient UI, you have set up the transaction launcher Customizing for the transaction PFCG. Start transaction CRMS_IC_CROSS_SYS in Customizing and create a new entry:

- Mapped logical system: OWNLOGSYS
- Logical system: *<logical system>*
- ITS client: *<ITS client>*
- ITS URL: /sap/bc/gui/sap/its/webgui/!?
~transaction=IC_LTX&~okcode=ICEXECUTE"http://<server>:
<port>/sap/bc/gui/sap/its/webgui/!~transaction=IC_LTX&~okcode=ICEXECUTE

i Note

<server> means the SAP server.

<port> means the port that is used for the HTTP protocol.

Both settings can be reviewed in transaction SMICM under **Goto** > **Services** .

For more information, see SAP Note [888931](#) .

Activities

Assigning an Authorization Role to a Business Role

The authorization role contains a certain authorization profile (PFCG profile) that defines the necessary authorizations. You assign this authorization role to your business role. You maintain authorizations in transaction PFCG.

Authorization roles have been enhanced with framework-specific authorizations that need to be added to existing authorization roles. Authorizations that you need to run the framework are part of the authorization role SAP_CRM_UIU_FRAMEWORK.

- In the WebClient UI, you can start the role maintenance in the business role overview page by clicking **Authorization Roles**.
- In SAP GUI, you define authorization roles for business roles in Customizing for **UI Framework** under **Business Roles** > **Define Authorization Role** .

Integration with SAP Fiori Launchpad

In addition to the standalone mode, WebClient UI Framework offers SAP Fiori launchpad integrated mode. In the SAP Fiori launchpad, you can create business roles and directly access WebClient UI applications in the integrated mode. All the WebClient UI applications needed for your business role are available in the SAP Fiori launchpad, which is the single point of entry.

Integration

Use

In the SAP Fiori launchpad integrated mode, you can directly access WebClient UI applications using tiles in SAP Fiori launchpad. You can also create front-end business roles.

Features

- SAP Fiori launchpad integrated mode can be used for applications that do not have dependencies on the L-shape of the classic WebClient UI. The L-shape is not displayed in this mode. Only the work area is displayed within the SAP Fiori launchpad shell.
- WebClient UI applications are embedded in the SAP Fiori launchpad shell.
- The saved searches and WebClient UI settings are available in the dashboard, which can be accessed with the dashboard icon in the system toolbar.
- The role configuration key **<DEFAULT>** is always used in SAP Fiori launchpad integrated mode.
- The back-end business role and related customizations are not applicable.
- **Help** link is available from the shell toolbar.

The following features are only supported in the standalone mode and not in the SAP Fiori launchpad integrated mode:

- **Apps** menu
- Navigation history and forward button
- Quick **Create** menu
- Direct link group
- **Recent Items**
- **Home** page
- **System News**
- Business role text
- Role selection
- System header

Procedure

Use SAP tools to set up SAP Fiori launchpad content as described in SAP Note [3170196](#) .

The WebClient application ID is the **Target ID** defined in Customizing for ► **UI Framework** ► **Technical Role Definition** ► **Define Work Area Component Repository** ►. The target ID uniquely identifies the component, window, and inbound plug for the navigation. The target ID corresponds to the unique combination of UI object type and object action.

When you assign catalogs with WCF Apps to the PFCG Role, the UIU services corresponding to the target IDs are automatically assigned to the catalog in the PFCG Role menu.

In addition to the target IDs in the catalog, the framework services required for the WCF Apps are added.

The PFCG integration for the WCF Apps is provided in SAP Note [2817664](#) ►.

More Information

For more information, see [Setting Up Launchpad Content](#).

Function Profiles

Use

With WebClient UI apps in SAP Fiori launchpad integrated mode, you can use the function profiles for different processes offered with function profiles. When using SAP Fiori launchpad integrated mode, you have a backend authorization role, which contains the catalogs for WebClient UI applications to navigate to certain target IDs.

Procedure

In Customizing ► **UI Framework** ► **Technical Role Definition** ► **Define Fiori Launchpad Catalogs** ►.

- Assign the function profiles to SAP Fiori launchpad catalogs.
- Create entries for the WebClient UI function profiles.
- Define the function profiles and values for SAP Fiori launchpad integrated mode for each catalog.
- Assign PFCG roles in the backend, which contains the catalogs.

i Note

If the user is assigned multiple catalogs, in the runtime, all the function profiles are loaded. A warning message is displayed if there are conflicting function profiles and the system uses the first value found.

Outbound Navigation

Use

With outbound mapping, you can navigate from WebClient UI applications directly to other apps available in SAP Fiori launchpad. These apps can be SAP Fiori apps, as well as applications based on Web Dynpro and SAP GUI for HTML technology. You must have access to these other apps from your SAP Fiori launchpad for the outbound mapping to work.

The internal navigation within WebClient UI works without any customizing, provided you have access to the applications.

Procedure

To use the outbound navigation, make entries in Customizing [►►UI Framework](#) [► Technical Role Definition](#) [► Define FLP Navigation Parameter Mapping](#) [►](#)

WebClient UI Framework Dashboard

Use

With the WebClient UI framework dashboard you can use features and settings specific to WebClient UI. The main settings like the theme, and the language are available in the SAP Fiori launchpad user settings.

The dashboard contains the following features:

- **General Settings:** Personalize your general settings specific to WebClient UI.
- **Performance Settings:** Manage performance settings specific to WebClient UI.
- **Shortcuts:** Personalize keyboard shortcuts specific to WebClient UI.
- **Favorites:** Access and maintain WebClient UI favorites.
- **Tags:** Access and maintain tag clouds.
- **Central Sharing Tools:** Share and display the objects that have been shared with you.
- **Saved Search:** To execute and maintain the saved searches.

Procedure

To access the WebClient UI dashboard, use the **My Dashboard** icon from the system toolbar of the WebClient UI page. You can personalize the content of the WebClient UI dashboard.

The WebClient UI dashboard features are available through function profiles. Link the required function profiles to the catalogs assigned to the user via the PFCG role.

- Function profile for **Central Sharing Tool:** CENTRAL_SHARING_TOOL
- Parameters for **Favorites:** FAVORITES_ALLOWED, FAVORITES_COMMUNITY
- Parameters for **Tags:** TAG_CLOUDS_ALLOWED, TAG_CLOUDS_COMMUNITY

Fact Sheet

Use

The fact sheet gives you an immediate and compressed overview of a particular business object.

The data displayed in the fact sheet can vary greatly. For this reason, you can configure the fact sheet and determine its layout yourself.

Features

- Format

The information contained in the fact sheet is displayed in assignment blocks that group together certain types of information.

- Availability

The fact sheet is available online. You can display the fact sheet and make changes to it such as showing table columns, if you have made the columns personalizable in Customizing. You can also print the fact sheet with your browser's print function.

- Configuration

You can determine which views you want to display in the fact sheet. You can choose between forms, tables, or views with graphics.

- Customizing

You can determine the way the fact sheet is laid out, provided that you have chosen the tile layout. With this layout, you can determine the number of rows and columns in which the tiles are arranged.

Example

The Account fact sheet is available in Account Management on the WebClient UI. On the overview page of the individual accounts, you can click [Fact Sheet](#) and receive the most important information about this account online.

This information can come from multiple sources, for example master data, statistical data, and operation data. It enables you to quickly and easily see the most essential details about your key customers and partners from the various business transactions.

For example, you want to visit one of your customers. You open the fact sheet about this customer and receive all the important information about this customer at a glance.

Configuring the Fact Sheet

Use

You can adjust the fact sheet to the special requirements of your company.

Prerequisites

- You can define the page layout of the fact sheet in Customizing for [UI Framework](#) under [UI Framework Definition](#) [Fact Sheet](#) [Maintain Fact Sheet](#). These settings are only valid for the page layout with tiles.
- You can change the title of the fact sheet and define placeholders by using the Business Add-In (BAdI) `BSP_DLC_FS_BADI` that belongs to the enhancement spot `BSP_DLC_FS`.

Features

- You can choose between different page layouts.

You have the choice between a single column page layout and a page layout that consists of multiple tiles.

- You can choose the available views and assign them to the fact sheet.

You can use the same view multiple times. For example, you can assign a view to a report from SAP NetWeaver Business Intelligence (BI) as well and then rename it.

❖ Example

You have created an account fact sheet that you want to use in a certain role. This account fact sheet includes the view **Opportunities** which is also displayed in the account fact sheet used in another role.

- You can choose between various load options for the fact sheet views.

Activities

Starting the Application Component

Enter transaction BSP_WD_CMPWB and the application component BSP_DLC_FS. In the **Browser Component Structure** under **Views**, select the view BSP_DLC_FS/fact sheet and click **Configuration**.

You can start the fact sheet configuration also in WebClient UI via hyperlink in the navigation bar. To make the fact sheet configuration accessible in your system administrator role in the WebClient UI, you have made the necessary settings in Customizing for **UI Framework**, by choosing **►Technical Role Definition ►Define Navigation Bar Profile ►**

Using the Role Configuration Key

Choose an existing role configuration key or use the default one. If you want to create a new configuration, you can choose between a single column layout and a layout that consists of tiles.

Choosing and Assigning Fact Sheet Views

- Single column layout

Use the **UP ARROW** and **DOWN ARROW** keys to move views between the **Available Fact Sheet Views** and **Assigned Fact Sheet Views** areas.

Assign a load option to each view in the **Assigned Fact Sheet Views** area.

- Page layout with tiles

Select a tile and use the arrow key to move the views that you want to display in the fact sheet from the **Available Fact Sheet Views** area to the **Assigned Fact Sheet Views** area.

Changing the Fact Sheet Title

If you want to change the title of the fact sheet, click **Properties**. You can use the default title or enter your own title in the **Config Title** field.

You can make further changes to the title and other fact sheet attributes by using a Business Add In (BAdI). Make these changes in the **BAdI Builder** transaction (SE18) and the enhancement spot BSP_DLC_FS. There is the default and example implementation BSP_DLC_FS_IMPL and the enhancement implementation BP_FACTSHEET for the Account fact sheet.

Displaying Technical Settings

Click **Show Technical Details** if you want to see the technical properties of the available and assigned fact sheet views.

Example

You can configure the Account fact sheet yourself. To do so, enter transaction BSP_WD_CMPWB and open the component BP_FACTSHEET.

Using Attributes for Assignment Blocks

Use

You can define attributes, which you can assign to your own application components.

Prerequisites

You have assigned attribute structures using the [Table View Maintenance](#) transaction (SM30) in table BSPC_DL_VIEWS.

Activities

1. Start transaction BSP_WD_CMPWB.
2. Start your application component and the required view.
3. Click [Configuration](#).
4. Select [More](#) [Attributes](#).
5. In the [Displayed Assignment Blocks](#) area, you can assign attributes to the views for which you have previously defined attribute structures in Customizing.

Adjusting the Navigation Bar Profile

Use

Create an entry for the fact sheet in the navigation bar profiles that you use in generic outbound plug mapping (OP mapping).

Prerequisites

Create the entry for the fact sheet in Customizing for [UI Framework](#) under [Technical Role Definition](#) [Define Navigation Bar Profile](#). Select the required navigation bar profile in the IMG activity and navigate to [Define Profile](#) [Define Generic OP Mapping](#). Create a new entry and enter FACTSHEET as the object type, DISPLAY as the object action and BSP_DLC_FS as the target ID.

Creating and Assigning Views

Use

You can create your own views and assign them to a fact sheet. You can choose between the following two options:

- Option 1:
 - Create your own component

- Create your views and enter a configuration key
- Option 2:
 - Use the framework enhancement concept
 - Create your views in an existing component, for example BP_FACTSHEET, and enter a configuration key

Activities

In both cases, proceed as follows:

1. Create an inbound plug and assign it to a window.

Using this process, you can assign multiple views to a window simultaneously. Alternatively, you can create a window for each view.

2. To calculate the property that is based on the BOL relation, for example, the business partner ID, write the program code in the inbound plug method. Alternatively, you can use the global data context.
3. Configure the views.
4. Assign the views to an existing fact sheet.

You make this assignment in Customizing for **UI Framework** under **UI Framework Definition > Fact Sheet > Maintain Fact Sheet**.

5. Configure the fact sheet component (BSP_DLC_FS) in transaction BSP_WD_CMPWB.

More Information

[How Do I Create Components and Views?](#)

[Framework Enhancement](#)

How Do I Create Components and Views?

Procedure

The following procedure corresponds to option 1, in which you create your own component, create your own views within this component, and define a configuration key.

Creating Components and Views

1. Enter the transaction code BSP_WD_CMPWB.
2. Enter a component name and click **Create**.
3. Create a new view with the following parameters:
 - Name
 - BOL entity
 - Model node
 - Model attribute

- View type: Table

Specifying the Configuration Key

1. Select your view under **Browser Component Structure > Views**.
2. Click **Structure**.
3. Double-click the *.htm file under **View Layout**.

You see the layout of the selected view.

4. Click **Page Attributes**.

You see the attributes that are assigned to the view.

Creating an Inbound Pug

1. Under **Structure > Inbound Plugs**, create a new inbound plug with the name DEFAULT.
2. Under **Runtime Repository Editor > Component > Windows**, select the **Main Window** to which you want to assign the inbound plug.
3. In the context menu, choose **Add Plug with Follow-Up Navigation**.

You see the page **Create Inbound Plug with Forwarding Navigational Link**.

4. Enter the appropriate values.
5. Under **Runtime Repository Editor > Component > Component Interface > Interface View**, add the inbound plug.
6. Under **Browser Component Structure > Component > Views**, select your view.
7. Double-click your inbound plug under **Structure > Inbound Plugs**.
8. In the **Class Builder**, enter the program code to calculate the property that is based on the BOL relation.

Assigning Views to an Existing Fact Sheet

1. Click **Configuration** and configure your views.
2. Assign the views to an existing fact sheet.

You make this assignment in Customizing for **UI Framework** under **UI Framework Definition > Fact Sheet > Maintain Fact Sheet**.

3. Add a new entry in Customizing for each view.

Configuring the Fact Sheet

1. Start component BSP_DLC_FS in transaction BSP_WD_CMPWB.
2. Under **Views**, choose the view BSP_DLC_FS/fact sheet.
3. Configure the fact sheet under **Configuration**.

More Information

[Creating and Assigning Views](#)

[Configuring the Fact Sheet](#)

Framework Enhancement

Use

The framework enhancement concept is based on component enhancements, which means that you enhance the function of standard components, standard views, and standard controllers that are delivered by SAP.

First, you verify whether you can use the UI configuration tool, for example, if you want to adjust SAP standard components to your business needs. If you want to make functional changes in a component, you need to use the enhancement concept.

You can also create your own components.

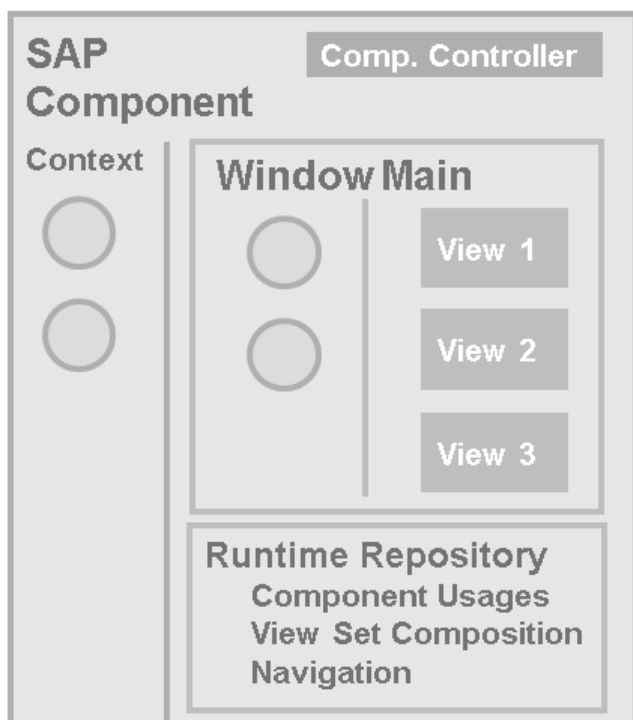
i Note

For more information about the framework enhancement concept, see SAP Note 1122248.

Features

UI Configuration or Component Enhancement?

You can adjust SAP standard components either with the UI configuration tool or with the component enhancement concept. You can use the UI configuration tool, for example, if you want to make specific layout changes in views. If you want to make functional changes in a component, for example, create new events, you need to use the component enhancement concept.



Configure Views

Save Clear Configuration Simple Configuration Show Item Set Show Technical Details Make 16 columns / 1 panel

Customer Default Configuration

To get field properties, ALT + mouse click. For multiple selection, CTRL + mouse click.

Add Caption to Left Panel

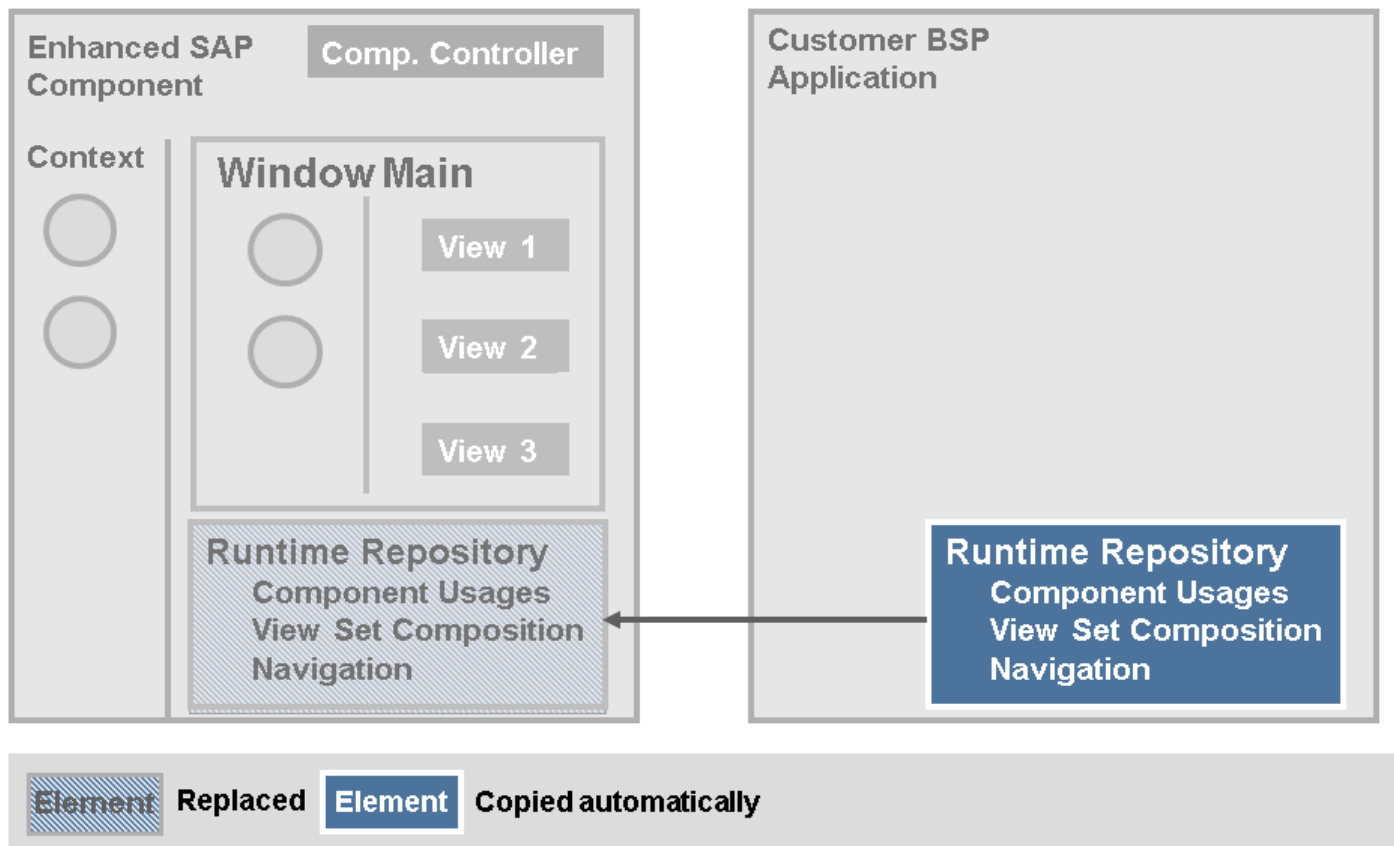
Apply:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	General Data								References							
2	Description								Account							
3	Location								Contact							
4	Start Date								Employee Responsible							
5	Time From/To								Reference							
6	Category								Private							
7	Direction															
8	Importance															
9	Status															

Enhance Component (wizard-supported)

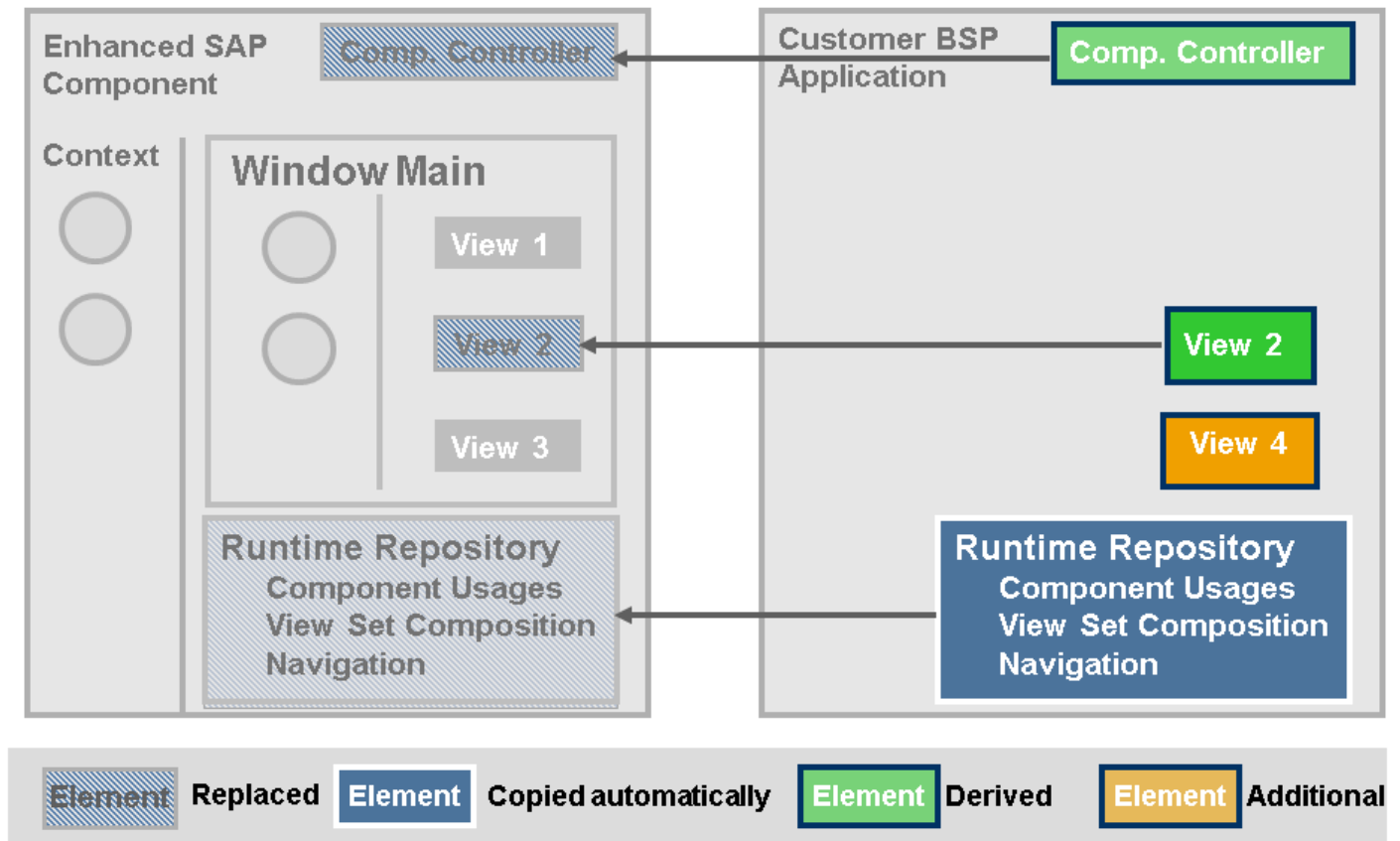
Component Enhancement: Step 1

If you use this wizard-supported approach, the system automatically creates a BSP application for the enhanced elements (runtime repository, derived controllers, and so on). It copies the runtime repository and maintains replacement information.

**Component Enhancement: Step 2**

When a component enhancement has been created, the SAP runtime repository is replaced by a customer version. Now it is possible to enhance views and controllers of the component.

If you need to enhance a view or controller, the wizard automatically guarantees that the required inheritance logic does not need to be changed in your system. The system automatically creates a BSP controller and view. It derives controller and context classes and maintains controller replacement information. Now the SAP version of the view or controller is replaced by the customer version.



Enhancement Set

Use

An enhancement set is used to group several component enhancements together.

Features

- An enhancement set contains the enhancement definition that shows the assigned BSP application and the assigned runtime repository.
- You can define as many enhancement sets as you like, but only one enhancement set is active at runtime.
- Based on the enhancement set, you can create enhancements in the BSP Component Workbench.

⚠ Caution

The enhancement set is not yet available as an attribute in the UI configuration tool. Therefore, you can only have one configuration for every enhancement. If a configuration is created for an enhanced object, some objects that are used in this configuration may not be available for the corresponding object that is not enhanced.

Activities

Define Enhancement Set

To define an enhancement set, use transaction SM34. Open the view cluster BSPWDVC_CMP_EXT and create the enhancement set that you can later use in the BSP Component Workbench. You only need to enter a name and a description, because the definition is automatically filled later when you create enhancements in the BSP Component Workbench.

Determine Active Enhancement Set

If the determination of an enhancement set at runtime is not only to be based on the client, but on other criteria, such as the user or the profile, you need to use a Business Add-In (BAI). You can use the [Object Navigator](#) (SE80) to open package CRM_BSP_WD. Open the enhancement spot COMPONENT_HANDLING with the BAI definition COMPONENT_LOADING and the DEFAULT implementation. This implementation uses the customizing entries from BSPWD_EHSET_ASGN to determine the enhancement based on a client. You can also use this BAI to implement other determination rules for an enhancement set.

Assign Enhancement Set to a Client

To assign an enhancement set to a client, use transaction SM30. Open table BSPWDV_EHSET_ASG to make the necessary assignment. You can only assign one enhancement set to a client.

Customer Namespace

Use

In the SAP standard, the prefix 'Z' is used for objects that are created in the customer namespace. In the Framework enhancement concept, this has been realized with Business Add-Ins (BAI). However, you can create and save your enhancements in your own namespace.

Features

- The Business Add-In BSP_WD_APPL_WB with the method GET_CUST_CLASSNAME_PROPOSAL provides you with a proposal for a name for customer-defined classes within the enhancement. The default implementation of this method replaces the first letter in the name with 'Z'. You can create your own implementation for this method to place all new class names in your own namespace / * / if necessary.
- The enhancement spot CLASS_GENERATION with the Business Add-In CLASS_NAME_BUILDER with five methods can be used to generate the class names for customer views and controllers. There is no default implementation. If you implement the BAI, you get a proposal for a name that you can change.

Activities

Use Your Own Namespace for Enhanced Controllers and Views

If you want to use your own namespace, you can use the [Object Navigator](#) (SE80) and open the package CRM_BSP_WD_TOOLS. Open the business add-in BSP_WD_APPL_WB. Under [Methods](#), you find the method GET_CUST_CLASSNAME_PROPOSAL.

Create Your Own Objects in Your Namespace

If you want to create your own objects in your own namespace, you can use the [Object Navigator](#) (SE80). Open the package CRM_BSP_WD_GEN_TOOLS. Under [Enhancements](#) [Enhancement Spots](#) you find the BAI CLASS_GENERATION with the BAI definition CLASS_NAME_BUILDER and the interface IF_BSP_WD_GEN_TOOLS.

i Note

With this BAdI, no default implementation is used.

Create Enhancements

Procedure

If you want to enhance components in the UI Component Workbench, proceed as follows:

1. Start transaction BSP_WD_CMPWB.
2. Choose the component that you want to enhance.
3. Choose the **Display** pushbutton.
4. Click the **Enhance Component** pushbutton.
5. Choose the enhancement set.
6. In the **BSP Application for Storing Enhancement Objects** dialog box, choose an existing BSP application or create a new one.

→ Recommendation

We recommend that you create your own BSP application in which you store the enhancements for each component. If you have not yet created enhancements for this component, enter a new name for the BSP application. For more information, see SAP Note [1122248](#) .

7. Select a view under **Browser Component Structure > Component > Views**.

i Note

All gray objects have not yet been enhanced. All black objects have already been enhanced. All blue objects are obsolete, since they are not used in the current release.

8. Choose **Enhance** in the context menu.

The object is created automatically in the customer namespace.

9. Select a workbench request.
10. Select a package.

You can create a new package in transaction SE 80.

11. Make the necessary enhancements.

i Note

The different colors of the icons to the left of the methods have the following meanings:

- Yellow means that the method is inherited
- Green means that the method is redefined
- Gray means that the method does not exist

i Note

You can delete enhancements by choosing the [Delete Component Enhancement](#) pushbutton.

Result

In the view cluster BSPWDVC_CMP_EXT, which you can open with transaction SM34, you can see the substitution.

More Information

[Enhancement Set](#)

UI Configuration Determination

Multiple Configurations

Use

The configuration of views, for example field labels, is based on configurations that have been created either with the UI Configuration Tool, in the design layer Customizing, or in the ABAP Dictionary.

Features

Multiple Configurations at Design Time

At design time, you can create multiple configurations using the different keys that are available for views. The available and supported keys are:

- Component (automatically set by the framework)
- View (automatically set by the framework)
- Role configuration key
- Component usage
- Object type
- Subobject type (depending on the object type)

Multiple Configurations at Runtime

At runtime, the multi-configuration is accessed using the following:

- Component (automatically set by the framework)
- View (automatically set by the framework)
- Role configuration key (automatically determined by the business role with which the user has logged on)
- Component usage (automatically set by the framework, as it is derived from the component usage itself)
- UI object type (set by the application)

- Subobject type (set by the application)

Determination of Different Keys

- You can define the role configuration key in Customizing for **UI Framework** under **►Technical Role Definition ► Define Role Configuration Key ►**.
- You can define the UI object type in Customizing for **UI Framework** under **►UI Framework Definition ► Define UI Object Types ►**.
- Subobject types are determined via the callback class, assigned to the UI object type in Customizing for **UI Framework** under **►UI Framework Definition ► Define UI Object Types ►**. The interface is `IF_BSP_DLC_OBJ_TYPE_CALLBACK~GET_OBJECT_SUB_TYPES`.
- UI object type and subobject type can be set by the application in method `DO_CONFIG_METHOD`.

Determination of Assigned Design Objects

For the determination of assigned design objects, the following keys are used:

- Component
- Component usage
- View
- Context node
- UI object type
- Attribute name

More Information

[UI Configuration Tool](#)

[Design Layer](#)

General Access Sequence

General Access Sequence

The figure below illustrates the general access sequence for configuration determination and field label determination:

Configuration Determination:

- 1 Configuration**
 - 1.1 Customertable**
 - 1.2 SAP table**

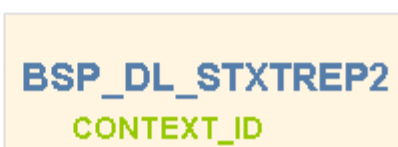
Field Label Determination:

- 1 Customertable**
 - 1.1 Textrepository**
 - 1.2 Design layer**
- 2 SAP table**
 - 2.1 Textrepository**
 - 2.2 Design layer**
- 3 ABAP dictionary**

There is a special handling for the assignment of block headers in overview pages which shows the technical name of the assignment block if the label does not exist. English values are shown in the case that the assignment block labels do not exist in a requested language.

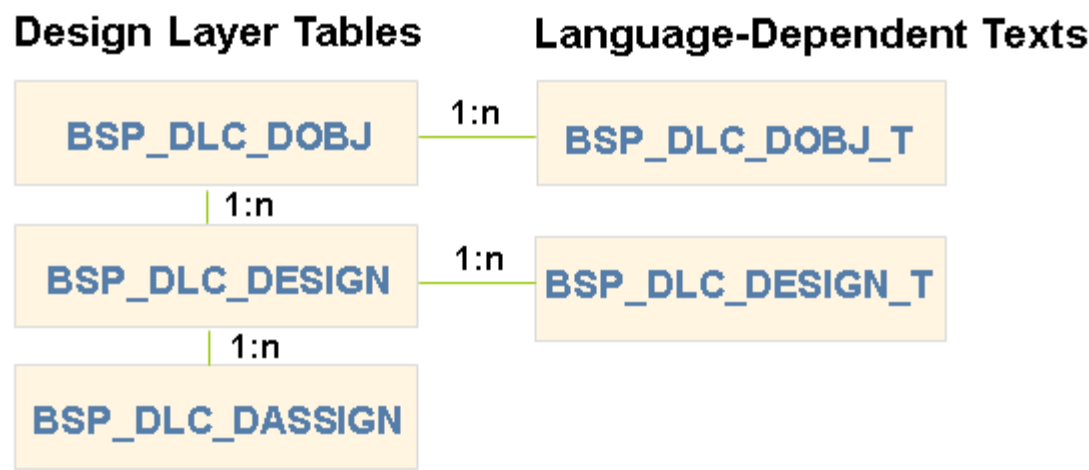
Data Model: UI Configuration/Text Repository

The figure below illustrates the data model of UI configuration and text repository:

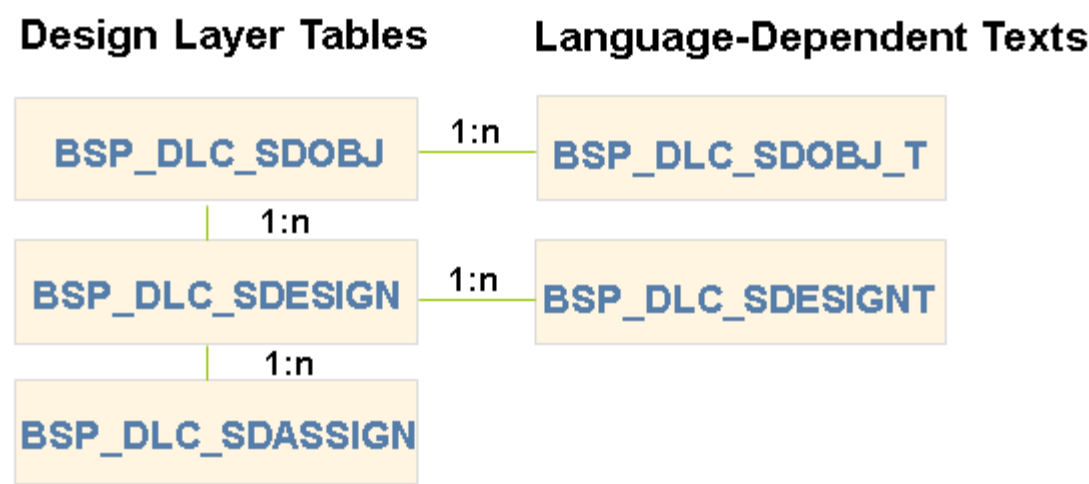
Customer Tables**Configuration****Language-Dependent Texts****SAP Tables****Configuration****Language-Dependent Texts**

The figure below illustrates the data model of the design layer:

Customer Tables



SAP Tables



Access Sequence Logic

Use

The determination takes place in two steps:

- Step 1: A configuration is determined, based on the search key.
- Step 2: Field labels from the text repository are determined, based on the successful key.

Step 1

The table below shows how the configuration is determined, based on the search key. The access sequence for customer tables and SAP tables is identical.

No.	Role Configuration Key	Component Usage	Object Type	Subobject Type	Configuration Table
1	X	X	X	X	Customer Table
2	X	X	X	<default>	Customer Table
3	X	<default>	X	X	Customer Table
4	<default>	X	X	X	Customer Table
5	<default>	<default>	X	X	Customer Table
6	X	<default>	X	<default>	Customer Table
7	<default>	<default>	X	<default>	Customer Table
8	X	X	<default>	<default>	Customer Table
9	<default>	X	<default>	<default>	Customer Table
10	X	<default>	<default>	<default>	Customer Table
11	<default>	<default>	<default>	<default>	Customer Table
12	X	X	X	X	SAP Table
13	X	X	X	<default>	SAP Table
14	X	<default>	X	X	SAP Table
15	<default>	X	X	X	SAP Table
16	<default>	<default>	X	X	SAP Table
17	X	<default>	X	<default>	SAP Table
18	<default>	<default>	X	<default>	SAP Table
19	X	X	<default>	<default>	SAP Table
20	<default>	X	<default>	<default>	SAP Table
21	X	<default>	<default>	<default>	SAP Table
22	<default>	<default>	<default>	<default>	SAP Table

Step 2

The table below shows how the field labels are determined from the text repository, based on the successful key. The search key is used again for the determination of the design object.

No.	Role Configuration Key	Component Usage	Object Type	Subobject Type	Attribute Name	Field Label Saved in
1	<successful key>				X	Customer Table (Text Repository)
2	<default>	<default>	<default>	<default>	X	Customer Table (Text Repository)
3		X	X		X	Customer Table (Design Layer)
4		X	X		<default>	Customer Table (Design Layer)
5		<default>	X		X	Customer Table (Design Layer)
6		<default>	X		<default>	Customer Table (Design Layer)
7		<default>	<default>		X	Customer Table (Design Layer)
8		<default>	<default>		<default>	Customer Table (Design Layer)
9	<successful key>				X	SAP Table (Text Repository)
10	<default>	<default>	<default>	<default>	X	SAP Table (Text Repository)
11		X	X		X	SAP Table (Design Layer)
12		X	X		<default>	SAP Table (Design Layer)
13		<default>	X		X	SAP Table (Design Layer)
14		<default>	X		<default>	SAP Table (Design Layer)
15		<default>	<default>		X	SAP Table (Design Layer)
16		<default>	<default>		<default>	SAP Table (Design Layer)
17					X	ABAP Dictionary

i Note

The gray color in the table means that these parameters exist neither in customer tables and/or SAP tables in the design layer, nor in the ABAP Dictionary. When you create a configuration, you can use the above-mentioned parameters. An X means that you have entered a parameter value in the configuration. If you do not use these parameters in the configuration, this is represented by <default>.

Features

To determine the field label, the configuration text repository is searched, based on the successful key. If no field label is found and the search is processed in the design layer, the design object determination is started with the search key.

If you use other properties of the design layer Customizing, for example, you have set a field to hidden, the same access sequence logic applies as for the field label determination.

When a design object that has a more specific search key than the related configuration exists, special design layer settings based on this search key are not immediately visible if you are looking at this configuration in the UI Configuration Tool.

❖ Example

Assume that for the account details view (BP_HEAD/AccountDetails), a configuration with the following parameters exists:

- Role configuration key = <DEFAULT>
- Component usage = <DEFAULT>

- Object type = <DEFAULT>
- Subobject type = <DEFAULT>

You can see this configuration in the UI Configuration Tool with the field labels that are based on this configuration, for example **Name 1**.

Assume that a design object with a more specific key combination exists, for example, object type = BP_ACCOUNT, renamed field label **Account** instead of **Name 1**, assigned to the account details view BP_HEAD/AccountDetails. At runtime, the renamed field label from the design layer is displayed. However, if you look at the default configuration in the UI Configuration Tool, you cannot see this specific key combination.

More Information

[Design Layer Customizing](#)

[UI Configuration Tool](#)

Example: Configuration with a Renamed Field Label

Design Time

The table below shows several configurations that have been entered in the system at design time. In the **Table Type** column, you can see in which layer the configuration exists and in which table (customer table or SAP table) the configuration is saved.

No.	Component	View	Role Configuration Key	Object Type	Subobject Type	Attribute Name	Field Label	Table Type
1	BT116H_SRVO	DetailsEF	<default>	<default>	<default>	DESCRIPTION	<default>	Configuration Customer Table
2	BT116H_SRVO	DetailsEF	<default>	SRVQ	<default>	DESCRIPTION	<default>	Configuration SAP Table
3	BT116H_SRVO	DetailsEF	<default>	SRVT	<default>	DESCRIPTION	<default>	Configuration SAP Table
4	BT116H_SRVO	DetailsEF		SRVT		DESCRIPTION	Information	Design Layer Customer Table
5						DESCRIPTION	Description	ABAP Dictionary

i Note

The gray color means that these parameters exist neither in the design layer, nor in the ABAP Dictionary.

Runtime: Configuration Determination

At runtime, the user works with the following component, view, and object type. The configuration is determined using this search key combination, which consists of:

- Component: BT116H_SRVO
- View: DetailsEF

- Object Type: SRVQ

First, the customer tables are checked. According to the general access sequence, the configuration with the default values in the customer configuration is determined by the system.

No.	Role Configuration Key	Component Usage	Object Type	Subobject Type	Configuration Table	
1	X	X	X	X	Customer Table	} Checked
2	X	X	X	<default>	Customer Table	
3	X	<default>	X	X	Customer Table	
4	<default>	X	X	X	Customer Table	
5	<default>	<default>	X	X	Customer Table	
6	X	<default>	X	<default>	Customer Table	
7	<default>	<default>	X	<default>	Customer Table	
8	X	X	<default>	<default>	Customer Table	
9	<default>	X	<default>	<default>	Customer Table	
10	X	<default>	<default>	<default>	Customer Table	
11	<default>	<default>	<default>	<default>	Customer Table	} Determined

Runtime: Determination of the Field Label Renamed in the Design Layer

Based on the successful key, the text repository is searched. If no matching entry was found in the text repository, the design layer is checked, but the search key is used again for the design object determination.

According to the general access sequence, the customer tables are checked first (see first part of the table), before the SAP tables are checked (see second part of the table).

First Part of the Table

Role Configuration Key	Component Usage	Object Type	Subobject Type	Attribute Name	UI Configuration Saved in
X	X	X	X	X	Customer Table (Text Repository)
X	X	X	<default>	X	Customer Table (Text Repository)
<default>	X	X	X	X	Customer Table (Text Repository)
X	<default>	X	X	X	Customer Table (Text Repository)
<default>	<default>	X	X	X	Customer Table (Text Repository)
<default>	X	X	<default>	X	Customer Table (Text Repository)
<default>	<default>	X	<default>	X	Customer Table (Text Repository)
X	X	<default>	<default>	X	Customer Table (Text Repository)
X	<default>	<default>	<default>	X	Customer Table (Text Repository)
<default>	X	<default>	<default>	X	Customer Table (Text Repository)
<default>	<default>	<default>	<default>	X	Customer Table (Text Repository)
	X	X		X	Customer Table (Design Layer)
	X	X		<default>	Customer Table (Design Layer)
	<default>	X		X	Customer Table (Design Layer)
	<default>	X		<default>	Customer Table (Design Layer)
	<default>	<default>		X	Customer Table (Design Layer)
	<default>	<default>		<default>	Customer Table (Design Layer)

Checked
 Determined

Second Part of the Table

Role Configuration Key	Component Usage	Object Type	Subobject Type	Attribute Name	UI Configuration Saved in
X	X	X	X	X	SAP Table (Text Repository)
X	X	X	<default>	X	SAP Table (Text Repository)
<default>	X	X	X	X	SAP Table (Text Repository)
X	<default>	X	X	X	SAP Table (Text Repository)
<default>	<default>	X	X	X	SAP Table (Text Repository)
<default>	X	X	<default>	X	SAP Table (Text Repository)
<default>	<default>	X	<default>	X	SAP Table (Text Repository)
X	X	<default>	<default>	X	SAP Table (Text Repository)
X	<default>	<default>	<default>	X	SAP Table (Text Repository)
<default>	X	<default>	<default>	X	SAP Table (Text Repository)
<default>	<default>	<default>	<default>	X	SAP Table (Text Repository)
	X	X		X	SAP Table (Design Layer)
	X	X		<default>	SAP Table (Design Layer)
	<default>	X		X	SAP Table (Design Layer)
	<default>	X		<default>	SAP Table (Design Layer)
	<default>	<default>		X	SAP Table (Design Layer)
	<default>	<default>		<default>	SAP Table (Design Layer)
				X	ABAP Dictionary

i Note

The gray color in the table means that these parameters exist neither in customer tables and/or SAP tables in the design layer, nor in the ABAP Dictionary. When you create a configuration, you can use the above-mentioned parameters. An X means that you have entered a parameter value in the configuration. If you do not use these parameters in the configuration, this is represented by <default>.

Enhancement Spot

Use

You can change the access sequence logic completely by implementing an enhancement spot. For a given set of four key fields, your implementation selects one of the available configurations.

Assume you want to implement a template approach regarding role-based configurations. The intention is to be able to replace the SAP fallback (role=<DEFAULT>) with the fallback of your customer-specific role. If no customer-specific configuration is available for the current role, you can select a special configuration for a “fallback” role. This means that you do not need to maintain all configurations for all available business roles.

Integration

You can find the enhancement spot in Customizing for **UI Framework** under **UI Framework Definition** > **Business Add-Ins (BAdIs)** > **Define Configuration Access**.